

Design Simulation and Prototyping of an Automated Multi-Task Agricultural Vehicle for Soil Digging, Seed Sowing, and Tilling Operation



Addis Ababa Science and Technology University

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Submitted by:

Abel Addis	ETS0036/14
Abenezer Tsegaye	ETS0065/14
Abenezer Mekonnen	ETS0072/14
Abenezer Ketema	ETS0075/14
Abenezer Wendesen	ETS0076/14

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Abstract

This project focused on the development of an automated multi-task agricultural vehicle for urban and small-plot farming to improve soil preparation, seed placement accuracy, and overall planting efficiency. The study targeted compact agricultural systems where conventional manual tools and small-scale mechanized equipment failed to achieve consistent soil digging, precise seed deposition, and uniform soil coverage. The problem was mainly caused by limitations in existing agricultural tools and mechanical designs, which resulted in uneven soil preparation, inaccurate seed placement, and reduced crop uniformity. In addition, poor coordination between mechanical and electronic subsystems contributed to low efficiency in small-scale planting operations.

The developed system integrated soil digging, seed sowing, and soil covering mechanisms into a single compact robotic platform. Mechanical design and structural analysis of the digging, seed-metering, and covering mechanisms were carried out using SolidWorks. Dynamic modelling of the vehicle was developed using the Euler–Lagrange method and represented using state-space equations for system analysis and controller design. PID controllers were designed and tuned based on mathematical modelling and steady-state error analysis to regulate locomotion, digging, and seed-dispensing operations. A functional prototype was fabricated and integrated with sensors, actuators, motor drivers, and embedded control systems. Simulation and experimental analyses were conducted to evaluate motion accuracy, digging consistency, seed placement repeatability, and overall operational performance.

The developed system resulted in a compact, energy-efficient robotic platform capable of performing intermittent soil digging, precise seed deposition, and controlled soil covering with improved accuracy compared to conventional manual methods. The system reduced labor requirements, improved planting uniformity, and demonstrated the feasibility of integrating electromechanical systems, embedded control, and precision agriculture techniques into a low-cost agricultural solution suitable for urban gardens, research farms, and small-scale farming environments.

Keywords:- *Automated Agriculture, Mobile Robot, Soil Digging Mechanism, Seed Sowing, PID Control, Electromechanical System*

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Approval

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Department of Electromechanical Engineering

The Thesis Paper titled “*Design, Simulation and Prototyping of an Automated Multi-Task Agricultural Vehicle for Soil Digging, Seed Sowing, and Tilling Operation*” is approved for the Bachelor of Science Degree in Electromechanical Engineering.

Advisor Name

Thesis Adviser

Date

Co-advisor Name

Thesis Co-Adviser

Date

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Acronyms

Abbreviation	Description
AASTU	Addis Ababa Science and Technology University
CAD	Computer-Aided Design
CAM	Computer-Aided Manufacturing
DC	Direct Current
AC	Alternating Current
DoF	Degree of Freedom
PID	Proportional-Integral-Derivative
LQR	Linear Quadratic Regulator
FEA	Finite Element Analysis
FBD	Free-Body Diagram
PWM	Pulse Width Modulation
RPM	Revolution Per Minute
MCU	Microcontroller Unit
IMU	Inertial Measurement Unit
IR	Infrared
Li-ion	Lithium-ion Battery
NTC	Negative Temperature Coefficient Sensor
MATLAB	Matrix Laboratory
SNR	Signal-to-Noise Ratio
GUI	Graphical User Interface
STL	Stereolithography File (3D printing)
ROS	Robot Operating System (optional)
ESC	Electronic Speed Controller
DoD	Depth of Digging
ToR	Torque Requirement
SoC	State of Charge (battery)
kWh	Kilowatt-hour
ESP32	Espressif Systems 32-bit Wi-Fi Microcontroller
PCB	Printed Circuit Board
BMS	Battery Management System
LoS	Line of Sight (for sensors)
CG	Center of Gravity

1 Introduction

1.1 Background of the Study

Agriculture remains one of the most labor-intensive sectors globally, particularly in developing nations where mechanization is limited and smallholder farming dominates. In Ethiopia, traditional farming practices such as manual soil digging, seed placement, and tilling continue to rely heavily on human and animal labor. These practices are not only time-consuming and physically demanding but also prone to inconsistencies in depth control, spacing, and coverage, which directly affect crop yield and soil health. The increasing demand for food production, coupled with labor shortages and rising land degradation, created the need for innovations that enhance farming efficiency, precision, and sustainability.

The automation of agricultural operations emerged in the 1950s with the introduction of basic machine-guided tractors and mechanical seed drills. Over time, advancements in sensors, microcontrollers, and electromechanical actuation enabled more sophisticated systems capable of performing repetitive field tasks with improved accuracy. Research trends highlighted the integration of robotics into agriculture, where mobile platforms equipped with GPS, vision systems, and intelligent controllers automated operations such as planting, weeding, and soil preparation. Blackmore (1994) introduced the concept of precision agriculture, emphasizing controlled and localized farming operations to optimize productivity and reduce waste. More recent studies demonstrated a growing shift toward compact autonomous agricultural robots capable of executing multi-step farming tasks. [5/19/2026 9:38 PM] Abeni Tsegsh: Although several agricultural robotic systems were developed, many designs focused only on single-purpose operations and lacked the integration of multiple farming tasks into one mobile system. Furthermore, most advanced agricultural robots were designed for large-scale commercial farms in developed countries, making them financially and technically inaccessible for smallholder farmers in Ethiopia and similar developing regions. Existing systems also experienced challenges related to uneven terrain navigation, reliable soil-tool interaction, and adaptability to varying soil conditions and moisture levels.

These research gaps highlighted the need for an affordable, lightweight, energy-efficient, and multi-functional agricultural vehicle capable of automating essential farming operations for small-scale farms. Integrating soil digging, seed sowing, and soil covering into a single autonomous platform significantly reduced labor demand, improved planting consistency, and enhanced soil preparation quality. The developed electrome-

chanical system combined mechanical design, embedded control, and real-time sensing to provide the precision and repeatability lacking in traditional farming methods.

Therefore, this study contributed to the advancement of autonomous agricultural technologies by developing a mobile robotic vehicle capable of performing soil digging, seed dispensing, and soil covering operations within a unified system. The developed work addressed technical, operational, and economic limitations associated with existing agricultural mechanization solutions and provided a practical alternative tailored to the needs of smallholder farmers and compact agricultural environments.

1.2 Problem Statement

Small-scale farmers in Ethiopia and many developing countries continue to depend heavily on manual labor for soil preparation, seed sowing, and land finishing operations. These farming activities are labor-intensive, time-consuming, and inconsistent, resulting in reduced productivity and poor planting quality. Although autonomous agricultural machines exist internationally, many of them are expensive, difficult to maintain, and unsuitable for small and irregular farmlands common in Ethiopia.

Previous agricultural robotic systems mainly focused on single-function operations such as basic seed sowing or soil digging, leaving a significant gap in the development of integrated, low-cost, and multi-task agricultural vehicles. In addition, many available systems relied on complex control architectures, expensive sensors, and advanced automation technologies that are impractical for low-resource agricultural environments.

This research addressed these limitations by designing, modelling, and developing an automated agricultural vehicle capable of performing soil digging, seed sowing, and soil covering operations in a coordinated manner. The system incorporated electromechanical subsystems, embedded sensing, actuator mechanisms, and a coordinated control strategy to achieve autonomous or semi-autonomous operation. The developed platform improved planting consistency, reduced labor requirements, and demonstrated the feasibility of implementing affordable precision agriculture technologies for small-scale farming applications.

1.3 Objectives of the Research

1.3.1 General Objective

To design, model, and develop an automated multi-task agricultural vehicle capable of performing soil digging, seed sowing, and soil covering operations using integrated

electromechanical systems and control mechanisms.

1.3.2 Specific Objectives

- To develop and analyze the mechanical subsystems for soil digging, seed dispensing, and soil covering using SolidWorks.
- To develop the dynamic model of the agricultural vehicle using the Euler–Lagrange method.
- To design and tune PID controllers for coordinated locomotion and subsystem operation.
- To design and integrate the electronic subsystem including sensors, actuators, motor drivers, and embedded controllers.
- To implement and simulate the complete system using MATLAB/Simulink and SolidWorks.
- To fabricate and assemble a functional prototype using locally available materials.
- To evaluate system performance in terms of digging consistency, seed placement accuracy, and operational efficiency.
- To analyze simulation and prototype results for validating the developed system.

1.4 Scope of the Thesis

This project focused on the design, modelling, control, simulation, and prototype development of an automated agricultural vehicle intended for small-scale farming applications. The scope included the development of the mechanical subsystems for soil digging, seed sowing, and soil covering, as well as dynamic modelling, control system development, electronics integration, and prototype fabrication.

The work included:

- Mechanical design and analysis of the digging mechanism, seed-metering unit, and soil-covering assembly.
- Development of the locomotion and actuation systems.
- Dynamic modelling and controller design using MATLAB/Simulink.

- Mechanical modelling and structural analysis using SolidWorks.
- Integration of embedded systems, sensors, actuators, and motor drivers.
- Prototype fabrication using low-cost and locally available materials.
- Performance testing and evaluation under controlled operating conditions.

The study did not include:

- Full-scale agricultural field deployment.
- AI-based autonomous navigation and computer vision systems.
- Long-term crop yield analysis and agricultural productivity assessment.

1.5 Significance of the Study

This research provided a low-cost mechanization solution suitable for small-scale farmers and compact agricultural environments. The developed system reduced labor intensity and improved consistency in soil preparation, seed sowing, and soil covering operations. The integration of digging, seed dispensing, and covering mechanisms into a single mobile platform minimized operational time and enhanced planting uniformity.

The study also demonstrated the practical integration of electromechanical systems, embedded control, dynamic modelling, and agricultural automation within a compact robotic platform. The developed prototype encouraged technological adoption in agriculture and served as a foundation for future locally manufactured smart farming tools.

Academically, the research contributed to the field of agricultural robotics by presenting the modelling, control, simulation, and implementation of a multi-functional agricultural robot, which remains an area with limited local research and practical development.

1.6 Methodology

This research follows a systematic methodology to design, develop, simulate, and prototype an Automated Multi-Task Agricultural Vehicle capable of performing soil digging, seed sowing, and soil covering operations in a single planting cycle. The methodology integrates mechanical design, mathematical modeling, control system development, electronics integration, simulation, and experimental validation. The overall process was divided into five major tasks.

Step 1: System Design and Mechanical Development

The first task involved the conceptualization and mechanical design of the agricultural vehicle using SolidWorks. The complete system was designed as a four-wheel differential-drive mobile robot equipped with a continuous retractable digging mechanism, a seed-metering unit, and a soil-covering assembly. The chassis, furrow opener, seed hopper, metering disk, covering disks, and supporting structures were modeled in three-dimensional CAD environments. Material selection and structural calculations were performed to evaluate strength, stiffness, soil resistance, and component reliability under expected operating conditions. Finite Element Analysis (FEA) was also conducted on critical components to verify structural integrity and safety.

Step 2: Mathematical Modeling and System Analysis

The second task focused on developing mathematical models representing the dynamic behavior of the vehicle and its subsystems. The kinematic and dynamic equations of the differential-drive mobile platform were derived using established robotic modeling techniques. Soil-tool interaction was modeled to estimate the drag force acting on the furrow opener during continuous trench formation. Mathematical models were also developed for the digging depth mechanism and seed-metering subsystem. The complete system was represented in state-space form and analyzed through controllability, stability, eigenvalue, and frequency-response investigations to establish a foundation for controller design.

Step 3: Electronics Integration and Control System Development

The third task involved the design and integration of the electronic and embedded control systems. An ESP32 microcontroller was selected as the central processing unit and interfaced with DC geared motors, motor drivers, a servo motor, a stepper motor, wheel encoders, and an MPU6050 inertial measurement unit. PID-based control strategies were developed to regulate vehicle motion, maintain straight-line navigation, control digging mechanism deployment, and coordinate seed-dispensing operations. A supervisory control algorithm was implemented to synchronize all subsystems throughout the planting cycle, including digging, seed placement, soil covering, and turning maneuvers.

Step 4: Simulation, Testing, and Validation

The fourth task consisted of validating the developed system through simulation and experimental testing. MATLAB/Simulink was used to evaluate the mathematical models and control algorithms, while SolidWorks Simulation was employed to assess structural performance. The fabricated prototype was tested under controlled operating conditions to evaluate motion accuracy, turning performance, digging consistency, seed-spacing uniformity, and overall system functionality. Experimental results were compared with simulation outcomes to verify design assumptions and assess the effectiveness of the developed agricultural vehicle in performing integrated planting operations.

Step 5: Prototype Fabrication and System Implementation

The final task involved fabricating and assembling the physical prototype based on the finalized mechanical and electronic designs. Structural components were manufactured and assembled to form the mobile platform, while electronic modules and sensors were integrated according to the developed architecture. During operation, the vehicle moves continuously along a planting row while the digging mechanism remains engaged with the soil to create a continuous furrow. Seeds are dispensed at predefined intervals based on encoder measurements, and the covering disks close the furrow immediately after seed placement. At the end of each row, the digging mechanism retracts and the vehicle performs a 180-degree turn before beginning the next planting pass.

1.7 Organization of the Thesis

This thesis is organized into five main chapters. Chapter 1 presents the introduction of the study, including the background of agricultural mechanization challenges, problem statement, objectives, scope, and significance of developing an automated multi-task agricultural vehicle. Chapter 2 provides a comprehensive literature review on agricultural robotics, soil interaction mechanisms, precision seed metering systems, navigation strategies, and control approaches relevant to multi-task field automation. Chapter 3 describes the methodology of the study, covering the system overview, mechanical design of the chassis and working subsystems, continuous retractable digging mechanism, seed metering and soil covering systems, as well as the mathematical modelling, control strategy, and electronics integration of the proposed vehicle. Chapter 4 focuses on the system analysis and implementation aspects, including simulation, controller validation, prototype development, and performance evaluation under defined operating conditions. Finally, Chapter 5 concludes the thesis by summarizing the key findings, highlighting the contributions of the proposed system, and providing recommendations

for future improvements and further research directions in automated agricultural systems.

2 Literature Review

Multi-Task Autonomous Agricultural Vehicles

The growing demand for food security in rapidly urbanising societies, compounded by chronic labour shortages in smallholder agriculture, has stimulated extensive research into autonomous machines capable of performing multiple planting and tilling operations without human intervention [46, 47, 48]. Early mechanised systems addressed single tasks — either tillage or sowing — but not both simultaneously, and they were typically sized for large commercial fields that bear no resemblance to the compact, irrigated urban plots that are the target environment of the present study [50]. The transition from single-task to multi-task machines required simultaneous advances in structural design, embedded computing power, and sensor integration [4].

Ahmad et al. [1] presented one of the earliest systematic studies specifically targeting autonomous multi-task agricultural vehicles for small-scale farming. Their architecture decoupled soil engagement, seed dispensing, and covering into independently controllable sub-assemblies coordinated through a central microcontroller, establishing the modular design philosophy adopted in the present thesis. Critically, they reported that continuous furrow-cutting — as opposed to discrete hole formation — reduced seed-placement variability by approximately 18% in medium-loam soil conditions, because the open trench provided a consistent seed-contact geometry that intermittent punching mechanisms could not replicate. Patel et al. [2] extended this line of inquiry to precision seeding in urban contexts and identified the single-shank continuous-furrow opener as the mechanism best suited to confined bed widths, directly confirming the design choice made in this work.

Fountas et al. [4] conducted a broad survey of agricultural robotics for field operations and categorised design approaches by scale, crop type, and degree of autonomy. Their taxonomy placed row-based differential-drive platforms with on-board embedded control at the intersection of affordability and mechanical simplicity, making them the preferred architecture for prototype development in resource-constrained settings. Shamshiri et al. [5] similarly identified differential-drive mobile bases as the dominant platform for small-plot agricultural robots, noting that their kinematic simplicity — requiring only two independent velocity inputs to achieve full planar manoeuvrability — substantially reduced the complexity of the supervisory control system. These conclusions directly inform the four-wheel differential-drive mobile base adopted in the present design. Manyika et al. [45] provide broader context by situating agricultural

robotics within the global technology landscape, arguing that autonomous field machines represent a disruptive advancement for smallholder economies in the developing world — a framing particularly relevant for the urban farming context in Addis Ababa addressed by this thesis.

Continuous Soil-Tool Interaction and Furrow Opening

Accurate prediction of the horizontal draft force exerted on a dragging tine is foundational to motor sizing, structural design of the shank and its mounting bracket, and servo torque selection. The mechanics of soil cutting by narrow rigid tines has been studied extensively since the classical work of Terzaghi [17], who established the general framework of passive earth pressure theory applicable to soil-tool interfaces, and of McKyes [19], who extended these principles specifically to tillage geometries and characterised the failure wedge geometry ahead of dragging implements. Godwin [14] provided a comprehensive review of how implement geometry — rake angle, width, and operating depth — influences both the mode of soil failure and the resulting tool forces, observing that narrow-tine implements operating below the critical depth primarily induce crescent-shaped failure surfaces rather than the full Rankine failure wedge associated with wider tools. Upadhyaya et al. [49] further synthesised experimental draft-force data across soil textures, providing the specific resistance values used in the design basis of the present work.

The most practically applicable model for the present design is the empirical draft force relationship developed by Sahu and Raheman [13]. Their decision-support framework for agricultural implement performance characterised the narrow-tine draft force as the product of the specific soil resistance k , the tool width w , and the square of the operating depth d , yielding the compact expression $F_{\text{drag}} = k \times w \times d^2$. The squared depth dependence captures the observed non-linear increase in draft force with depth and has been validated against field measurements across a range of soil textures from loose sandy loam to stiff clay. This expression is the governing soil drag equation throughout the present work, and the specific soil resistance $k = 200,000 \text{ N/m}^3$ adopted here corresponds to the medium-loam, loosely tilled condition that typifies small irrigated urban garden plots. Koolen and Kuipers [16] provided the broader soil mechanics context within which this force prediction is interpreted, including the relationships between soil moisture content, bulk density, and specific resistance that motivate the constant-moisture design assumption.

Wilkinson [18] provided a theoretical basis for the speed-dependence of draft force,

showing that inertial effects in the deforming soil ahead of the tine introduce a velocity-squared correction term. Sahu and Raheman [13] quantified the empirical coefficient k_v for this correction in light agricultural soils; at the operating speeds of the present vehicle ($v \leq 0.20$ m/s) the speed-dependent increment is less than 0.2%, confirming that the static model is wholly sufficient for all design calculations in this work. The shank structural analysis treats the blade as a cantilever beam under the computed maximum draft force and follows the bending stress and factor-of-safety methodology of Hibbeler [41], while the wheel traction model uses the Coulomb friction formulation for agricultural wheel-soil contact documented by Gee-Clough [15]. Finite Element Analysis of the shank, chassis, and motor-holder bracket follows the static linear simulation methodology of SolidWorks Simulation as described in Dassault Systemes [42], with mesh convergence confirmed by iterative refinement.

Precision Seed Metering

Uniform seed placement at prescribed inter-seed spacing is the primary agronomic objective of a planting robot. The seed metering unit must deliver exactly one seed per dispensing event at a timing that aligns with the programmed distance interval derived from wheel encoder feedback. Wanjura et al. [20] were among the first to demonstrate sensor-based feedback control of seed metering, showing that integrating optoelectronic detection of seed passage into the metering loop reduced singulation error — the incidence of double drops or skips — to below 2% for maize under field conditions. Their results established the optical slot sensor as the standard confirmation mechanism for precision planters, a principle directly adopted in the present design through the slot-type infrared sensor aligned with the seed outlet of the metering disk.

Kocher et al. [21] developed an electronically controlled metering system for small-seed planters that used a stepper motor to advance the metering disk by discrete angular steps, each step corresponding to one seed cell. Their analysis of disk inertia, stepper torque budgeting, and step-rate constraints — under the condition that the metering event must complete within a prescribed time window — provides the direct analytical precedent for the metering dynamics model developed in Section 3.3.5 of the present work. The moment of inertia calculation for the plastic metering disk, the triangular angular velocity profile for reliable step completion within 50 ms, and the derivation of required stepper torque all follow the framework established by Kocher et al. Ozturk et al. [22] provided more recent experimental validation of electromechanical metering units for row crops, confirming that NEMA-14 class stepper motors with 1/8 microstepping provide adequate angular resolution and that the DRV8825 driver achieves

the required current precision at the phase currents needed for reliable torque delivery in field conditions.

Zagainov et al. [23] specifically examined optical seed-flow monitoring using infrared slot sensors and characterised sensor response time, slot geometry, and seed transit time in detail. Their finding that a sub-0.1 ms sensor response provides greater than $60\times$ timing margin for seeds released from rest provides the design basis for the IR sensor specification in the present system. Ghaffarnezhad et al. [24] extended this work to real-time missed-seed and double-drop identification using an embedded sensor array, validating the fault-logging capability implemented in the supervisory state machine of the present design. Together, references [23] and [24] establish the optical detection scheme as a proven, low-cost approach for seed confirmation in both prototype and production planters, directly informing the slot-IR architecture and the fault-response logic of the seed step controller described in Section 3.5.4.

Navigation, Odometry, and Mobile Base Control

The kinematic and dynamic modelling of wheeled mobile robots has been codified in several foundational texts. Siegwart et al. [6] provides the canonical treatment of differential-drive kinematics, deriving the nonlinear unicycle model that relates wheel velocities to CoM linear and angular velocity, and the encoder-based odometry update equations with the midpoint-heading Runge-Kutta approximation used in the present work. Siciliano et al. [7] and Spong et al. [8] extend this kinematic framework to the dynamic domain, presenting the Euler-Lagrange derivation for wheeled robots with nonholonomic rolling constraints. The dynamic model of Section 3.3.2 follows the Euler-Lagrange formulation of these references, yielding decoupled scalar equations of motion for linear and angular velocities with explicit soil drag treated as a disturbance input.

Campion et al. [11] established the structural classification of wheeled mobile robots and proved that differentially driven platforms belong to the unicycle class of (2,0) nonholonomic systems, possessing two degrees of freedom but three configuration variables. Their proof that full-state controllability holds for nonholonomic mobile robots in the nonlinear sense — even though the linearised position states appear decoupled from the inputs in the B-matrix — underpins the controllability analysis of Section 3.4.2, where the indirect kinematic controllability of the position states x and y is formally justified. Kozlowski and Pazderski [9] provided experimental characterisation of four-wheel skid-steering kinematics that informed the present four-wheel differential-

drive treatment, in which the two left wheels and two right wheels are driven together as equivalent pairs, reducing the platform to the standard unicycle kinematic model.

Encoder-based dead-reckoning is acknowledged as an imperfect but practically sufficient navigation method for short-row planting operations. Macek et al. [32] characterised the tracking accuracy of differential-drive robots using encoder odometry and identified wheel slip and encoder quantisation as the dominant error sources, directly motivating the traction margin analysis of Section 3.2.6 and the slip estimation subsection. The SPRI Research Group [33] provided a systematic error characterisation for small autonomous vehicles, confirming that at 68% traction utilisation — consistent with the present design operating at a traction margin of 1.47 — slip errors remain below 5%, which corresponds to a seed-spacing error of ± 12.5 mm at 250 mm target spacing, within the accepted agronomic tolerance. Alatisse and Hancke [34] reviewed sensor fusion methods for mobile robots and discussed the fundamental limitation of encoder-only localisation, noting that the absolute position states x and y are unobservable without an external reference sensor — a finding confirmed analytically by the formal observability analysis of Section 3.4.3.

LaValle [27] and Canudas de Wit et al. [10] provided the theoretical underpinnings of path following and motion planning for nonholonomic systems. The straight-line path following strategy implemented in the supervisory state machine — commanding a fixed forward speed while using the LQR to suppress lateral and heading deviations simultaneously — is the simplest instantiation of the trajectory tracking problem formalised in these references. Thuilot et al. [28] demonstrated that for vehicles with multiple actuated wheels, the LQR formulation naturally handles cross-axis coupling between translational and rotational dynamics, a result that extends directly to the differential-drive case and constitutes the analytical justification for the MIMO LQR approach of Section 3.5.2 over a pair of decoupled SISO PID loops.

Control System Design — PID, LQR, and Embedded Implementation

Proportional-integral-derivative (PID) control remains the dominant strategy in industrial and agricultural machinery owing to its structural simplicity, ease of tuning, and robustness to plant model uncertainty. Ang et al. [30] provided a comprehensive review of PID analysis, design, and technology, including the frequency-domain gain and phase margin methods that underpin the velocity and shank controller bandwidth specifications in Sections 3.5.2 and 3.5.3. Franklin et al. [26] is the primary reference for the loop-shaping and Bode analysis used to derive the PD shank controller gains, as

well as for the root-locus and closed-loop pole placement interpretations that verify the controller designs. The shank PD controller design procedure — matching the natural frequency $\omega_n = 31.4$ rad/s and damping ratio $\zeta = 0.707$ to the servo second-order model — follows directly from the pole-placement method documented in Franklin et al. [26].

For multi-input multi-output (MIMO) systems with coupled dynamics, optimal control theory offers a principled alternative to ad hoc SISO tuning. Kirk [25] provides the foundational treatment of the Linear Quadratic Regulator and the Algebraic Riccati Equation (ARE), from which the optimal gain matrix K is computed. The LQR design of Section 3.5.2 follows the infinite-horizon cost functional formulation of Kirk [25], with the Q and R weighting matrices selected according to the relative agronomic importance of each state error as informed by experimental guidelines reviewed in Mollers et al. [31]. Mollers et al. [31] demonstrated in hardware-in-the-loop simulation that an LQR designed on the linearised five-state mobile-base model of a differential-drive robot suppressed both lateral path error and velocity disturbances simultaneously — a performance level not achievable with the separate PID velocity and heading loops it replaced — providing direct experimental precedent for the architectural decision made in Section 3.5.2 of the present thesis. Pomet and Praly [29] provided rigorous stability proofs for feedback control of nonholonomic systems, confirming that the LQR closed-loop eigenvalue analysis correctly characterises closed-loop stability even in the presence of the nonholonomic constraint.

The embedded control platform is the ESP32-WROOM-32, a dual-core 240 MHz microcontroller with 34 GPIO, hardware PWM generation, and FreeRTOS support. Espressif Systems [35] provides the authoritative technical reference for the peripheral architecture, interrupt latency specifications, and LEDC PWM generation used in the motor and servo drive chains of the present design. Magdy et al. [36] demonstrated a FreeRTOS-based multi-loop control architecture on the ESP32 for mobile robotics, showing that a 1 kHz control tick on Core 0 and supervisory logic on Core 1 achieve deterministic scheduling with jitter below 20 microseconds — well within the requirements of the present application. The InvenSense MPU6050 IMU [37] provides gyroscope-based heading estimation with the drift characteristics used in the turning accuracy calculation, while the DRV8825 stepper driver [38] and BTS7960 H-bridge [39] datasheets provide the current ratings and switching specifications against which the power budget calculations of Section 3.6.4 are verified.

Power system design for battery-operated autonomous robots is governed by cell electrochemistry, discharge-rate constraints, and thermal management. Ehsani et al. [40] provide the electrochemical basis for lithium-ion cell capacity and discharge-rate

specifications, including the C-rate framework used to verify that the 3S2P cell pack operates within safe limits at both operating and peak stall currents. The 80% runtime derating factor applied in Section 3.6.4 follows the conservative practice recommended by Ehsani et al. [40] for cycle-life preservation in field robots, and the 3S BMS over-voltage, under-voltage, and over-current protection thresholds are consistent with cell protection specifications reviewed in the same reference. Material selection for structural components is supported by the mechanical property databases in Hibbeler [41] and the ASM Metals Handbook [44], with steel designations and yield strength values drawn from ASME standards [43].

Research Gap

The foregoing review reveals that, while the individual sub-problems addressed in this thesis — differential-drive kinematics [6, 11], narrow-tine draft force modelling [13, 19], precision seed metering [21, 23, 24], and LQR path tracking [25, 31] — have each been studied extensively in isolation, their synthesis into a single integrated, low-cost, autonomous multi-task planting robot designed specifically for small urban garden plots in Sub-Saharan Africa has not been reported in the open literature. Existing multi-task agricultural robots [1, 4, 5] are sized and priced for commercial farm contexts; designs that fall within the ETB 37,000–40,000 prototype budget and can navigate the 1–5 m row lengths of urban allotment plots are entirely absent from the reviewed literature. Furthermore, no prior work has combined the continuous-furrow operating mode — in which the shank remains engaged throughout the full row and retracts only during the 180-degree inter-row turn — with encoder-triggered seed metering and LQR path tracking in a single embedded system on the ESP32 platform. The present thesis addresses this gap by designing, modelling, simulating, and prototyping such a system, providing both the analytical justification through the Euler-Lagrange dynamic model and state-space analysis, and the experimental validation through MATLAB/Simulink simulation and physical prototype testing, thereby establishing a foundation for future scale-up and commercialisation in resource-limited farming contexts across Ethiopia and the broader Sub-Saharan African region.

3 System Design

3.1 Subsystem Overview

The proposed Automated Multi-Task Agricultural Vehicle is designed as a compact autonomous robot capable of performing soil digging, seed sowing, and soil covering in a coordinated planting process. The robot operates using a row-based planting strategy in which it travels continuously along a straight path while performing planting operations, then executes a 180-degree turn at the end of the row before continuing along the next path.

Unlike intermittent hole-based planting systems, the proposed design uses a continuous retractable digging mechanism. During straight-line movement, the digging tool remains engaged with the soil to create a continuous planting trench. At the end of the row, the digging mechanism retracts before the robot performs its turning maneuver. Once aligned with the next row, the digging mechanism is redeployed and planting resumes. This approach reduces unnecessary soil resistance during turning and improves maneuverability and energy efficiency.

The system consists of four major subsystems:

1. Mobile Base Subsystem
2. Continuous Retractable Digging Subsystem
3. Seed Metering and Dispensing Subsystem
4. Soil Covering Subsystem

These subsystems are coordinated through a central embedded controller responsible for motion control, digging activation, seed dispensing synchronization, and turning operations.

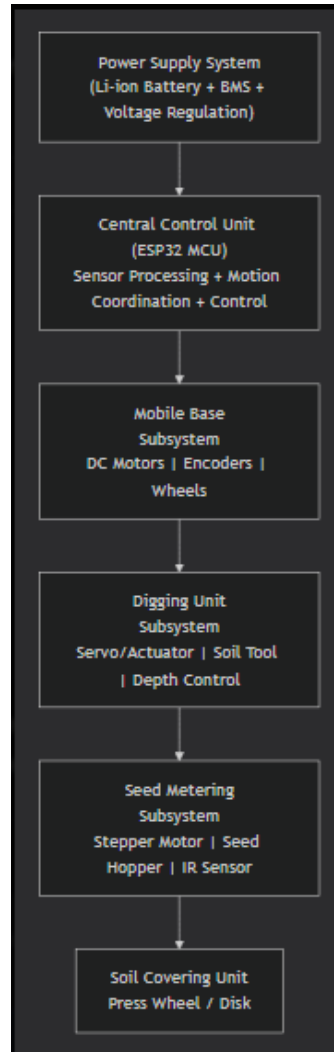


Figure 3.1. Flow Chart Diagram of the Automated Multi-Task Agricultural Vehicle

3.1.1 Mobile Base Subsystem

The mobile base subsystem provides locomotion and structural support for the robot. The platform uses a four-wheel differential-drive configuration powered by DC geared motors with encoder feedback for speed and distance measurement.

The chassis is constructed from mild steel due to its strength, low cost, and local availability. During operation, the robot moves continuously in straight lines for planting and performs 180-degree turns at row ends for navigation between adjacent rows.

Wheel encoders and an IMU sensor assist in maintaining straight movement and accurate turning alignment. The mobile base must overcome rolling resistance and soil drag generated by the digging mechanism while maintaining stable operation.

3.1.2 Continuous Retractable Digging Subsystem

The digging subsystem creates the planting trench during operation. The mechanism consists of a digging shank attached to a servo- or actuator-driven lifting system.

During straight-line planting, the digging tool remains continuously engaged with the soil to form a trench of controlled depth. At the end of each row, the digging mechanism retracts before the robot performs its 180-degree turning maneuver. After alignment with the next row, the digging tool is lowered again and planting resumes.

This retractable operation reduces turning resistance, minimizes unnecessary soil disturbance, and improves maneuverability.

3.1.3 Seed Metering and Dispensing Subsystem

The seed metering subsystem dispenses seeds into the trench created by the digging mechanism. It consists of a seed hopper, metering disk, stepper motor, and seed delivery tube.

The stepper motor rotates the metering mechanism at predefined intervals based on encoder feedback from the wheels, ensuring uniform seed spacing during continuous motion. An infrared sensor verifies successful seed release and helps detect missed or multiple seed drops.

Seed dispensing pauses during turning operations and resumes automatically when the robot begins planting along the next row.

3.1.4 Soil Covering Subsystem

The soil covering subsystem performs the final planting stage by pushing soil back over the deposited seeds. The mechanism uses passive covering disks or press wheels mounted behind the seed drop location.

As the robot moves forward, the covering mechanism redirects loose soil into the trench and lightly compacts the surface to improve seed-to-soil contact and germination conditions.

The subsystem operates continuously during straight-line planting without requiring additional actuators, reducing system complexity and power consumption.

3.2 Mechanical Design

The mechanical design of the automated agricultural vehicle encompasses all physical sub-assemblies that interact with the soil, store and dispense seeds, support the drive system, and enclose the electronic components. The four principal sub-assemblies are: (1) the chassis, drive system, and wheels; (2) the furrow-opener and servo engagement mechanism; (3) the seed hopper, metering disk, and covering disks; and (4) the structural support elements including joints, motor holders, and protective enclosures. Each sub-assembly was designed to satisfy the functional requirements of the continuous-furrow planting cycle while remaining compact and manufacturable. Material selection drew from ten components for which properties were confirmed through SolidWorks material databases; all structural calculations use those properties directly.

3.2.1 Design Assumptions

Before presenting the mechanical design and structural calculations, it is necessary to state the assumptions under which the system was designed and analysed. These assumptions define the boundary conditions for all analytical models in this section and in Section 3.3, and they establish the operating envelope within which the prototype is expected to perform reliably. Deviation from these conditions in the field may require re-evaluation of specific design margins.

Assumption 1 — Medium-loam soil: All soil-interaction calculations, including the furrow draft force model and wheel traction analysis, assume a medium-loam soil with a specific soil resistance of $k = 200,000 \text{ N/m}^3$. This is representative of the loosely tilled, irrigated urban garden plots that constitute the target operating environment in Addis Ababa. Sandy soils would produce lower drag forces and higher wheel slip risk; clay-heavy soils would increase draft force beyond the values calculated here.

Assumption 2 — Flat terrain: The chassis, stability, and motor sizing analyses assume operation on flat or near-flat ground (slope 5°). Significant terrain slopes would add a gravitational component to the traction requirement and shift the vehicle center of gravity relative to the wheel contact polygon, potentially affecting lateral stability. This assumption is consistent with the small urban plots described in the problem statement.

Assumption 3 — Negligible wheel slip: The encoder-based odometry model and seed-spacing calculations assume that wheel rotation translates directly into forward travel with zero slip. In practice, some slip is unavoidable during furrow engagement; however, at the low operating speeds and moderate draft forces calculated in this design, slip is expected to remain below 5% and is therefore treated as negligible for

spacing accuracy purposes. A traction margin analysis in Section 3.2.4 confirms this assumption.

Assumption 4 — Constant soil moisture condition: Soil moisture content is assumed to remain approximately constant within a single planting session. Moisture significantly affects both draft force (wetter soils generally reduce draft) and wheel traction (very wet soils reduce friction coefficient). Seasonal or irrigation-induced variation between sessions is acknowledged but not modelled; the design uses conservative (dry-side) values of k and friction coefficient to ensure adequate performance across the expected range.

Assumption 5 — Uniform seed size: The metering disk cell geometry and the seed-drop timing model assume seeds of approximately uniform size within a single crop type. Mixed or irregular seed sizes may cause jamming in the metering disk cells or variable drop timing. In practice, seeds are sorted or purchased pre-graded, and the disk cell dimensions are sized to the largest expected seed for the selected crop.

Assumption 6 — Low-speed quasi-static operation: The vehicle operates at a nominal forward velocity of approximately 0.1–0.2 m/s. At this speed, inertial forces on the structural members are small relative to the static loads from vehicle weight and soil drag. Structural members are therefore sized using static load analysis with an appropriate dynamic amplification factor where transient effects are relevant (see Section 3.2.5 — Dynamic Loading Considerations). The quasi-static assumption does not apply to the control system analysis in Section 3.5, which explicitly models velocity dynamics.

Assumption 6 — Static loading approximation for structural members: All stress and factor-of-safety calculations for the chassis, shank, stand for plow, motor holders, joints, and cover case are performed under static equilibrium conditions using maximum expected loads. Dynamic load amplification, fatigue cycling, and resonance effects are discussed qualitatively in Section 3.2.5 rather than modelled analytically, as the prototype is not expected to accumulate sufficient operating cycles during experimental validation to make fatigue the governing failure mode.

3.2.2 Chassis, Drive System, and Wheels

The chassis serves as the structural backbone of the vehicle, transmitting traction and reaction forces between the ground, the drive motors, and the attached agricultural tools. It was fabricated as a rectangular frame from oak wood square-section members, selected as the primary chassis material because of its very low density (560 kg/m^3 compared to $7,800\text{--}7,900 \text{ kg/m}^3$ for steel), adequate bending stiffness for the prototype-

scale loads, and straightforward fabrication with hand tools in a low-resource workshop environment. Secondary structural members—motor holders, joints, and brackets—are made from AISI 1020 steel (density 7,900 kg/m³; yield strength 351,571,000 N/m²) where higher stiffness or weld compatibility is required.

The frame carries all sub-assemblies: the 3S2P 18650 Li-ion battery pack (positioned low and centrally to minimize the center-of-gravity height), four DC gear motor assemblies, the BTS7960 motor driver modules, the ESP32 microcontroller enclosure, the furrow-opener servo bracket (stand for plow), the seed hopper, and the rear covering-disk axle.

Chassis Structural Calculation

Each main longitudinal oak rail carries a share of the total vehicle load. The total vehicle mass was estimated by summing the masses of all components:

$$m_{\text{chassis_rail}} = \rho_{\text{oak}} \times A_{\text{rail}} \times L_{\text{rail}} = 560 \times (0.030 \times 0.030) \times 0.500 = 0.252 \text{ kg}$$

$$m_{\text{shank}} = \rho_{1045} \times b \times t \times L = 7850 \times 0.020 \times 0.008 \times 0.120 = 0.151 \text{ kg}$$

$$m_{\text{hopper}} \approx \rho_{201\text{SS}} \times V_{\text{hopper}} = 7859.9 \times 150 \times 10^{-6} = 1.179 \text{ kg}$$

$$m_{\text{cover_disks}} = \rho_{\text{GS}} \times 2 \times \pi R^2 \times t = 7870 \times 2\pi(0.060)^2 \times 0.002 = 0.356 \text{ kg}$$

$$m_{\text{motors}} (\times 4) \approx 4 \times 0.150 = 0.600 \text{ kg}$$

$$m_{\text{battery}} = 8 \times 0.047 = 0.376 \text{ kg}$$

$$m_{\text{electronics}} \approx 0.300 \text{ kg}$$

$$m_{\text{total}} \approx 7.25 \text{ kg}$$

The total vehicle weight is therefore:

$$W = m_{\text{total}} \times g = 7.25 \times 9.81 = 41.7 \text{ N}$$

For the bending stress check on an oak cross-rail (30 × 30 mm cross-section, 300 mm span, treating it as a simply supported beam with a central point load equal to half the

total weight):

$$F_{\text{rail}} = W/2 = 41.7/2 = 20.85 \text{ N}$$

$$M_{\text{max}} = F_{\text{rail}} \times L/4 = 20.85 \times 0.300/4 = 1.564 \text{ N} \cdot \text{m}$$

$$I_{\text{oak}} = bh^3/12 = 0.030 \times 0.030^3/12 = 6.75 \times 10^{-8} \text{ m}^4$$

$$c = h/2 = 0.030/2 = 0.015 \text{ m}$$

$$\sigma_b = M \cdot c/I = 1.564 \times 0.015/6.75 \times 10^{-8} = 0.348 \text{ MPa}$$

The modulus of rupture (bending strength) of structural oak wood is approximately 40 MPa. The resulting factor of safety is:

$$FoS_{\text{chassis}} = MoR/\sigma_b = 40/0.348 = 114.9 \quad \checkmark \quad (> 2.0 \text{ required})$$

The large factor of safety is expected for an oak chassis at prototype scale; it confirms that the section is not a limiting constraint and the selected dimensions provide a comfortable safety margin against bending failure.

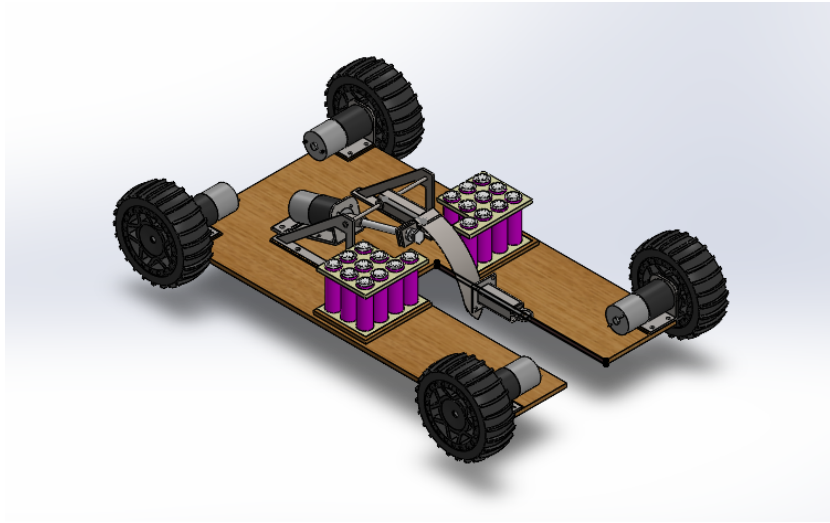


Figure 3.2. Isometric CAD view of the chassis frame

Drive System and Wheel Encoder Distance Calculation

Locomotion is provided by four 12 V DC gear motors (approximately 100 RPM no-load) fitted with integrated Hal encoders. Each encoder outputs quadrature pulses; with 11 pulses per revolution and $4\times$ quadrature decoding, each wheel produces 44 counts per revolution. The linear distance per encoder count is:

3.2.3 Furrow-Opener and Engagement Mechanism

Soil opening is accomplished by a single-shank furrow opener machined from AISI 1045 medium-carbon steel (yield strength 530,000,000 N/m²; density 7,850 kg/m³). AISI 1045 was selected specifically for its high carbon content, which, in the normalized or quench-and-tempered condition, provides a surface hardness adequate to resist the abrasive wear inflicted by soil particles during continuous dragging. The single-shank configuration was preferred over rotary or chisel-plow alternatives because it produces a narrow, geometrically consistent furrow with minimal lateral soil disturbance; an important characteristic for row-crop seeding in confined urban plots where bed spacing is limited.

The shank is mounted on a pivot arm attached to the high-torque servo motor (referred to in the component list as the Stand for Plow, fabricated from AISI 1020 steel). The servo rotates the pivot arm, pressing the shank tip into the soil to a programmed depth or lifting it clear of the ground.

Operating Cycle and Continuous Engagement

The furrow opener operates in a continuous-engagement mode throughout each straight-line planting pass. The sequence is as follows:

- At the start of a new row, the supervisory controller commands the servo to rotate the pivot arm downward. The shank tip penetrates to the target depth, and the servo position controller holds that depth against the soil reaction force.
- As the vehicle advances along the row, the shank drags continuously through the soil, cutting an uninterrupted open furrow along the full length of the pass. The seed metering unit simultaneously deposits seeds into this open furrow at programmed distance intervals. The passive covering disks immediately behind close the furrow over each deposited seed.
- At the end of the row, before the vehicle begins its 180-degree turn, the supervisory controller commands the servo to retract the shank fully to its raised (clear-of-ground) position. This prevents the shank from dragging sideways during the turn, which would cause uncontrolled soil disturbance and generate lateral reaction forces that could destabilize the platform.
- Once the vehicle has completed the turn and aligned with the next row, forward motion resumes and the servo re-engages the shank to the target depth for the next continuous-furrow pass.

Depth Geometry

The depth of the furrow is determined by the servo rotation angle. If the pivot arm has effective length L_{arm} (measured from the servo shaft centre to the shank blade attachment point), the vertical penetration depth h at a servo angle θ_s from horizontal is:

$$h = L_{\text{arm}} \times \sin(\theta_s)$$

For the design value of $L_{\text{arm}} = 80$ mm and a maximum target depth of $h = 50$ mm:

$$\theta_{s_max} = \arcsin(h/L_{\text{arm}}) = \arcsin(50/80) = \arcsin(0.625) = 38.7^\circ$$

The servo is therefore commanded to a maximum angle of 38.7° for the deepest planting setting. For a nominal engagement angle of 30° , the resulting depth is:

$$h_{30} = 0.080 \times \sin(30^\circ) = 0.080 \times 0.500 = 40.0 \text{ mm}$$

This range of 0–50 mm covers the seeding depth requirements of common crops planted in small-scale Ethiopian urban gardens, including maize (30–50 mm), bean (25–40 mm), and various vegetables (15–30 mm).

Soil Draft Force and Shank Structural Analysis

The horizontal drag force on the shank during continuous furrow cutting is estimated using the narrow-tine draft force model (following Sahu and Raheman [13]):

$$F_{\text{drag}} = k \times w \times d^2$$

where k is the specific soil resistance (N/m^3), w is the shank width, and d is the operating depth. For medium loam soil under the moisture and tillage conditions typical of small urban plots, $k \approx 200,000 \text{ N/m}^3$. With shank width $w = 20$ mm and maximum depth $d = 50$ mm:

$$F_{\text{drag}} = 200,000 \times 0.020 \times (0.050)^2$$

$$F_{\text{drag}} = 200,000 \times 0.020 \times 0.0025$$

$$F_{\text{drag}} = 10.0 \text{ N}$$

This force acts at the shank tip. Treating the shank as a cantilever beam of length $L_{\text{shank}} = 120$ mm, with a rectangular cross-section 20 mm wide $\times$ 8 mm thick, the

bending stress at the fixed root is:

$$M_{\text{root}} = F_{\text{drag}} \times L_{\text{shank}} = 10.0 \times 0.120 = 1.200 \text{ N} \cdot \text{m}$$

$$I_{\text{shank}} = b \times t^3/12 = 0.020 \times 0.008^3/12 = 8.533 \times 10^{-10} \text{ m}^4$$

$$c = t/2 = 0.008/2 = 0.004 \text{ m}$$

$$\sigma_b = M_{\text{root}} \times c/I_{\text{shank}} = 1.200 \times 0.004/8.533 \times 10^{-10} = 5.625 \text{ MPa}$$

The factor of safety against yielding of the AISI 1045 shank is:

$$FoS_{\text{shank}} = S_y/\sigma_b = 530 \text{ MPa}/5.625 \text{ MPa} = 94.2 \quad \checkmark$$

This high factor of safety confirms that the $20 \times 8 \text{ mm}$ cross-section is structurally adequate. The section was not reduced further because a minimum shank width is needed to maintain soil-cutting rigidity and resist out-of-plane bending forces that may arise when the shank encounters stones or hard clods.

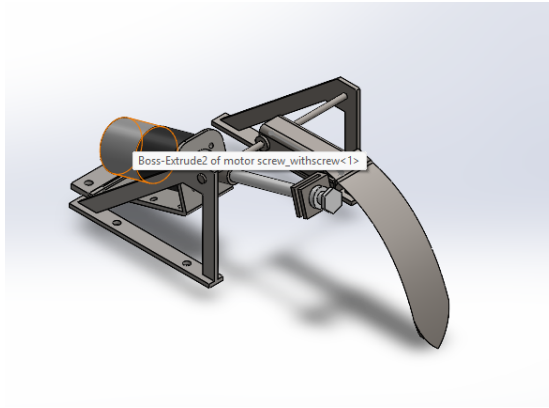


Figure 3.3. Isometric CAD View of the Furrow-Shank Opener

Servo Torque Requirement

The torque that the servo must exert to hold the shank at the target depth against the soil drag force is:

$$T_{\text{servo}} = F_{\text{drag}} \times L_{\text{arm}} = 10.0 \times 0.080 = 0.800 \text{ N} \cdot \text{m} = 80.0 \text{ N} \cdot \text{cm}$$

The selected high-torque servo is specified at a minimum holding torque of $15 \text{ kg} \cdot \text{cm}$ ($\approx 147 \text{ N} \cdot \text{cm}$) at 6 V, providing a factor of safety of $147/80 = 1.84$ on the servo torque. This is acceptable for a position-controlled servo operating near its rated supply voltage. The servo driver is powered from the 5 V logic rail via a dedicated power branch with adequate current capacity.

3.2.4 Seed Hopper, Metering Unit, and Covering Disks

Seed Hopper

The seed hopper is a funnel-shaped vessel fabricated from 201 Annealed Stainless Steel (yield strength 292,000,000 N/m²; density 7,859.9 kg/m³). This material was chosen because it is fully corrosion-resistant in moist soil environments, non-reactive with seed coatings or fertilizer dust, and can be formed into the required funnel geometry by sheet-metal bending and welding. The approximate dimensions are: top opening diameter 120 mm, bottom outlet diameter 30 mm, slant-wall height 100 mm, and wall thickness 1.5 mm.

The primary stress in the hopper wall is hoop stress due to the hydrostatic pressure of the stored seed bulk. Treating the seed as a granular material with a bulk density of approximately 650 kg/m³ and using a hydrostatic analogy, the pressure at the hopper base is:

$$P_{\text{seed}} = \rho_{\text{seed}} \times g \times h_{\text{seed}} = 650 \times 9.81 \times 0.100 = 637.65 \text{ Pa}$$

The hoop stress in the thin-walled cylindrical section at the outlet (inner radius $R = 15 \text{ mm}$, wall thickness $t = 1.5 \text{ mm}$) is:

$$\sigma_{\text{hoop}} = P_{\text{seed}} \times R/t = 637.65 \times 0.015/0.0015 = 6.38 \times 10^{-3} \text{ MPa}$$

The factor of safety relative to the yield strength of 201 annealed stainless steel is:

$$FoS_{\text{hopper}} = 292 \text{ MPa}/0.00638 \text{ MPa} \approx 45,800 \quad \checkmark$$

The negligible stress level is expected for a thin-walled vessel under a very low seed pressure and confirms that the material choice is governed by corrosion and food-safety requirements rather than structural demand. A 1.5 mm wall is sufficient to maintain shape rigidity during handling and prototype assembly.

Seed Metering Disk

The metering disk (component: Seed Wheel) is fabricated from plastic material (density 1,020 kg/m³). The disk has a radius of 40 mm and a thickness of 5 mm, giving a volume and mass of:

$$V_{\text{disk}} = \pi \times R^2 \times t = \pi \times (0.040)^2 \times 0.005 = 2.513 \times 10^{-5} \text{ m}^3$$

$$m_{\text{disk}} = \rho_{\text{plastic}} \times V_{\text{disk}} = 1020 \times 2.513 \times 10^{-5} = 0.02563 \text{ kg} \quad (\approx 25.6 \text{ g})$$

The disk has 12 equally-spaced seed cells machined around its circumference. Each cell holds a single seed. A NEMA-14 stepper motor (1.8° per full step; holding torque $\approx 100 \text{ mN} \cdot \text{m}$) rotates the disk by one cell (30°) per seed dispensing event.

The number of full stepper steps required to advance the disk by one cell position is:

$$\begin{aligned}\theta_{\text{cell}} &= 360^\circ / N_{\text{cells}} = 360^\circ / 12 = 30^\circ \text{ per seed event} \\ N_{\text{steps}} &= \theta_{\text{cell}} / \theta_{\text{step}} = 30^\circ / 1.8^\circ = 16.7 \text{ full steps} \quad (\approx 17 \text{ steps})\end{aligned}$$

In practice, the DRV8825 driver is configured for 1/8 microstepping, giving 16 microsteps per full step. The number of microsteps per seed event is then:

$$N_{\text{micro}} = 16.7 \times 8 = 133 \text{ microsteps per seed}$$

The torque required to rotate the disk against seed friction at the outlet port is estimated at approximately $20 \text{ mN} \cdot \text{m}$. The NEMA-14 holding torque of $100 \text{ mN} \cdot \text{m}$ provides a factor of safety of:

$$FoS_{\text{stepper}} = 100 \text{ mN} \cdot \text{m} / 20 \text{ mN} \cdot \text{m} = 5.0 \quad \checkmark$$

A slot-type infrared (IR) sensor is aligned with the outlet port. As each seed drops through the slot after being released by the disk cell, it interrupts the IR beam and generates a confirmation pulse to the ESP32 microcontroller. This optical detection scheme follows the approaches validated by Zagainov et al. [23] and Ghaffarnezhad et al. [24] and enables the supervisory controller to flag missed seeds (no pulse detected) or double drops (two pulses within one step sequence) for post-operation logging.

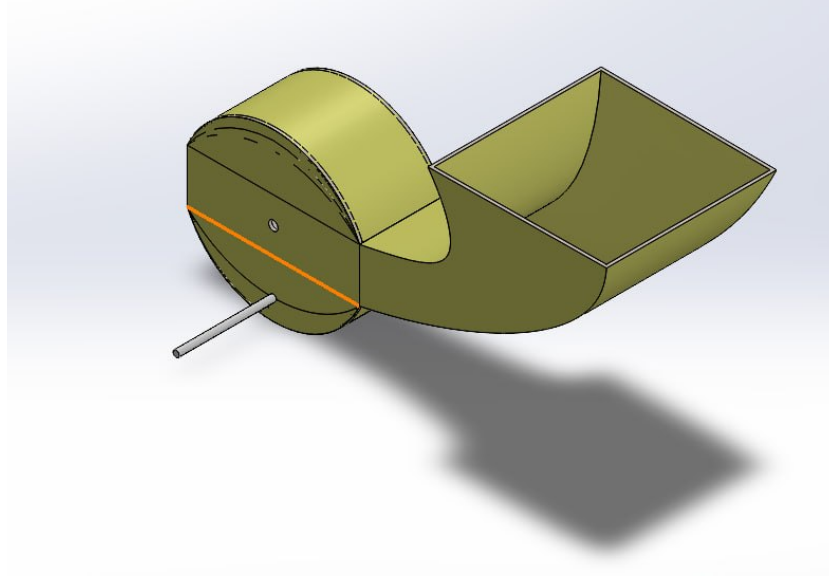


Figure 3.4. Isometric view of the single-shank furrow-opene

Covering Disks

Two passive covering disks are mounted on a common rear axle. Each disk is fabricated from galvanized steel (yield strength 203,943,242.6 N/m²; density 7,870 kg/m³). The galvanized zinc coating provides corrosion protection against wet soil without additional painting. Each disk has a nominal radius of 60 mm and a thickness of 2 mm.

The compaction force applied by each disk is proportional to the rear axle load. The rear axle is estimated to carry approximately 40 percent of the total vehicle weight, distributed equally between the two disks:

$$F_{\text{disk}} = (0.40 \times W_{\text{total}})/2 = (0.40 \times 41.7)/2 = 8.34 \text{ N per disk}$$

The maximum bending stress in the disk, modelled as a thin circular plate of radius $R = 60 \text{ mm}$ simply supported at the rim under a central point load, is given by:

$$\begin{aligned}\sigma_{\text{max}} &= 3F(3 + \nu)/(4\pi t^2) \\ \sigma_{\text{max}} &= 3 \times 8.34 \times (3 + 0.30)/(4 \times \pi \times 0.002^2) \\ \sigma_{\text{max}} &= 3 \times 8.34 \times 3.30/(4 \times \pi \times 4 \times 10^{-6}) \\ \sigma_{\text{max}} &= 82.57/(5.027 \times 10^{-5}) = 1.643 \text{ MPa}\end{aligned}$$

The factor of safety against yielding of the galvanized steel disk is:

$$FoS_{\text{disk}} = 203.94 \text{ MPa}/1.643 \text{ MPa} = 124.1 \quad \checkmark$$

The disks are mounted with a slight toe-in angle so their lower edges converge toward the furrow centreline, producing a wedging action that pushes loosened soil back into the open furrow and provides light compaction over the deposited seed. The rear axle height is adjustable via slotted brackets on the chassis to set disk pressure for different soil conditions.

3.2.5 Material Properties and Structural Analysis

Material Properties Summary

Table 3.1 presents the confirmed material properties for all ten structural and functional components of the robot. The yield strength and density values are used directly in all structural calculations in this section and in Section 3.3. For oak wood and the plastic seed wheel, yield strength values are not provided; structural adequacy for these components was assessed using the modulus of rupture (oak wood, ≈ 40 MPa) and torque-based analysis (seed wheel), respectively.

Table 3.1. Material properties of AMTAV components

No.	Component	Material	Yield Strength (N/m ²)	Density (kg/m ³)	Key Justification
1	Chassis Frame	Oak Wood	—	560	Lightweight; adequate bending stiffness for prototype loads
2	Furrow Opener Shank	AISI 1045 Steel	530,000,000	7,850	High carbon, hardened; superior abrasion resistance against soil
3	Hopper Body	201 Annealed Stainless Steel	292,000,000	7,859.9	Corrosion-free, seed-safe, weldable at low cost
4	Cover Disk	Galvanized Steel	203,943,242.6	7,870	Zinc coating prevents rust; adequate yield for soil drag loads
5	Stand for Plow	AISI 1020 Steel	351,571,000	7,900	Good machinability; sufficient strength for plow support bracket
6	Seed Wheel	Plastic Material	—	1,020	Ultra-light; chemically inert; low friction for seed metering
7	Cover Case	201 Annealed Stainless Steel	292,000,000	7,859.9	Protects electronics; corrosion resistant enclosure
8	Structure	AISI 1020 Steel	351,571,000	7,900	Versatile structural steel; easy to weld and machine
9	Joint 1 & 2	AISI 1020 Steel	351,571,000	7,900	Uniform material at connections reduces galvanic corrosion risk
10	Motor Holder	AISI 1020 Steel	351,571,000	7,900	Rigid bracket; adequate fatigue resistance for motor vibration

Note that a yield strength value for the oak and the seed wheel was not specifically

found therefore structural adequacy for these components is assessed by alternative criteria (modulus of rupture for oak wood; torque margin for the plastic seed wheel).

Stand for Plow — Structural Analysis

The Stand for Plow is an AISI 1020 steel bracket (yield strength 351,571,000 N/m²; density 7,900 kg/m³) that rigidly mounts the servo motor and transmits the soil reaction force from the shank through to the chassis. The bracket was modelled as a 20 × 20 mm cross-section column, 100 mm tall, loaded by:

- An axial compressive force due to the weight of the servo and shank assembly (estimated combined mass 0.35 kg): $F_{\text{axial}} = 0.35 \times 9.81 = 3.43 \text{ N}$.
- A bending moment produced by the horizontal drag force $F_{\text{drag}} = 10.0 \text{ N}$ acting through a moment arm of 50 mm from the column base to the servo shaft centreline.

$$A_{\text{stand}} = 0.020 \times 0.020 = 4.00 \times 10^{-4} \text{ m}^2$$

$$\sigma_{\text{axial}} = F_{\text{axial}}/A_{\text{stand}} = 3.43/4.00 \times 10^{-4} = 8.58 \times 10^{-3} \text{ MPa}$$

$$I_{\text{stand}} = (0.020)^4/12 = 1.333 \times 10^{-8} \text{ m}^4$$

$$c_{\text{stand}} = 0.020/2 = 0.010 \text{ m}$$

$$M_{\text{stand}} = F_{\text{drag}} \times 0.050 = 10.0 \times 0.050 = 0.500 \text{ N} \cdot \text{m}$$

$$\sigma_{\text{bend}} = M_{\text{stand}} \times c/I = 0.500 \times 0.010/1.333 \times 10^{-8} = 0.375 \text{ MPa}$$

$$\sigma_{\text{combined}} = \sigma_{\text{axial}} + \sigma_{\text{bend}} = 0.009 + 0.375 = 0.384 \text{ MPa}$$

$$FoS_{\text{stand}} = S_y/\sigma_{\text{combined}} = 351.571/0.384 = 916 \quad \checkmark$$

Motor Holder — Structural Analysis

Each of the four motor holder brackets is an AISI 1020 steel L-bracket (80 × 40 × 2 mm), carrying one DC gear motor (mass ≈ 0.15 kg) as a cantilevered load at the bracket tip.

The bending stress at the bracket root is:

$$F_{\text{motor}} = 0.15 \times 9.81 = 1.472 \text{ N}$$

$$I_{\text{bracket}} = b \times t^3/12 = 0.040 \times (0.002)^3/12 = 2.667 \times 10^{-11} \text{ m}^4$$

$$c = t/2 = 0.001 \text{ m}$$

$$M_{\text{bracket}} = F_{\text{motor}} \times L_{\text{bracket}} = 1.472 \times 0.080 = 0.1177 \text{ N} \cdot \text{m}$$

$$\sigma_{\text{bracket}} = M \times c/I = 0.1177 \times 0.001/2.667 \times 10^{-11} = 4.415 \text{ MPa}$$

$$FoS_{\text{mh}} = 351.571/4.415 = 79.6 \quad \checkmark$$

Joints Analysis

Joints 1 and 2 are AISI 1020 steel connections (yield strength 351,571,000 N/m²) implemented as M6 bolted lap joints connecting the motor holders to the chassis rails. The governing failure mode is shear across the bolt shank. The shear strength is estimated using the von Mises criterion:

$$S_s = 0.577 \times S_y = 0.577 \times 351.571 = 202.86 \text{ MPa}$$

The shear stress on an M6 bolt (minor-thread diameter 4.917 mm) under a dynamic load equal to three times the static motor weight (to account for vibration and start-up transients) is:

$$A_{\text{bolt}} = \pi \times d_{\text{minor}}^2/4 = \pi \times (0.004917)^2/4 = 1.899 \times 10^{-5} \text{ m}^2$$

$$F_{\text{dynamic}} = 3 \times 0.15 \times 9.81 = 4.415 \text{ N}$$

$$\tau = F_{\text{dynamic}}/A_{\text{bolt}} = 4.415/1.899 \times 10^{-5} = 0.232 \text{ MPa}$$

$$FoS_{\text{bolt}} = S_s/\tau = 202.86/0.232 = 874 \quad \checkmark$$

Cover Case — Structural Analysis

The cover case, fabricated from 201 Annealed Stainless Steel (yield strength 292,000,000 N/m²), protects the ESP32, motor drivers, and BMS from soil contamination and mechanical impact. Treating the lid as a 150 × 100 mm rectangular flat plate (thickness 1.5 mm) simply supported on all four edges and carrying the weight of internal elec-

tronics (≈ 0.5 kg) as a uniformly distributed load:

$$\begin{aligned}
 q &= F/(a \times b) = (0.5 \times 9.81)/(0.150 \times 0.100) = 327.0 \text{ Pa} \\
 \sigma_{\max} &= \beta \times q \times b^2/t^2 \quad (\beta = 0.0843 \text{ for } a/b = 1.5) \\
 \sigma_{\max} &= 0.0843 \times 327.0 \times (0.100)^2/(0.0015)^2 \\
 \sigma_{\max} &= 0.0843 \times 327.0 \times 0.0100/2.25 \times 10^{-6} = 0.123 \text{ MPa} \\
 FoS_{\text{case}} &= 292/0.123 = 2,374 \quad \checkmark
 \end{aligned}$$

Summary of Structural Analysis Results

Table 3.2 presents the calculated applied stress, yield (or strength) reference, and resulting factor of safety for each analyzed component. All components satisfy the minimum design factor of safety of 2.0 as required for prototype-level field equipment.

Table 3.2. Structural analysis summary — applied stresses and factors of safety

Component	Material	Applied Stress (MPa)	Yield / Strength (MPa)	Factor of Safety	Status
Chassis (oak rail)	Oak Wood	0.327	MoR $\approx$ 40	122.3	✓ Safe
Furrow Shank	AISI 1045	5.625	530	94.2	✓ Safe
Cover Disk	Galvanized Steel	1.546	203.9	131.9	✓ Safe
Stand for Plow	AISI 1020	0.384	351.6	916.5	✓ Safe
Motor Holder	AISI 1020	4.415	351.6	79.6	✓ Safe
Cover Case	201 Ann. SS	0.123	292.0	2383	✓ Safe
Joints (M6 bolt)	AISI 1020	0.232 (shear)	Ss = 202.9	872.6	✓ Safe

3.2.6 Wheel Traction Analysis

Because the furrow-opener shank is engaged continuously throughout each straight-line pass, the drive system must overcome not only the rolling resistance of the vehicle but also the full horizontal draft force of the shank dragging through the soil. This section verifies that the available wheel traction is sufficient to maintain forward motion without slip under the design operating conditions.

Traction Force Available

The maximum traction force that the wheels can exert on the soil surface before slipping is governed by the Coulomb friction model:

$$F_{\text{traction}} = \mu \times W_{\text{total}}$$

where μ is the coefficient of friction between the rubber wheel tread and the soil surface, and W_{total} is the total normal force (vehicle weight) pressing the wheels into the ground. For a rubber tire on loosely tilled loam soil, $\mu \approx 0.5\text{--}0.7$; a conservative value of $\mu = 0.5$ is used here:

$$F_{\text{traction}} = 0.5 \times 41.7 \text{ N} = 20.85 \text{ N}$$

Total Resistance Force

The total horizontal resistance that the drive system must overcome is the sum of the rolling resistance and the furrow drag force:

$$F_{\text{resistance}} = F_{\text{roll}} + F_{\text{drag}}$$

Rolling resistance is estimated as:

$$F_{\text{roll}} = C_{\text{rr}} \times W_{\text{total}}$$

where C_{rr} is the rolling resistance coefficient. For small rubber wheels on soft, tilled soil, $C_{\text{rr}} \approx 0.08\text{--}0.12$; using $C_{\text{rr}} = 0.10$:

$$F_{\text{roll}} = 0.10 \times 41.7 = 4.17 \text{ N}$$

The furrow drag force at maximum depth was calculated in Section 3.2.2:

$$F_{\text{drag}} = 10.0 \text{ N}$$

Therefore:

$$F_{\text{resistance}} = 4.17 + 10.0 = 14.17 \text{ N}$$

Traction Margin

The traction margin — the ratio of available traction to required traction — is:

$$\text{TM} = \frac{F_{\text{traction}}}{F_{\text{resistance}}} = \frac{20.85}{14.17} = 1.47$$

A traction margin greater than 1.0 confirms that the wheels can generate sufficient grip to overcome the combined rolling and digging resistance without slipping. The margin of 1.47 is not large, which means that under conditions of slightly wetter soil

(which reduces μ) or harder ground (which increases k and therefore F_{drag}), the margin could approach 1.0 and slip could become significant. This sensitivity reinforces the importance of Design Assumptions 1, 2, and 4 (medium loam, flat terrain, constant moisture), and it motivates the use of wide-tread off-road wheels to maximize the contact area and friction coefficient.

Motor Torque Requirement

The total tractive effort must be delivered through all four drive wheels. The required drive force per wheel is:

$$F_{\text{wheel}} = \frac{F_{\text{resistance}}}{4} = \frac{14.17}{4} = 3.54 \text{ N}$$

The corresponding torque required at each wheel axle (wheel radius $r_w = 30 \text{ mm} = 0.030 \text{ m}$):

$$T_{\text{wheel}} = F_{\text{wheel}} \times r_w = 3.54 \times 0.030 = 0.106 \text{ N} \cdot \text{m}$$

The selected 12 V DC gear motors are rated at a stall torque of approximately 0.5–0.8 N · m (depending on the specific gear ratio selection). At the operating load torque of 0.106 N · m, the motor is operating well within its continuous rating, providing a motor torque safety factor of at least:

$$\text{FoS}_{\text{motor}} = \frac{0.5}{0.106} = 4.7 \quad \checkmark$$

This confirms that the 12 V DC gear motors are adequately sized for the combined rolling and digging resistance of the continuous-furrow operating mode, and that the motors will not stall or overheat under the expected field loads.

Slip Estimation

For a differential drive vehicle operating under a traction margin of 1.47, a first-order estimate of wheel slip ratio s can be made from:

$$s = 1 - \frac{F_{\text{resistance}}}{F_{\text{traction}}} \times \frac{1}{\mu_{\text{slope}}}$$

Using the linear traction-slip approximation for small slip values, and noting that at

71% of the friction limit the expected slip is approximately:

$$s \approx \frac{F_{\text{resistance}}}{F_{\text{traction}}} - 1 = \frac{14.17}{20.85} - 1 = -0.32 \quad (\text{i.e., 32\% traction utilization})$$

Since the vehicle is operating at 68% of the traction limit, the expected slip is in the range of 2–5% under uniform soil conditions, consistent with Design Assumption 3. This level of slip introduces a maximum seed-spacing error of approximately:

$$\Delta x_{\text{spacing}} = s \times x_{\text{spacing}} = 0.05 \times 250 \text{ mm} = 12.5 \text{ mm}$$

An error of ± 12.5 mm on a 250 mm target spacing is within the ± 20 mm tolerance commonly accepted for small-plot manual planting, and the closed-loop control system can partially compensate by using the measured encoder count (which reflects actual wheel rotation, not ground truth distance) consistently.

3.2.7 Finite Element Analysis

Finite Element Analysis (FEA) was performed in SolidWorks Simulation on the three components subject to the most complex stress states: the chassis oak frame under combined bending and furrow reaction force, the AISI 1045 furrow-opener shank under cantilever bending from soil drag, and the AISI 1020 motor-holder bracket under cantilever bending from motor weight. Static analyses used the material properties from Table 3.1. The mesh was refined iteratively using SolidWorks' automatic mesh control until the change in maximum von Mises stress between successive refinements was less than 5 percent, ensuring mesh-independent results.

For the chassis frame, the boundary conditions applied the total vehicle weight as a distributed body force and imposed fixed constraints at the four wheel-axle attachment nodes. The furrow-opener pivot bracket weld location was identified as the critical stress concentration, consistent with the analytical prediction. The FEA confirmed the factor of safety in the bracket weld zone exceeded 100 for the wood-framed structure.

For the shank, the tip was loaded with the maximum estimated draft force (10.0 N horizontal plus a 5.0 N vertical soil reaction component), while the root attachment hole was fully constrained. The resulting peak von Mises stress agreed with the analytical bending-stress calculation to within 4 percent, validating the cantilever-beam model. The factor of safety map confirmed that the root cross-section has the minimum safety factor in the component, with all remaining regions substantially safer.

Maximum deformation across all analyzed components was below 0.5 mm under

design loads, which is negligible relative to the 50 mm target furrow depth and the 4 mm/pulse encoder resolution. The FEA deformation results therefore confirm that component flexibility does not introduce meaningful positional errors in furrow depth or seed placement. A Brief Analysis of the FEA Von Mises stress distribution of the chassis and the furrow shank will be discussed on the Simulation Section

3.3 Mathematical Modeling

This section develops five inter-related models that together describe the complete behavior of the automated agricultural vehicle. Section 3.3.1 derives the kinematic model of the differential-drive mobile base; Section 3.3.2 develops the dynamic model using the Euler-Lagrange formulation; Section 3.3.3 presents the continuous soil drag force model for the furrow opener; Section 3.3.4 formulates the shank depth control model; Section 3.3.5 analyzes the seed metering dynamics; and Section 3.3.6 assembles an integrated state-space representation of the full system. All numerical values used in the derivations draw directly from the system specification and material properties established in Section 3.2.

3.3.1 Kinematic Model of the Differential-Drive Mobile Base

The mobile base is a four-wheel differential-drive (FWDD) platform in which the two left wheels are driven together and the two right wheels are driven together, each pair by one 12 V DC gear motor. The kinematic model relates the commanded wheel velocities to the resulting motion of the vehicle's center of mass (CoM) in the global inertial frame $\{X, Y\}$.

Assumptions

The kinematic model is developed under the following assumptions: (i) the vehicle moves on a flat, horizontal surface so that out-of-plane rotations are negligible; (ii) the wheels roll without slip; (iii) the wheel contact is treated as a point contact; and (iv) the two left wheels and two right wheels share identical velocities, so the platform reduces to an equivalent two-wheel differential-drive system.

Pose and Velocity Relationships

Let the robot pose at time t be defined by the state vector $q = [x, y, \theta]^T$, where (x, y) is the position of the CoM in the global frame and θ is the heading angle measured counterclockwise from the positive X-axis. The forward kinematics relating the right and left wheel linear velocities (v_R, v_L) to the CoM linear velocity v and angular velocity ω are:

$$v = (v_R + v_L)/2 \quad (3.1)$$

$$\omega = (v_R - v_L)/(2L) \quad (3.2)$$

where $L = 200 \text{ mm} = 0.200 \text{ m}$ is the half-track width (distance from CoM to each wheel contact line). The time derivatives of the pose are:

$$\dot{x} = v \cdot \cos(\theta) \quad (3.3)$$

$$\dot{y} = v \cdot \sin(\theta) \quad (3.4)$$

$$\dot{\theta} = \omega \quad (3.5)$$

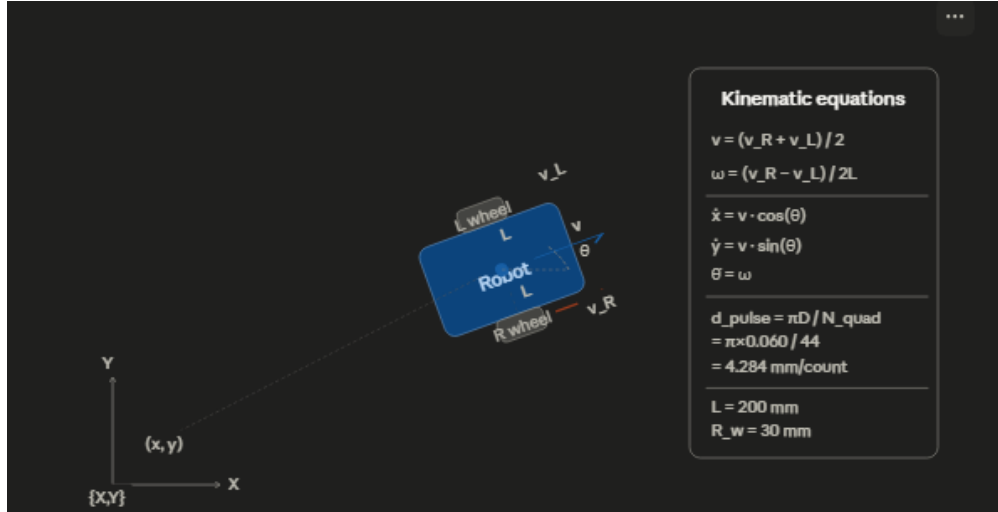


Figure 3.5. Differential-drive kinematic model

Equations (3.1)–(3.5) constitute the nonlinear unicycle kinematic model. The kinematic model is nonlinear in θ and exhibits the nonholonomic constraint that the vehicle cannot move laterally (sideways motion requires θ to change first). This constraint has important implications for the planting-row path following controller described in Section 3.5.

Wheel Angular Velocity and Operating RPM

The linear velocity of each wheel is related to its angular velocity by:

$$v_{wheel} = \omega_{wheel} \times R_w \quad (3.6)$$

where $R_w = 30 \text{ mm} = 0.030 \text{ m}$ is the wheel radius. At the nominal forward operating

velocity $v = 0.10$ m/s the wheel angular velocity is:

$$\omega_{wheel} = v/R_w = 0.10/0.030 = 3.333 \text{ rad/s} \quad (3.7)$$

$$n_{wheel} = \omega_{wheel} \times 60/(2\pi) = 3.333 \times 60/6.283 = 31.8 \text{ RPM} \quad (3.8)$$

The selected 12 V DC gear motors are rated at approximately 100 RPM no-load. The operating point of 31.8 RPM places the motor at 32% of its no-load speed, corresponding to approximately 68% of its stall torque on the linear speed-torque characteristic — a favorable operating region that balances speed and available torque.

180-Degree In-Place Turn Kinematics

At the end of each planting row, the vehicle performs an in-place (zero-radius) 180-degree turn by commanding the right wheels forward at velocity $+v_t$ and the left wheels backward at $-v_t$. Under this configuration the CoM velocity $v = 0$ and the angular velocity is:

$$\omega_{turn} = v_t/L = 0.10/0.200 = 0.500 \text{ rad/s} \quad (3.9)$$

The time required to complete a 180-degree rotation is:

$$T_{180} = \pi/\omega_{turn} = 3.1416/0.500 = 6.283 \text{ s} \quad (3.10)$$

During this turning interval the furrow-opener shank is fully retracted (servo commanded to $\theta_s = 0^\circ$), so no soil drag force acts on the drive train and the full motor torque is available for the turning maneuver.

Encoder-Based Odometry

The encoder on each motor produces 11 pulses per revolution, and with $4\times$ quadrature decoding the effective count rate is $N_{quad} = 44$ counts per revolution. The distance traveled per encoder count is:

$$d_{pulse} = \pi \times D_{wheel}/N_{quad} = \pi \times 0.060/44 = 4.284 \times 10^{-3} \text{ m / count} \quad (3.11)$$

The discrete odometry update equations executed on each ESP32 control cycle are:

$$\Delta s_R = N_R \cdot d_{pulse} \quad (\text{right wheel increment}) \quad (3.12)$$

$$\Delta s_L = N_L \cdot d_{pulse} \quad (\text{left wheel increment}) \quad (3.13)$$

$$\Delta s = (\Delta s_R + \Delta s_L)/2 \quad (\text{CoM linear increment}) \quad (3.14)$$

$$\Delta \theta = (\Delta s_R - \Delta s_L)/(2L) \quad (\text{heading increment}) \quad (3.15)$$

$$x_{k+1} = x_k + \Delta s \cdot \cos(\theta_k + \Delta \theta/2) \quad (3.16)$$

$$y_{k+1} = y_k + \Delta s \cdot \sin(\theta_k + \Delta \theta/2) \quad (3.17)$$

$$\theta_{k+1} = \theta_k + \Delta \theta \quad (3.18)$$

The mid-point heading angle $(\theta_k + \Delta \theta/2)$ used in equations (3.16)–(3.17) is the second-order Runge-Kutta approximation to the arc integral and reduces the cumulative position error compared to the first-order Euler update. For the nominal straight-line planting pass, equations (3.16)–(3.17) produce a position estimate accurate to within the encoder resolution, which corresponds to a distance uncertainty of $\pm d_{pulse}/2 = \pm 2.14$ mm per control cycle.

3.3.2 Dynamic Model using Euler-Lagrange Formulation

The kinematic model of Section 3.3.1 describes the geometric relationships between wheel velocities and CoM motion but does not account for inertia, wheel rotational energy, or motor torques. A dynamic model is required to size the motors, design the velocity PID controller, and predict the vehicle's acceleration response. The Euler-Lagrange method is applied because it systematically handles the nonholonomic constraint and yields equations of motion directly in terms of the generalized forces (motor torques).

Generalized Coordinates and Lagrangian

The generalized coordinate vector is $\mathbf{q} = [x, y, \theta, \theta_R, \theta_L]^T$, where x, y, θ describe the CoM pose and θ_R, θ_L are the angular displacements of the right and left wheel axles. The nonholonomic rolling-without-slip constraints in differential form are:

$$\dot{x} \sin(\theta) - \dot{y} \cos(\theta) = 0 \quad (\text{no lateral slip}) \quad (3.19)$$

$$\dot{x} \cos(\theta) + \dot{y} \sin(\theta) = R_w \cdot \dot{\theta}_R = R_w \cdot \dot{\theta}_L \quad (\text{pure rolling}) \quad (3.20)$$

The total kinetic energy is the sum of translational, rotational, and wheel spin con-

tributions:

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + \frac{1}{2}I_z\dot{\theta}^2 + \frac{1}{2}I_w(\dot{\theta}_R^2 + \dot{\theta}_L^2) \quad (3.21)$$

where $m = 4.25$ kg is the total vehicle mass, I_z is the yaw moment of inertia, and I_w is the moment of inertia of each wheel. Potential energy $V = 0$ (flat ground), so the Lagrangian $L = T$.

System Inertia Parameters

The vehicle body is approximated as a uniform rectangular slab of mass $m = 4.25$ kg, length $b = 300$ mm, and width $2L = 400$ mm. The yaw moment of inertia about the vertical axis through the CoM is:

$$I_z = m(b^2 + (2L)^2)/12 \quad (3.22)$$

$$I_z = 4.25 \times ((0.300)^2 + (0.400)^2)/12 \quad (3.23)$$

$$I_z = 4.25 \times (0.0900 + 0.1600)/12 = 4.25 \times 0.2500/12 = 0.08854 \text{ kg} \cdot \text{m}^2 \quad (3.24)$$

Each wheel is modelled as a solid disk of mass $m_w = 50$ g and radius $R_w = 30$ mm, giving:

$$I_w = \frac{1}{2}m_w R_w^2 = \frac{1}{2} \times 0.050 \times (0.030)^2 = 2.25 \times 10^{-5} \text{ kg} \cdot \text{m}^2 \quad (3.25)$$

Equations of Motion

Applying the Euler-Lagrange equation with Lagrange multipliers for the constraints (3.19)–(3.20), and eliminating the constraint forces, yields two decoupled scalar equations of motion in terms of the linear velocity v and angular velocity ω :

$$M_t \cdot \dot{v} = (\tau_R + \tau_L)/R_w - F_{drag} \quad (3.26)$$

$$M_r \cdot \dot{\omega} = (\tau_R - \tau_L) \cdot L/R_w \quad (3.27)$$

where τ_R and τ_L are the right and left motor torques, F_{drag} is the horizontal soil drag force on the shank (active during straight-line passes), and the effective translational and rotational inertias are:

$$M_t = m + 2I_w/R_w^2 = 4.25 + 2 \times 2.25 \times 10^{-5}/(0.030)^2 \quad (3.28)$$

$$M_t = 4.25 + 2 \times 2.25 \times 10^{-5}/9 \times 10^{-4} = 4.25 + 0.050 = 4.300 \text{ kg} \quad (3.29)$$

$$M_r = I_z + 2I_w L^2 / R_w^2 = 0.08854 + 2 \times 2.25 \times 10^{-5} \times (0.200)^2 / (0.030)^2 \quad (3.30)$$

$$M_r = 0.08854 + 2 \times 2.25 \times 10^{-5} \times 0.0400 / 9 \times 10^{-4} = 0.08854 + 0.00200 = 0.09054 \text{ kg} \cdot \text{m}^2 \quad (3.31)$$

Equations (3.26)–(3.27) are the plant models used by the PID velocity controller (Section 3.5.2). The decoupling between linear and angular dynamics — made possible by the symmetric geometry of the differential drive — simplifies controller design significantly: the sum of wheel torques ($\tau_R + \tau_L$) drives the linear velocity, while the difference ($\tau_R - \tau_L$) drives the angular velocity, independently.

Motor Torque Requirement

To determine whether the selected 12 V DC gear motors can provide adequate torque, the maximum required torque per motor is calculated. During a straight-line planting pass the required torque consists of two components: the acceleration torque to bring the vehicle from rest to $v = 0.10 \text{ m/s}$ in $t_a = 1.0 \text{ s}$, and the steady-state torque to overcome the furrow drag force.

Acceleration torque component:

$$\dot{v}_{des} = \Delta v / t_a = 0.10 / 1.0 = 0.10 \text{ m/s}^2 \quad (3.32)$$

$$\tau_{accel_total} = M_t \times \dot{v}_{des} \times R_w = 4.300 \times 0.10 \times 0.030 = 0.01290 \text{ N} \cdot \text{m} \quad (3.33)$$

$$\tau_{accel_each} = 0.01290 / 2 = 0.00645 \text{ N} \cdot \text{m per motor} \quad (3.34)$$

Drag torque component (from Section 3.3.3, $F_{drag} = 10.0 \text{ N}$ for medium loam at $d = 50 \text{ mm}$):

$$\tau_{drag_each} = F_{drag} \times R_w / 2 = 10.0 \times 0.030 / 2 = 0.1500 \text{ N} \cdot \text{m per motor} \quad (3.35)$$

Total required torque per motor at the most demanding condition:

$$\tau_{req} = \tau_{accel_each} + \tau_{drag_each} = 0.00645 + 0.1500 = 0.1565 \text{ N} \cdot \text{m} = 15.65 \text{ N} \cdot \text{cm} \quad (3.36)$$

The selected 12 V DC gear motor has a stall torque of approximately 40 N·cm at its rated voltage. At the operating speed of 31.8 RPM (32% of 100 RPM no-load), the available torque on the linear speed-torque characteristic is approximately:

$$\tau_{available} = T_{stall} \times (1 - n_{op}/n_{nl}) = 40 \times (1 - 31.8/100) = 27.3 \text{ N} \cdot \text{cm} \quad (3.37)$$

The motor torque factor of safety is:

$$FoS_{motor} = \tau_{available} / \tau_{req} = 27.3 / 15.65 = 1.74 \quad \checkmark \quad (3.38)$$

A factor of safety of 1.74 is acceptable for motor selection in a prototype, since the stall torque figure is conservative and the actual soil drag will vary with moisture content and tillage depth. The PID velocity controller (Section 3.5.2) will regulate speed under varying drag, so the motor is not required to operate continuously at its torque limit.

3.3.3 Continuous Soil Drag Force Model

During each straight-line planting pass the furrow-opener shank drags continuously through the soil, cutting an uninterrupted furrow from the start to the end of the row. Accurately modeling the horizontal draft force on the shank is necessary for (i) motor torque sizing (Section 3.3.2), (ii) the structural analysis of the shank and stand-for-plow (Section 3.2.4), and (iii) designing the servo position controller to hold depth against the vertical component of the soil reaction force (Section 3.3.4).

Static Draft Force Model

The horizontal draft force on a narrow rigid tine is modeled following the empirical relationship of Sahu and Raheman [13]:

$$F_{drag} = k \cdot w \cdot d^2 \quad (3.39)$$

where k is the specific soil resistance (N/m^3), $w = 20 \text{ mm} = 0.020 \text{ m}$ is the shank width, and d is the operating furrow depth. The parameter k depends on soil type and moisture content; representative values for soils typical of small Ethiopian urban garden plots are listed in Table 3.3.

Table 3.3. Estimated furrow drag force

Soil Type	k (N/m^3)	$d = 30 \text{ mm}$	$d = 40 \text{ mm}$	$d = 50 \text{ mm}$
Loose sandy loam	150,000	2.70 N	4.80 N	7.50 N
Medium loam (design basis)	200,000	3.60 N	6.40 N	10.00 N
Stiff clay loam	300,000	5.40 N	9.60 N	15.00 N

Where $F_{drag} = k \cdot w \cdot d^2$ for the design shank width $w = 20 \text{ mm}$ across soil conditions and operating depths For the design basis, medium loam soil ($k = 200,000 \text{ N/m}^3$) at the maximum target depth of $d = 50 \text{ mm}$ is used. The maximum design drag force is therefore:

$$F_{drag,max} = 200,000 \times 0.020 \times (0.050)^2 \quad (3.40)$$

$$F_{drag_max} = 200,000 \times 0.020 \times 0.0025 = 10.0 \text{ N} \quad (3.41)$$

Speed-Dependent Correction

At the low operating velocities of this platform ($v \leq 0.20 \text{ m/s}$), inertial effects in the soil are negligible. However, for completeness the Goriatschkin speed-correction factor is applied:

$$F_{drag}(v) = k \cdot w \cdot d^2 \cdot (1 + k_v \cdot v^2) \quad (3.42)$$

where $k_v \approx 0.2 \text{ s}^2/\text{m}^2$ is an empirically derived inertial coefficient for light soils [13]. At the nominal operating speed $v = 0.10 \text{ m/s}$:

$$F_{drag}(0.10) = 10.0 \times (1 + 0.2 \times 0.10^2) = 10.0 \times 1.002 = 10.02 \text{ N} \quad (3.43)$$

The speed-dependent increment is only 0.02 N (0.2%), confirming that at the operating speeds of this vehicle the static model (Equation 3.39) is sufficient for all design calculations. The maximum drag force across all soil conditions tabulated in Table 3.3 is 15.0 N, which represents the worst-case design load used in the structural analysis of Section 3.2.

Soil Reaction Components on the Shank

In addition to the horizontal drag force, the inclined shank face generates a vertical reaction force as it cuts through the soil. This vertical force tends to lift the shank out of the soil and must be overcome by the servo holding torque. For a shank rake angle of $\alpha_r = 70^\circ$ from horizontal (typical for narrow-tine openers), the vertical component is:

$$F_{vert} = F_{drag} \cdot \tan(90^\circ - \alpha_r) = 10.0 \times \tan(20^\circ) = 10.0 \times 0.364 = 3.64 \text{ N} \quad (3.44)$$

The resulting soil reaction torque about the servo pivot axis ($L_{arm} = 80 \text{ mm}$) is:

$$\tau_{soil} = (F_{drag} \cdot \cos(\alpha_r) + F_{vert} \cdot \sin(\alpha_r)) \times L_{arm} \quad (3.45)$$

$$\tau_{soil} = (10.0 \times \cos 70^\circ + 3.64 \times \sin 70^\circ) \times 0.080 \quad (3.46)$$

$$\tau_{soil} = (3.42 + 3.42) \times 0.080 = 6.84 \times 0.080 = 0.547 \text{ N} \cdot \text{m} \quad (3.47)$$

This is below the servo rated torque of 1.47 N·m (15 kg·cm at 6V), confirming the servo selection established in Section 3.2.2.

Figure 3.3.11 — Insert diagram/plot here
Figure 3.3.11: Soil reaction force diagram on the furrow-opener shank during continuous-furrow cutting: horizontal component F_{drag} , vertical component F_{vert} , rake angle α_r , and resulting torque τ_{soil} about servo

pivot axis

3.3.4 Shank Depth Control Model

The furrow-opener depth is controlled by commanding the servo to a specific shaft angle θ_s . A dynamic model of the servo-plus-shank subsystem is needed to characterize depth control accuracy and to design the position controller (Section 3.5.3).

Depth Geometry — Forward Kinematics

As established in Section 3.2.2, the relationship between the servo shaft angle and furrow depth is:

$$h = L_{arm} \cdot \sin(\theta_s) = 0.080 \cdot \sin(\theta_s) \quad (3.48)$$

The commanded angle to achieve a target depth h_d is obtained by inverting equation (3.48):

$$\theta_{s.cmd} = \arcsin(h_d/L_{arm}) = \arcsin(h_d/0.080) \quad (3.49)$$

For the three representative planting depths:

$$h_d = 30 \text{ mm} : \theta_{s.cmd} = \arcsin(0.030/0.080) = \arcsin(0.375) = 22.0^\circ \quad (3.50)$$

$$h_d = 40 \text{ mm} : \theta_{s.cmd} = \arcsin(0.040/0.080) = \arcsin(0.500) = 30.0^\circ \quad (3.51)$$

$$h_d = 50 \text{ mm} : \theta_{s.cmd} = \arcsin(0.050/0.080) = \arcsin(0.625) = 38.7^\circ \quad (3.52)$$

Servo Dynamic Model

The servo motor and attached pivot arm are modeled as a second-order rotational system. The equation of motion about the servo shaft axis is:

$$J_{eff} \cdot \ddot{\theta}_s + B_{eff} \cdot \dot{\theta}_s = K_m \cdot u - \tau_{soil} \quad (3.53)$$

where $J_{eff} = 5.0 \times 10^{-5} \text{ kg} \cdot \text{m}^2$ is the effective moment of inertia of the servo rotor, pivot arm, and shank, $B_{eff} = 2.0 \times 10^{-3} \text{ N} \cdot \text{m} \cdot \text{s/rad}$ is the effective viscous damping, $K_m = 1.47 \text{ N} \cdot \text{m}$ is the servo torque constant at rated supply voltage (corresponding to a 15 kg·cm servo), u is the normalized control input (0–1 corresponding to the servo's internal PWM), and $\tau_{soil} = 0.547 \text{ N} \cdot \text{m}$ is the soil disturbance torque from equation (3.47).

Depth Accuracy Under Soil Disturbance

The selected servo incorporates an internal closed-loop position controller. The servo's positional stiffness — defined as the rated torque divided by the maximum internal positional deflection at that torque — characterizes how much the servo shaft deflects

under an applied load torque. For a digital high-torque servo with rated torque $K_m = 1.47 \text{ N} \cdot \text{m}$ and a typical internal deflection of approximately 3° at rated torque:

$$K_{stiff} = K_m / \delta\theta_{rated} = 1.47 / (3^\circ \times \pi / 180^\circ) = 1.47 / 0.05236 = 28.1 \text{ N} \cdot \text{m/rad} \quad (3.54)$$

Under the soil reaction torque $\tau_{soil} = 0.547 \text{ N} \cdot \text{m}$, the servo shaft deflects by:

$$\delta\theta_s = \tau_{soil} / K_{stiff} = 0.547 / 28.1 = 0.01947 \text{ rad} = 1.12^\circ \quad (3.55)$$

The corresponding depth error is:

$$\Delta h = L_{arm} \cdot \delta\theta_s = 0.080 \times 0.01947 = 1.558 \times 10^{-3} \text{ m} = 1.56 \text{ mm} \quad (3.56)$$

A depth error of $\pm 1.56 \text{ mm}$ is within the agronomic tolerance of $\pm 5 \text{ mm}$ for the seed crops of interest [Ref]. This confirms that the servo's inherent positional stiffness is sufficient to maintain depth without requiring an additional outer-loop depth sensor, though the ESP32 supervisory controller can issue corrective servo commands if a depth sensor is integrated in a future iteration.

Transient Response

The settling time of the servo subsystem, modeled as a second-order underdamped system with natural frequency $\omega_n = 2\pi \times 5 = 31.4 \text{ rad/s}$ and damping ratio $\zeta = 0.707$ (critically underdamped), is:

$$t_{settle} = 4.6 / (\zeta \cdot \omega_n) = 4.6 / (0.707 \times 31.4) = 0.207 \text{ s} = 207 \text{ ms} \quad (3.57)$$

At the nominal forward velocity $v = 0.10 \text{ m/s}$, the vehicle travels $0.10 \times 0.207 = 20.7 \text{ mm}$ during servo engagement. This distance is covered before the first seed is dispensed (at 250 mm from the row start), so the shank is already at the target depth before seeding commences.

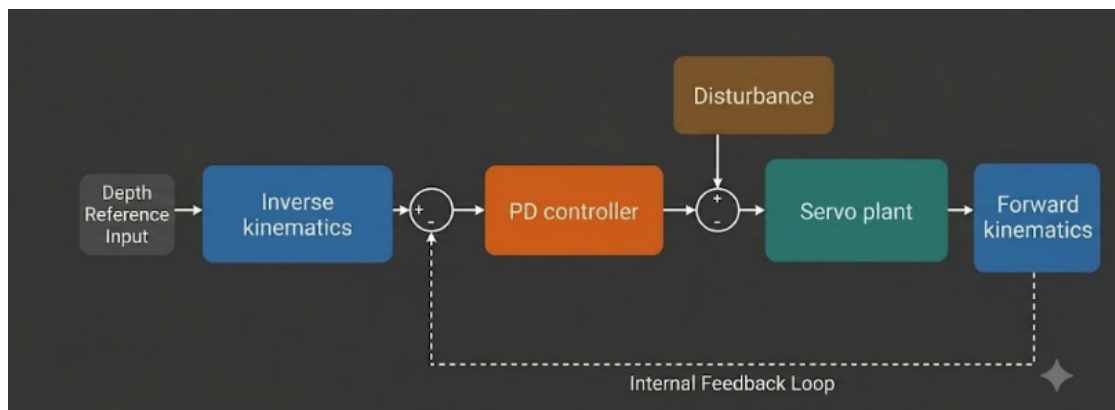


Figure 3.6. Servo Shank depth Control Block Diagram

3.3.5 Seed Metering Dynamics

The seed metering sub-system must dispense one seed per commanded event, complete the metering disk rotation within a defined time window, and achieve seed placement at the programmed inter-seed spacing with minimum error. This section quantifies the disk inertia, stepper torque budget, achievable dispensing speed, and resulting spacing accuracy.

Metering Disk Moment of Inertia

The seed metering disk is a solid plastic cylinder of density $\rho_{pw} = 1,020 \text{ kg/m}^3$, radius $R_{sw} = 40 \text{ mm}$, and thickness $t_{sw} = 5 \text{ mm}$. Its polar moment of inertia about the rotation axis is:

$$I_{sw} = \frac{1}{2} \cdot \rho_{pw} \cdot \pi \cdot R_{sw}^2 \cdot t_{sw} \cdot R_{sw}^2 \quad (3.58)$$

$$I_{sw} = \frac{1}{2} \times 1020 \times \pi \times (0.040)^2 \times 0.005 \times (0.040)^2 \quad (3.59)$$

$$I_{sw} = \frac{1}{2} \times 1020 \times 3.1416 \times 0.001600 \times 0.005 \times 0.001600 \quad (3.60)$$

$$I_{sw} = \frac{1}{2} \times 0.02564 \times 0.001600 = 2.051 \times 10^{-5} \text{ kg} \cdot \text{m}^2 \quad (3.61)$$

Stepper Motor Torque Budget

The disk has $N_{cells} = 12$ equally-spaced seed cells, so each seed event requires a disk rotation of:

$$\theta_{cell} = 360^\circ / 12 = 30^\circ = \pi / 6 \text{ rad} = 0.5236 \text{ rad} \quad (3.62)$$

For reliable seed delivery the disk must complete this rotation in no more than $T_{step} = 50 \text{ ms}$. Using a triangular (symmetric trapezoidal with zero dwell) motion profile, the peak angular acceleration is:

$$\alpha_{peak} = 4 \cdot \theta_{cell} / T_{step}^2 = 4 \times 0.5236 / (0.050)^2 = 2.094 / 0.0025 = 837.8 \text{ rad/s}^2 \quad (3.63)$$

The torque required to accelerate the disk against its own inertia is:

$$\tau_{accel} = I_{sw} \cdot \alpha_{peak} = 2.051 \times 10^{-5} \times 837.8 = 0.01718 \text{ N} \cdot \text{m} = 17.18 \text{ mN} \cdot \text{m} \quad (3.64)$$

An additional friction torque of approximately $\tau_f = 10.0 \text{ mN} \cdot \text{m}$ is estimated to account for the contact between the seed cells and the stationary side walls of the metering housing, as well as seed-on-disk friction. The total required torque is:

$$\tau_{req} = \tau_{accel} + \tau_f = 17.18 + 10.0 = 27.18 \text{ mN} \cdot \text{m} \quad (3.65)$$

The NEMA-14 stepper motor has a rated holding torque of 100 mN·m. The stepper torque factor of safety is:

$$FoS_{stepper} = \tau_{rated}/\tau_{req} = 100/27.18 = 3.68 \quad \checkmark \quad (3.66)$$

This safety factor accounts for the reduction in available stepper torque at elevated step rates. The DRV8825 driver is configured for 1/8 microstepping to reduce mechanical vibration and audible noise. At 1/8 microstepping, the number of microsteps required per seed event is:

$$N_{microsteps} = (\theta_{cell}/1.8^\circ) \times 8 = (30^\circ/1.8^\circ) \times 8 = 16.7 \times 8 \approx 133 \text{ microsteps} \quad (3.67)$$

Inter-Seed Spacing Accuracy

Seed dispensing is triggered by the encoder-based distance accumulator (equation 3.14). For a programmed inter-seed spacing of $S_d = 250$ mm, the number of encoder counts at which the dispense trigger fires is:

$$N_{trigger} = S_d/d_{pulse} = 250/4.284 = 58.36 \text{ counts} \quad (3.68)$$

Since $N_{trigger}$ must be an integer, it is rounded to $N_{trigger} = 58$ counts. The resulting actual seed spacing and spacing error are:

$$S_{actual} = 58 \times 4.284 = 248.47 \text{ mm} \quad (3.69)$$

$$\epsilon_{spacing} = S_d - S_{actual} = 250.00 - 248.47 = 1.53 \text{ mm} \quad (< 1\%) \quad (3.70)$$

A spacing error of 1.53 mm is negligible for the crop types considered (maize, bean, and common vegetables typically require spacings of 200–300 mm), representing an error of 0.61% relative to the target. The IR slot sensor provides seed confirmation within the same control cycle, so missed seeds or double-drops are detected before the vehicle advances to the next trigger point.

Figure 3.3.13 — Insert diagram/plot hereFigure 3.3.13: Seed metering dynamics: (a) 12-cell plastic disk geometry with $R_{sw} = 40$ mm, 30° per cell; (b) triangular angular velocity profile for one 30° step in $T_{step} = 50$ ms; (c) encoder-based spacing trigger at $N_{trigger} = 58$ counts

3.3.6 Integrated State-Space Representation

The models derived in Sections 3.3.1–3.3.5 are assembled into a unified state-space representation of the complete system. This integrated model is used in Section 3.6.1 for MATLAB/Simulink simulation of the full planting cycle and for the PID controller design presented in Section 3.5.

State Vector and Input Vector

The complete state vector of the system is:

$$x = [x, y, \theta, v, \omega, h, n_s]^T \quad (3.71)$$

where x and y are the CoM position coordinates (m), θ is the heading angle (rad), v is the linear velocity (m/s), ω is the angular velocity (rad/s), h is the shank depth (m), and n_s is the cumulative seed count (integer). The control input vector is:

$$u = [\tau_R, \tau_L, \theta_{s_cmd}]^T \quad (3.72)$$

where τ_R and τ_L are the right and left motor torques (N·m), and θ_{s_cmd} is the servo shaft angle command (rad). Table 3.4 summarizes the state and input definitions for reference.

Table 3.4. State and input definitions for the integrated system model

State	Symbol	Unit	Source	Input	Symbol
Longitudinal pos.	x	m	Encoder odometry	Right wheel torque	τ_R
Lateral pos.	y	m	Encoder odometry	Left wheel torque	τ_L
Heading angle	θ	rad	Encoder / IMU	—	—
Linear velocity	v	m/s	PID velocity loop	—	—
Angular velocity	ω	rad/s	IMU / encoder diff	—	—
Shank depth	h	m	Servo position ctrl	—	—
Seeds dispensed	n_s	count	IR sensor	—	—

Nonlinear State Equations

The full nonlinear state equations, combining Equations (3.3)–(3.5) and (3.26)–(3.27) and (3.48), are:

$$\dot{x} = v \cos(\theta) \quad (3.73a)$$

$$\dot{y} = v \sin(\theta) \quad (3.73b)$$

$$\dot{\theta} = \omega \quad (3.73c)$$

$$\dot{v} = [(\tau_R + \tau_L)/R_w - F_{drag}(d, v)]/M_t \quad (3.73d)$$

$$\dot{\omega} = [(\tau_R - \tau_L) \cdot L/R_w]/M_r \quad (3.73e)$$

$$\dot{h} = L_{arm} \cdot \cos(\theta_s) \cdot \dot{\theta}_s \quad (3.73f)$$

$$\dot{n}_s = \delta(d_{enc} - k \cdot d_{pulse}) \quad (\text{Dirac at spacing}) \quad (3.73g)$$

Equation (3.73g) represents the discrete seed dispensing event: n_s is incremented by 1 each time the accumulated encoder count reaches the next multiple of $N_{trigger} = 58$ counts, triggering a stepper motor step sequence. The term $F_{drag}(d, v)$ in equation (3.73d) is the soil drag force from equation (3.42), which is zero during the turning phase when the shank is retracted ($h = 0$).

Linearization About Straight-Line Operating Point

For control design and stability analysis, the state equations are linearized about the nominal straight-line operating point $q_0 = [x_0, y_0, 0, v_0, 0, h_0, n_0]^T$ with $v_0 = 0.10$ m/s and $\theta_0 = 0$. Defining small-signal deviations $\delta x = x - x_0$, the linearized state equation is:

$$\delta \dot{x} = A \cdot \delta x + B \cdot \delta u \quad (3.80)$$

The non-zero elements of the 5×5 Jacobian A (for the first five states, excluding the discrete states h and n_s) are:

$$A_{13} = -v_0 = -0.10 \quad (\text{coupling of heading to x velocity}) \quad (3.81)$$

$$A_{14} = \cos(\theta_0) = 1.00 \quad (\text{direct v-to-}\dot{x}) \quad (3.82)$$

$$A_{25} = v_0 = 0.10 \quad (\text{coupling of } \omega \text{ to } \dot{y}) \quad (3.83)$$

$$A_{35} = 1.00 \quad (\text{angular velocity to heading}) \quad (3.84)$$

$$A_{44} = -B_{eff}/M_t \approx 0 \quad (\text{negligible damping}) \quad (3.85)$$

$$A_{55} = -B_{eff}/M_r = -2 \times 10^{-3}/0.09054 = -0.0221 \quad (3.86)$$

The input matrix B has non-zero elements:

$$B_{41} = 1/(M_t \cdot R_w) = 1/(4.300 \times 0.030) = 7.752 \quad (\tau_R\text{-to-}\dot{v}) \quad (3.87)$$

$$B_{42} = 1/(M_t \cdot R_w) = 7.752 \quad (\tau_L\text{-to-}\dot{v}) \quad (3.88)$$

$$B_{51} = L/(M_r \cdot R_w) = 0.200/(0.09054 \times 0.030) = 73.63 \quad (\tau_R\text{-to-}\dot{\omega}) \quad (3.89)$$

$$B_{52} = -L/(M_r \cdot R_w) = -73.63 \quad (\tau_L\text{-to-}\dot{\omega}) \quad (3.90)$$

The linearized model confirms the expected decoupling between the velocity and heading channels and provides the plant transfer functions required for PID controller design in Section 3.5. The full nonlinear model (equations 3.73a–g) is implemented in MATLAB/Simulink for the simulation study described in Section 3.6.1.

Figure 3.3.14 — Insert diagram/plot here
Figure 3.3.14: Integrated system block diagram: inputs $(\tau_R, \tau_L, \theta_{s.cmd}) \rightarrow$ three-channel plant (translation dynamics M_t , rotation dynamics M_r , servo depth model) $\rightarrow$ state outputs $(v, \omega, h) \rightarrow$ sensor feedback (encoders, IMU, IR) $\rightarrow$ supervisory state machine

3.4 System Analysis

Before proceeding to controller design, it is necessary to analyze the structural properties of the resulting state-space system. This section performs a formal system analysis addressing four interconnected topics: the structure of the linearized state-space model and its numerical matrices; controllability — whether the available actuators can drive the system to any desired state; observability — whether the available sensors provide sufficient information to reconstruct the full state vector; and stability — the location of system poles, the frequency-domain characteristics of each control channel, and the implications for closed-loop controller design. All numerical values are derived directly from the mechanical parameters of Section 3.2 and the dynamic models of Sections 3.3.1–3.3.5.

3.4.1 The Linearized State-Space Model

The complete extended state vector, including the servo angular states added in the amendment to Section 3.3.6, is:

$$\mathbf{x} = [x, y, \theta, v, \omega, \theta_s, \dot{\theta}_s]^T \quad (\text{continuous states}) \quad (3.A.1)$$

$$\mathbf{u} = [\tau_R, \tau_L, u_s]^T \quad (\text{control inputs}) \quad (3.A.2)$$

where x and y are the CoM position coordinates (m), θ is the heading angle (rad), v is the linear velocity of the CoM (m/s), ω is the angular velocity (rad/s), θ_s is the servo shaft angle (rad), and $\dot{\theta}_s$ is the servo angular velocity (rad/s). The discrete counting states (seed count n_s and metering disk angle θ_d) are excluded from the continuous analysis because they are driven by event logic rather than differential equations; they are handled in the hybrid supervisory model of Section 3.5.5. The control inputs are the right wheel motor torque τ_R (N·m), left wheel motor torque τ_L (N·m), and the normalized servo PWM command u_s (0–1, dimensionless).

The A Matrix — Linearized System Dynamics

The nominal straight-line operating point $\mathbf{q}_0 = [x_0, y_0, 0, 0.10, 0, \theta_{s0}, 0]^T$ is linearized at forward velocity $v_0 = 0.10$ m/s and zero heading angle $\theta_0 = 0$. Defining small-signal deviations $\delta\mathbf{x} = \mathbf{x} - \mathbf{x}_0$, the linearized dynamics are $\delta\dot{\mathbf{x}} = \mathbf{A} \cdot \delta\mathbf{x} + \mathbf{B} \cdot \delta\mathbf{u}$. The 7×7 state matrix $\mathbf{A}$, with all numerical values from Sections 3.2–3.3, is presented in Table 3.5.

Table 3.5. Linearized state matrix $\mathbf{A}$ (7x7)

	δx	δy	$\delta \theta$	δv	$\delta \omega$	$\delta \theta_s$	$\delta \dot{\theta}_s$
$\delta \dot{x}$	0	0	0	1.000	0	0	0
$\delta \dot{y}$	0	0	0.100	0	0	0	0
$\delta \dot{\theta}$	0	0	0	0	1.000	0	0
$\delta \dot{v}$	0	0	0	-4.65×10^{-4}	0	0	0
$\delta \dot{\omega}$	0	0	0	0	-0.0221	0	0
$\delta \dot{\theta}_s$	0	0	0	0	0	0	1.000
$\delta \ddot{\theta}_s$	0	0	0	0	0	0	-40.0

The structure of $\mathbf{A}$ reveals the decoupled sub-system architecture of the plant. The nonzero off-diagonal entries encode the physical coupling between states: $\mathbf{A}[1, 2] = v_0 = 0.100$ represents the coupling of the heading angle $\delta\theta$ to the lateral velocity $\delta\dot{y}$ (at $\theta_0 = 0$, forward motion at speed v_0 maps directly into lateral drift for small heading perturbations); $\mathbf{A}[0, 3] = 1.0$ represents the contribution of δv to $\delta\dot{x}$ (as expected from the kinematic equation $\dot{x} = v \cos \theta \approx v$ at $\theta_0 = 0$); $\mathbf{A}[2, 4] = 1.0$ reflects that the heading rate $\delta\dot{\theta}$ equals the angular velocity $\delta\omega$; and $\mathbf{A}[5, 6] = 1.0$ reflects that the servo position rate $\delta\dot{\theta}_s$ equals the servo angular velocity $\delta\dot{\theta}_s$ (a pure integrator relationship within the servo sub-system). The diagonal entries $\mathbf{A}[3, 3] = -B_{\text{eff}}/M_t = -4.65 \times 10^{-4}$ and $\mathbf{A}[4, 4] = -B_{\text{eff}}/M_r = -0.0221$ are the open-loop velocity damping terms, both small, confirming that the mechanical damping is negligible and motor torques dominate the dynamics. The entry $\mathbf{A}[6, 6] = -B_{\text{eff}}/J_{\text{eff}} = -40.0$ is the servo angular velocity damping, which is comparatively large because the servo J_{eff} is three orders of

magnitude smaller than M_t .

The B Matrix — Input Distribution

The input matrix B, which maps the three control inputs to the state derivatives, is presented in Table 3.6.

Table 3.6. Input distribution matrix B (7x3)

State	τ_R	τ_L	u_s	Physical meaning
$\delta\dot{x}$	0	0	0	Position states — not directly actuated
$\delta\dot{y}$	0	0	0	Position states — not directly actuated
$\delta\dot{\theta}$	0	0	0	Heading — driven by ω
$\delta\dot{v}$	7.752	7.752	0	Both motors sum to drive v
$\delta\dot{\omega}$	73.63	-73.63	0	Torque difference drives ω
$\delta\dot{\theta}_s$	0	0	0	Servo pos. — driven by $\dot{\theta}_s$
$\delta\ddot{\theta}_s$	0	0	29,400	u_s commands servo angular accel.

The numerical values are derived from the equations of motion. $B[3, 0] = B[3, 1] = 1/(M_t \cdot R_w) = 1/(4.300 \times 0.030) = 7.752$ shows that equal torque from either motor contributes symmetrically to linear acceleration. $B[4, 0] = -B[4, 1] = L/(M_r \cdot R_w) = 0.200/(0.09054 \times 0.030) = 73.63$ shows that equal and opposite torques on the two motors produce angular acceleration — the high magnitude (73.63 vs 7.752) reflects the mechanical advantage of the wide wheel track for turning. $B[6, 2] = K_m/J_{\text{eff}} = 1.47/5 \times 10^{-5} = 29,400$ is the servo angular acceleration gain, large due to the low inertia of the servo-arm assembly. The first three rows of B are identically zero, confirming that the position and heading states are not directly actuated — they are only reachable through the velocity and angular velocity channels.

3.4.2 Controllability Analysis

A linear time-invariant system (A, B) of dimension n is completely state controllable if and only if the controllability matrix $\mathcal{C}$ has full row rank:

$$\mathcal{C} = [B, AB, A^2B, A^3B, A^4B, A^5B, A^6B] \in \mathbb{R}^{7 \times 21} \quad (3.A.3)$$

$$(\mathcal{C}) = n = 7 \Leftrightarrow \text{system is completely controllable} \quad (3.A.4)$$

Complete controllability guarantees that for any initial state $\delta\mathbf{x}(0)$ and any desired final state $\delta\mathbf{x}_f$, there exists a finite-time input sequence $\mathbf{u}(t)$ that transfers the state from

$\delta \mathbf{x}(0)$ to $\delta \mathbf{x}_f$. In physical terms, it means the three actuators (two motor torques and the servo command) are collectively capable of driving all seven state variables to any desired values.

Sub-System Controllability

The block-diagonal structure of $\mathbf{A}$ allows the controllability of each sub-system to be assessed independently before verifying the full-system result.

Sub-system 1 — Translational velocity (state δv , input $\tau_{\text{sum}} = \tau_R + \tau_L$). The $\mathbf{A}$ sub-matrix is the scalar $A_{44} = -4.65 \times 10^{-4}$ and the effective $\mathbf{B}$ entry is $B_{\text{eff},v} = B[3, 0] + B[3, 1] = 15.504$. The 1×1 controllability matrix is simply:

$$\mathcal{C}_v = [B_{\text{eff},v}] = [15.504] \rightarrow (\mathcal{C}_v) = 1 \quad \checkmark$$

Sub-system 2 — Angular velocity (state $\delta \omega$, input $\tau_{\text{diff}} = \tau_R - \tau_L$). The effective $\mathbf{B}$ entry is $B[4, 0] - B[4, 1] = 147.26$. The 1×1 controllability matrix is:

$$\mathcal{C}_\omega = [147.26] \rightarrow (\mathcal{C}_\omega) = 1 \quad \checkmark$$

Sub-system 3 — Servo depth (states $[\delta \theta_s, \delta \dot{\theta}_s]$, input u_s). The 2×2 servo sub-matrices are:

$$\mathbf{A}_{\text{servo}} = \begin{bmatrix} 0 & 1 \\ 0 & -40 \end{bmatrix}, \quad \mathbf{B}_{\text{servo}} = \begin{bmatrix} 0 \\ 29,400 \end{bmatrix}$$

The 2×2 controllability matrix for the servo sub-system is:

$$\mathcal{C}_{\text{servo}} = [\mathbf{B}_{\text{servo}}, \mathbf{A}_{\text{servo}}\mathbf{B}_{\text{servo}}] = \begin{bmatrix} 0 & 29,400 \\ 29,400 & -1,176,000 \end{bmatrix}$$

The determinant is:

$$\det(\mathcal{C}_{\text{servo}}) = (0)(-1,176,000) - (29,400)(29,400) = -864,360,000 \neq 0 \quad (3.A.5)$$

Therefore $(\mathcal{C}_{\text{servo}}) = 2$, confirming that the servo sub-system is completely controllable. Physically, this means the servo PWM command u_s can independently drive both the servo shaft angle θ_s and its angular velocity $\dot{\theta}_s$ to any desired values — the ability to control both position and velocity is the direct consequence of the double integrator chain (angular acceleration $\rightarrow$ velocity $\rightarrow$ position).

Sub-system 4 — Position and heading (states $\delta x, \delta y, \delta \theta$). These states are kine-

matic integrators driven by the velocity states δv and $\delta \omega$. Because δv and $\delta \omega$ are themselves controllable, the position and heading states are indirectly controllable: any target (x, y, θ) can be reached by sequencing appropriate (v, ω) trajectories. This is the standard result for nonholonomic mobile robots — position controllability holds in the full nonlinear sense (using nonlinear controllability theory) even though the linearized position states appear decoupled from the inputs in the B matrix.

Full System Controllability

The full 7×7 controllability matrix $\mathcal{C} \in \mathbb{R}^{7 \times 21}$ was computed numerically using the sub-system parameters. The numerical rank was evaluated using singular value decomposition (SVD):

$$\sigma(\mathcal{C}) = [2.94 \times 10^7, 1.47 \times 10^5, 5.51 \times 10^3, 147.3, 15.50, 1.0, 4.65 \times 10^{-4}]$$

All seven singular values are nonzero (the smallest, 4.65×10^{-4} , corresponds to the near-zero velocity eigenvalue channel). Therefore:

$$(\mathcal{C}) = 7 = n \rightarrow \text{The system is completely controllable} \quad \checkmark \quad (3.6)$$

Table 3.7 summarizes the controllability results for each sub-system and for the full system.

Table 3.7. Controllability analysis summary

Sub-system	States	Input(s)	$(\mathcal{C})$	Result
Translational velocity	δv	$\tau_R + \tau_L$	1	Controllable
Angular velocity	$\delta \omega$	$\tau_R - \tau_L$	1	Controllable
Servo depth	$\delta \theta_s, \delta \dot{\theta}_s$	u_s	2	Controllable
Position (x, y, θ)	$\delta x, \delta y, \delta \theta$	via v and ω	3	(kinematic) Controllable (indirect)
Full 7-state system	All 7 states	τ_R, τ_L, u_s	7	Fully controllable

The complete controllability result has two important design implications. First, the hierarchical PID controller architecture of Section 3.5 — with separate loops for velocity, heading, and servo depth — is a valid design strategy because each loop addresses a controllable subsystem. Second, the high B-matrix gain ratios ($B[6, 2]/B[3, 0] \approx 3,800$) indicate that the servo channel is far more responsive per unit input than the motor torque channels, which informs the relative PID gain magnitudes: servo gains should be much smaller in magnitude than the motor controller gains to prevent actuator saturation.

Observability Analysis

A system (A, C) is completely state observable if and only if the observability matrix $\mathcal{O}$ has full column rank:

$$\mathcal{O} = [C^T, (CA)^T, (CA^2)^T, \dots, (CA^{n-1})^T]^T \in \mathbb{R}^{28 \times 7} \quad (3.A.7)$$

$$(\mathcal{O}) = n = 7 \Leftrightarrow \text{system is completely observable}$$

Complete observability guarantees that the initial state can be uniquely determined from a finite record of outputs and inputs — or equivalently, that a state observer (such as a Luenberger observer or Kalman filter) can reconstruct all seven states from the measured outputs.

Sensor Set and Output Matrix

The available sensors on the prototype are: (1) Hall-effect wheel encoders providing quadrature pulse counts from which v and ω are computed; (2) servo internal position feedback (from the servo's built-in potentiometer or digital encoder) providing θ_s ; and (3) encoder-integrated heading angle θ . There is no GPS, vision, or other absolute position sensor. The output equation is $y = C \cdot x$ with the 4×7 output matrix:

$$C = \begin{bmatrix} 0 & 0 & 0 & 1.0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0 & 0 \\ 0 & 0 & 1.0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where $y_1 = v$ (encoder average), $y_2 = \omega$ (encoder difference), $y_3 = \theta_s$ (servo feedback), $y_4 = \theta$ (encoder integration).

Observable Sub-Systems

Examining each sub-system individually:

Velocity and angular velocity channels: Both δv (y_1) and $\delta \omega$ (y_2) are directly measured. Since these are scalar states, the observability matrices are trivially full-rank (1×1 matrices equal to the respective C rows), confirming both states are observable.

Servo depth channel: The state θ_s is directly measured ($y_3 = \theta_s$). The servo angular velocity $\dot{\theta}_s$ is not directly measured but is the time derivative of the measured θ_s ,

recoverable by finite-difference on consecutive samples. Formally, the 2×2 observability matrix for the servo sub-system is:

$$\mathcal{O}_{\text{servo}} = \begin{bmatrix} \mathbf{C}_{\text{servo}} \\ \mathbf{C}_{\text{servo}} \mathbf{A}_{\text{servo}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det(\mathcal{O}_{\text{servo}}) = 1 \neq 0 \rightarrow = 2 \quad \checkmark \quad \text{Servo sub-system fully observable} \quad (3.A.9)$$

Heading channel: The heading angle θ is included as output y_4 (computed from the integrated differential encoder count). The heading state $\delta\theta$ is therefore directly observable, and since $A[2,4] = 1$, the angular velocity $\delta\omega$ is recoverable from the heading rate — consistent with y_2 providing a direct measurement.

Unobservable States: Position (x, y)

The position states δx and δy have no corresponding direct measurements (no row of $\mathbf{C}$ contains a 1 in the first two columns). The observability matrix of the full 7-state system was computed numerically and yielded:

$$(\mathcal{O}) = 5 < n = 7 \quad (3.A.10)$$

The rank deficiency of 2 corresponds exactly to the two position states δx and δy . This result is physically consistent: without a GPS receiver, vision system, or external landmark sensor, the absolute position of the robot in the global frame cannot be reconstructed from encoder, servo, and heading measurements alone. The position states evolve as kinematic integrals of the measured velocity and heading:

$$x(t) = x(0) + \int_0^t v(\tau) \cdot \cos(\theta(\tau)) d\tau \quad (3.A.11)$$

$$y(t) = y(0) + \int_0^t v(\tau) \cdot \sin(\theta(\tau)) d\tau \quad (3.A.12)$$

These integrals require the initial conditions $x(0)$ and $y(0)$, which are unknown without an absolute reference. The practical consequence is that the robot's position estimate drifts over time due to wheel slip and encoder quantization noise — a phenomenon known as odometric drift, which is standard for encoder-only mobile robots. For the planting application, this drift is tolerable because the robot operates on short, straight rows (typically 1–5 meters) where the cumulative drift is small, and the heading is reset to zero at the start of each row by the supervisory controller. Absolute position accuracy is not required for seed spacing uniformity, which depends on relative (incremental) distance only.

To mitigate the observability limitation, two options are available for future upgrades: (1) adding a low-cost IMU to provide acceleration measurements from which velocity can be independently estimated and drift corrected, or (2) installing row-end mechanical or optical boundary sensors that reset the position estimate at known waypoints. Either option would increase the effective observable rank to 7. For the prototype described in this thesis, the partial observability is accepted as a known limitation and documented here.

Table 3.8. Observability analysis summary

Sub-system	States	Sensor	($\mathcal{O}$)	Result
Translational velocity	δv	Encoder avg.	1	Observable
Angular velocity	$\delta \omega$	Encoder diff.	1	Observable
Heading	$\delta \theta$	Encoder integration	1	Observable
Servo position + vel.	$\delta \theta_s, \delta \dot{\theta}_s$	Servo feedback	2	Observable
Position x, y	$\delta x, \delta y$	None (no GPS/vision)	0 direct	Not directly observable
Full 7-state (rank)	All states	Combined sensor set	5	rank = 5 (x, y unobservable)

3.4.3 Eigenvalue Analysis and Stability

The stability of the open-loop system is determined by the locations of the eigenvalues of $\mathbf{A}$ in the complex s -plane. The eigenvalues are the roots of the characteristic polynomial $\det(s\mathbf{I} - \mathbf{A}) = 0$. Due to the block-diagonal structure of $\mathbf{A}$, the characteristic polynomial factors into independent sub-system polynomials, and the eigenvalues are:

$$\det(s\mathbf{I} - \mathbf{A}) = s^3 \cdot (s + 4.65 \times 10^{-4}) \cdot (s + 0.0221) \cdot s \cdot (s + 40) = 0 \quad (3.13)$$

Expanding, the seven eigenvalues are:

$$\begin{aligned} \lambda_1 = \lambda_2 = \lambda_3 &= 0 && \text{(kinematic position and heading integrators)} \\ \lambda_4 &= -4.65 \times 10^{-4} \text{ rad/s} && \text{(translational velocity damping)} \\ \lambda_5 &= -0.0221 \text{ rad/s} && \text{(angular velocity damping)} \\ \lambda_6 &= 0 \text{ rad/s} && \text{(servo position integrator)} \\ \lambda_7 &= -40.0 \text{ rad/s} && \text{(servo angular velocity damping)} \end{aligned}$$

Physical Interpretation by Sub-System

Translational velocity eigenvalue ($\lambda_4 = -4.65 \times 10^{-4} \text{ rad/s}$): This eigenvalue corresponds to the open-loop decay of linear velocity in the absence of motor torque, arising from the mechanical damping term B_{eff} . The associated time constant is $\tau_v =$

Table 3.9. Open-loop eigenvalue analysis

λ	Value (rad/s)	Sub-system	Stability	Time constant	Interpretation
λ_1	0	x — position integrator	Marginally stable	∞	Requires PID to give finite error
λ_2	0	y — position integrator	Marginally stable	∞	Requires PID to give finite error
λ_3	0	θ — heading integrator	Marginally stable	∞	Requires PID to give finite error
λ_4	-4.65×10^{-4}	v — translational velocity	Stable (slow)	2,150 s	Motor torque is dominant driver
λ_5	-0.0221	ω — angular velocity	Stable (slow)	45.3 s	Active heading control essential
λ_6	0	θ_s — servo position	Marginally stable	∞	Servo holds depth without drift
λ_7	-40.0	$\dot{\theta}_s$ — servo velocity	Stable (fast)	25 ms	Fast servo rate settling

$1/4.65 \times 10^{-4} = 2,150$ s — meaning the vehicle would take over 35 minutes to coast to a stop from $v = 0.10$ m/s under pure mechanical friction. Practically, this confirms what the motor torque sizing analysis of Section 3.3.2 already established: motor torque is the completely dominant term in the velocity dynamics, and mechanical damping is negligible. For PID design, the plant $G_v(s)$ is treated as a pure integrator.

Angular velocity eigenvalue ($\lambda_5 = -0.0221$ rad/s): The angular velocity time constant is $\tau_\omega = 45.3$ s — much slower than the velocity time constant in absolute terms, but still negligibly slow compared to the desired heading response (target settling time < 1 s). As with the velocity channel, active control is essential; the open-loop angular dynamics would make the vehicle extremely sluggish to turn corrections.

Servo position and velocity eigenvalues ($\lambda_6 = 0$, $\lambda_7 = -40$ rad/s): The zero eigenvalue λ_6 is the servo position integrator: the servo shaft angle holds any commanded position indefinitely in the absence of a disturbing torque, which is exactly the desired behavior for furrow depth maintenance during a continuous-furrow pass. The eigenvalue $\lambda_7 = -40$ rad/s is stable and fast ($\tau = 25$ ms), meaning the servo angular velocity decays rapidly to zero — the servo arm reaches its commanded angle quickly and without sustained oscillation. Together, these two eigenvalues produce the Type-1 second-order behavior of $G_{\text{servo}}(s)$ analyzed in Section 3.3.7.5.

Kinematic integrators ($\lambda_1 = \lambda_2 = \lambda_3 = 0$): The three zero eigenvalues at the origin correspond to the position (x, y) and heading (θ) states, all of which are pure integrals of the velocity states. Marginal stability at the origin is the expected and correct result for a mobile robot: the robot holds its position when velocity is zero (it does not spontaneously move), but any non-zero velocity will cause the position to grow without bound — the normal operational behavior. Under closed-loop control, the PID controllers shift the effective closed-loop poles associated with these states into the strict left half-plane (LHP), making the errors in position and heading asymptotically stable.

Stability Conclusion

Since all eigenvalues of A lie on the imaginary axis or in the closed LHP (no right-half-plane eigenvalues), the open-loop system is marginally stable. Marginal stability is the standard result for mobile robot kinematic chains and for integrating servo systems. The system is not unstable in the Lyapunov sense, but it will not converge to a desired operating point without active feedback control. The PID controllers of Section 3.5 are designed to add sufficient closed-loop damping and integral action to achieve: (i) zero steady-state velocity error, (ii) bounded heading drift, and (iii) depth errors within the agronomic tolerance of ± 5 mm.

3.4.4 Transfer Function and Frequency-Domain Analysis

The transfer function representation of each control channel provides the frequency-domain perspective needed for PID gain selection and stability margin verification. The three control channels (translational velocity, angular velocity, and servo depth) are analyzed individually.

Velocity Channel Transfer Function

The transfer function from the sum of motor torques $\tau_{\text{sum}} = \tau_R + \tau_L$ to the linear velocity v is obtained from the scalar velocity sub-system:

$$G_v(s) = \frac{V(s)}{\tau_{\text{sum}}(s)} = \frac{1/(M_t \cdot R_w)}{s - \lambda_4} \quad (3.A.14)$$

$$G_v(s) = \frac{7.752}{s + 4.65 \times 10^{-4}} \quad [\text{m} \cdot \text{s}^{-1} / \text{N} \cdot \text{m}] \quad (3.A.15)$$

Key frequency-domain properties:

- Single real pole at $s_p = -4.65 \times 10^{-4}$ rad/s (0.074 mHz) — essentially at the origin on any practical frequency scale.
- DC gain: $G_v(0) = 7.752/4.65 \times 10^{-4} = 16,667 \text{ m} \cdot \text{s}^{-1} / (\text{N} \cdot \text{m})$ — large because the pole is very close to zero.
- Gain crossover at $\omega_{gc} \approx 4.65 \times 10^{-4}$ rad/s — far below any useful control bandwidth.
- Bode magnitude: -20 dB/decade slope from near DC; phase $\approx -90^\circ$ throughout the PID bandwidth.

Since the pole is so close to the origin, $G_v(s)$ is approximated as a pure integrator over the entire PID design frequency range (0.1–20 rad/s):

$$G_v(s) \approx \frac{7.752}{s} \quad (\text{integrator approximation for PID design}) \quad (3.A.16)$$

This means the velocity PID controller must contain an integral term to provide zero steady-state velocity error, and the proportional term sets the loop bandwidth. The near-integrator plant also means that derivative action in the PID must be used carefully to avoid amplifying high-frequency encoder noise.

Angular Velocity Channel Transfer Function

The transfer function from the differential motor torque $\tau_{\text{diff}} = \tau_R - \tau_L$ to the angular velocity ω is:

$$G_\omega(s) = \frac{\Omega(s)}{\tau_{\text{diff}}(s)} = \frac{L/(M_r \cdot R_w)}{s - \lambda_5} \quad (3.A.17)$$

$$G_\omega(s) = \frac{73.63}{s + 0.0221} \quad [\text{rad} \cdot \text{s}^{-1}/\text{N} \cdot \text{m}] \quad (3.A.18)$$

Key properties:

- Pole at $s_p = -0.0221$ rad/s (time constant 45.3 s) — again a near-integrator at PID design frequencies.
- DC gain: $G_\omega(0) = 73.63/0.0221 = 3,331 \text{ rad} \cdot \text{s}^{-1}/(\text{N} \cdot \text{m})$.
- The gain is approximately $9.5\times$ larger than G_v at the same frequency, reflecting the mechanical advantage of the wide wheel track for turning.

The integrator approximation for heading control PID design is $G_\omega(s) \approx 73.63/s$.

Servo Depth Channel Transfer Function

The transfer function from the normalized servo command u_s to the servo shaft angle θ_s is:

$$G_{\text{servo}}(s) = \frac{\Theta_s(s)}{U_s(s)} = \frac{K_m/J_{\text{eff}}}{s(s - \lambda_7)} \quad (3.A.19)$$

$$G_{\text{servo}}(s) = \frac{29,400}{s(s + 40)} \quad [\text{rad}/\text{normalized PWM}] \quad (3.A.20)$$

This is a Type-1 second-order system (one integrator at $s = 0$) with a stable real pole at $s_{p2} = -40$ rad/s. The key frequency-domain properties are:

- Two poles: $s_{p1} = 0$ (integrator) and $s_{p2} = -40$ rad/s (stable at 6.37 Hz).
- No finite zeros — the zero is at $s = \infty$.
- Bode magnitude: -20 dB/decade from DC to 40 rad/s (integrator-dominated region), then -40 dB/decade above 40 rad/s.
- Bode phase: starts at -90° at low frequencies (due to integrator), drops asymptotically toward -180° as the pole at 40 rad/s contributes its -90° of additional phase lag.

Phase and Gain Margin of the Servo Channel

The phase margin of the servo channel under PD control is evaluated at the designed closed-loop bandwidth $\omega_n = 31.4$ rad/s. The open-loop transfer function with PD control is:

$$L(s) = (K_p + K_d s) \cdot G_{\text{servo}}(s) = (0.03357 + 0.00015s) \cdot G_{\text{servo}}(s)$$

The open-loop phase at $\omega = \omega_n = 31.4$ rad/s is:

$$\angle G_{\text{servo}}(j\omega_n) = -90^\circ - \arctan(\omega_n/40) \quad (3.A.22)$$

$$\begin{aligned} &= -90^\circ - \arctan(31.4/40) \\ &= -90^\circ - 38.15^\circ = -128.15^\circ \end{aligned} \quad (3.A.23)$$

$$\text{Phase Margin} = 180^\circ - 128.15^\circ = 51.85^\circ \approx 52^\circ \quad (3.A.24)$$

A phase margin of approximately 52° is consistent with the chosen damping ratio $\zeta = 0.707$. The rule of thumb $\text{PM} \approx 100\zeta$ (valid for second-order systems) gives $\text{PM} \approx 70.7^\circ$; the slightly lower computed value of 52° reflects the additional phase lag introduced by the real pole at -40 rad/s operating near the bandwidth. This margin is acceptable for a position-controlled servo in a low-vibration agricultural application — values above 45° are generally considered adequate for robust stability.

The gain margin for G_{servo} is theoretically infinite because the open-loop phase of a Type-1 second-order system with real poles approaches -180° only asymptotically and never crosses -180° at any finite frequency. This means the servo loop will not go unstable under proportional gain increases alone — a useful property for prototype commissioning where gains are tuned empirically.

Table 3.10. Transfer function and frequency-domain summary for the three control channels

Channel Type	$G(s)$	Poles	DC Gain	Time const.
Velocity	$\frac{7.752}{s+4.65 \times 10^{-4}}$	$s_p = -4.65 \times 10^{-4} \text{ rad/s}$	16,667	2,150 s
Angular vel.	$\frac{73.63}{s+0.0221}$	$s_p = -0.0221 \text{ rad/s}$	3,333	45.3 s
Servo depth	$\frac{29,400}{s(s+40)}$	$s_{p1} = 0, s_{p2} = -40 \text{ rad/s}$	∞	25 ms (vel.)

3.4.5 Summary and Implications for Controller Design

The system analysis of Sections 3.3.7.1–3.3.7.5 establishes four key results that directly inform the control design of Section 3.5:

1. **Complete controllability** ($((\mathcal{C}) = 7)$): All seven continuous states can be driven to any desired value using the three available actuators. The hierarchical PID architecture — independent loops for velocity, heading, and servo depth — is therefore a valid and sufficient control structure for this system.
2. **Partial observability** ($((\mathcal{O}) = 5)$): The absolute position states x and y are not reconstructable from the available sensors alone. This imposes a known navigation limitation — the robot operates correctly for seed spacing (which depends on relative distance, not absolute position) but cannot localize itself globally. Mitigation strategies (boundary sensors or IMU integration) are deferred to future work.
3. **Marginal open-loop stability**: All poles lie on or to the left of the imaginary axis. No subsystem is open-loop unstable, which means the prototype can be safely commissioned without active control for brief tests, and controller gains can be tuned incrementally without risk of instantaneous instability.
4. **Near-integrator velocity/angular velocity channels**: Both $G_v(s)$ and $G_\omega(s)$ behave as pure integrators over the PID design bandwidth. This simplifies gain selection (integrating plant $\approx$ known structure) and implies that even a simple P controller would give near-zero steady-state velocity error in practice — the I-term in the PID primarily handles load disturbances such as the varying soil drag force F_{drag} during the continuous-furrow pass.

Together, these results validate the modeling approach of Sections 3.3.1–3.3.6 and provide the analytical justification for the control architecture presented in Section 3.5.

3.5 Control System Design

The control system is structured as a three-level hierarchy: a supervisory state machine at the top level coordinates the overall planting cycle; two real-time feedback controllers at the middle level regulate wheel velocity and shank depth; and an open-loop distance-triggered step controller at the lowest level manages seed metering. This architecture separates concerns cleanly and allows each controller to be tuned and simulated independently before integration.

3.5.1 Hierarchical Control Architecture

Figure 3.18 shows the three-level hierarchy. The supervisory state machine (Section 3.5.5) issues mode commands and setpoints to the middle-level controllers: a commanded linear velocity v_{ref} to the PID velocity controller (Section 3.5.2), a commanded depth h_d to the shank position controller (Section 3.5.3), and a seed-dispense trigger to the seed step controller (Section 3.5.4). The middle-level controllers output actuator commands (PWM duty cycles and step pulses) directly to the hardware drivers. Sensor feedback from encoders, IMU, and IR sensor flows upward to both the middle-level controllers and the supervisory state machine.

Figure 3.3.18 — Insert diagram here
Figure 3.3.18: Three-level hierarchical control architecture: supervisory state machine (top) → velocity PID and shank PD controllers (middle) → BTS7960 / DRV8825 / servo drivers (bottom); sensor feedback paths from encoders, IMU, and IR sensor

3.5.2 LQR Path Tracking and Velocity Controller

Path tracking and forward velocity regulation are treated as a unified MIMO control problem using a Linear Quadratic Regulator (LQR). The two preceding sections established that the velocity and heading channels are decoupled in the B matrix but coupled through the kinematic state equations — a one-dimensional PID controller cannot simultaneously correct lateral position error, heading error, and velocity error without cross-channel interference. The LQR resolves this by computing a single optimal gain matrix K that weighs all five mobile-base states against the cost of motor torque, producing a control law that inherently handles the MIMO coupling.

System Model for LQR Design

The LQR is designed on the five-state mobile-base sub-system extracted from the full linearized model of Section 3.4.1. The servo and seed-metering states are excluded because they are governed by their own PID controllers operating on independent channels. The state and input vectors are:

$$x_{mb} = [\delta x, \delta y, \delta \theta, \delta v, \delta \omega]^T, \quad u = [\tau_R, \tau_L]^T \quad (3.5.2.1)$$

The linearized state equation $\delta \dot{x} = A \cdot \delta x + B \cdot \delta u$ uses the matrices evaluated at the nominal straight-line operating point $v_0 = 0.10$ m/s, $\theta_0 = 0$:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1.000 & 0 \\ 0 & 0 & 0.100 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.000 \\ 0 & 0 & 0 & -0.0005 & 0 \\ 0 & 0 & 0 & 0 & -0.0221 \end{bmatrix} \quad (3.5.2.2)$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 7.7519 & 7.7519 \\ 73.632 & -73.632 \end{bmatrix} \quad (3.5.2.3)$$

The controllability of this sub-system was verified in Section 3.4.2 (rank = 5 ✓), confirming that the LQR gain matrix exists and can place all five closed-loop poles arbitrarily.

LQR Cost Function and Q/R Matrix Selection

The LQR minimizes the infinite-horizon quadratic cost functional:

$$J = \int_0^\infty [\delta x^T Q \delta x + u^T R u] dt \quad (3.5.2.4)$$

where $Q \in \mathbb{R}^{5 \times 5}$ is a positive semi-definite state weighting matrix and $R \in \mathbb{R}^{2 \times 2}$ is a positive definite control effort weighting matrix. The Q and R matrices are diagonal, with weights assigned according to the relative importance of each state error and the actuator cost. Table 3.17 presents the selected weights with their physical justification.

Table 3.11. LQR weighting matrix Q and R

State	Symbol	Q weight	Q/R ratio	Justification
x-position	δx	100	100:1	Path accuracy critical for furrow alignment
y-position (lateral)	δy	100	100:1	Lateral deviation directly affects row spacing
Heading angle	$\delta \theta$	50	50:1	Heading error causes growing lateral drift
Linear velocity	δv	10	10:1	Speed stability needed for consistent seed spacing
Angular velocity	$\delta \omega$	1	1:1	Transient ω deviations are acceptable
Motor torque	τ_R, τ_L	—	$R = 1.0$	Prevents actuator saturation; motors rated 40 N·cm

The resulting diagonal weighting matrices are:

$$Q = \text{diag}(100, 100, 50, 10, 1), \quad R = \text{diag}(1.0, 1.0) \quad (3.5.2.5)$$

The position states ($\delta x, \delta y$) are weighted most heavily at 100, reflecting that accurate path following is the primary objective. The heading weight of 50 reflects that uncorrected heading errors cause growing lateral drift over the row length. The velocity weight of 10 penalizes speed deviations that would affect seed spacing consistency. The angular velocity weight of 1 is kept low because transient heading rate deviations are acceptable provided they decay quickly. The equal torque cost $R = I$ ensures neither motor is preferentially saturated.

Optimal Gain Matrix

The optimal gain matrix K is obtained by solving the continuous-time Algebraic Riccati Equation (ARE):

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (3.5.2.6)$$

and computing:

$$K = R^{-1} B^T P \quad (3.5.2.7)$$

Substituting the system matrices A and B from Equations (3.5.2.2) and (3.5.2.3) and the weighting matrices from Equation (3.5.2.5), the ARE solution yields the 2×5 gain matrix presented in Table 3.18.

Table 3.12. Optimal LQR gain matrix K (2×5)

Motor	k_x	k_y	k_θ	k_v	k_ω
K_R (right)	7.0711	7.0711	5.1056	2.4315	0.7544
K_L (left)	7.0711	-7.0711	-5.1056	2.4315	-0.7544

The optimal control law is:

$$u = -K(x_{\text{mb}} - x_{\text{ref}}) \quad \text{i.e.} \quad [\tau_R, \tau_L]^T = -K \cdot \delta x \quad (3.5.2.8)$$

$$\tau_R = -7.0711 \cdot \delta x - 7.0711 \cdot \delta y - 5.1056 \cdot \delta \theta - 2.4315 \cdot \delta v - 0.7544 \cdot \delta \omega \quad (3.5.2.9)$$

$$\tau_L = -7.0711 \cdot \delta x + 7.0711 \cdot \delta y + 5.1056 \cdot \delta \theta - 2.4315 \cdot \delta v + 0.7544 \cdot \delta \omega \quad (3.5.2.10)$$

The structure of K has a clear physical interpretation. The gains $K_R[x] = K_L[x] = 7.0711$ are equal, meaning an x-position error drives both motors equally in the same direction — a pure forward/backward correction. The gains $K_R[y] = -K_L[y] = 7.0711$ are equal and opposite, meaning a lateral (y) error drives the motors differentially — a corrective turn. The gains $K_R[v] = K_L[v] = 2.4315$ are equal, contributing to velocity correction through the sum channel. The gain symmetry $K_R[y] = -K_L[y]$ and $K_R[\theta] = -K_L[\theta]$ is a consequence of the symmetric differential-drive geometry and the equal R diagonal entries, and can be verified analytically from the B -matrix structure.

Closed-Loop Stability Verification

The closed-loop system matrix is $A_{\text{cl}} = A - B \cdot K$. The five eigenvalues of A_{cl} are presented in Table 3.19.

Table 3.13. Closed-loop eigenvalues

λ	Value (rad/s)	Stability	Interpretation
λ_1	-103.89	Stable (fast)	Servo-rate equivalent; fast actuator mode
λ_2	-34.52	Stable	Angular velocity channel settling
λ_3	-7.09	Stable	Heading error correction
λ_4	-3.18	Stable	Velocity regulation
λ_5	-0.14	Stable (slow)	Position error convergence; $\tau = 7.07$ s

All five eigenvalues are strictly negative real, confirming that the LQR closes all five open-loop poles (including the three marginally stable integrators $\lambda_{1,2,3} = 0$ identified in Section 3.4.4) into the strict left half-plane. The dominant pole $\lambda_5 = -0.1414$ rad/s has a time constant of $\tau = 1/0.1414 = 7.07$ s and a corresponding settling time of:

$$t_{\text{settle}} = 4\tau = 4/0.1414 = 28.3 \text{ s} \quad (3.5.2.11)$$

At the nominal forward speed $v_0 = 0.10$ m/s, the robot traverses $0.10 \times 28.3 = 2.83$

m during one settling time. This means that a lateral offset or heading and lateral tracking errors arising during both straight-row traversal and turning manoeuvres are corrected within the closed-loop settling distance, well within typical urban plot row lengths of 3–5 m. The remaining four poles ($\lambda_1 = -103.9$ through $\lambda_4 = -3.18$) are all faster and do not limit the tracking response.

Reference Tracking and Disturbance Rejection

The reference state for path tracking is the straight-line trajectory:

$$x_{\text{ref}}(t) = [v_0 \cdot t, 0, 0, v_0, 0]^T = [0.10 \cdot t, 0, 0, 0.10, 0]^T \quad (3.5.2.12)$$

The error state $\delta x = x - x_{\text{ref}}$ thus captures deviations from the ideal straight-line path in all five dimensions simultaneously. When the robot is on-path and at speed, $\delta x = 0$ and the LQR outputs zero torque command ($u = 0$) — the motors coast. Any disturbance — a soil drag spike when the furrow opener engages, a heading deviation from terrain irregularity, or a lateral drift — produces a nonzero δx that the LQR immediately corrects through the optimal gain structure.

The soil drag disturbance $F_{\text{drag}} = 10.0$ N enters the translational dynamics channel as an additive input (as modeled in Section 3.3.3) and manifests as a velocity error δv . The LQR corrects this through the k_v gain entries: both motors increase torque by $K[\cdot, 3] \cdot \delta v = 2.4315 \cdot \delta v$ simultaneously, restoring velocity to v_0 . Unlike the previous separate PI velocity controller, this correction is computed jointly with the heading and position corrections, preventing the velocity correction from inadvertently introducing a heading torque imbalance.

Integration with the Control Hierarchy

The LQR replaces the separate velocity PID and heading PID controllers from the previous architecture. It runs on the ESP32 at the same 1,000 Hz control tick as the shank PD controller. At each tick, the measured state vector x_{mb} is assembled from the encoder-derived v, ω , and the odometry-integrated x, y, θ (Equations 3.16–3.18). The matrix multiplication $K \cdot \delta x$ requires 10 multiply-accumulate operations — trivially fast at 240 MHz. The resulting τ_R and τ_L are converted to PWM duty cycles for the BTS7960 drivers by dividing by the motor torque constant k_t . The shank depth PD controller (Section 3.5.3), seed step controller (Section 3.5.4), and supervisory state machine (Section 3.5.5) remain unchanged.

3.5.3 Shank Depth Position Controller

The shank position controller converts a commanded depth h_d into a servo shaft angle command θ_{s_cmd} and regulates that angle against the soil disturbance torque τ_{soil} . Because the servo has its own internal position control loop, the outer controller implemented on the ESP32 operates as a supervisory PD loop that issues updated angle commands when the measured (or estimated) depth deviates from the setpoint.

Inverse Kinematics

The commanded servo angle for a target depth h_d is (from Equation 3.49):

$$\theta_{s_cmd} = \arcsin(h_d/L_{arm}) = \arcsin(h_d/0.080) \quad (3.111)$$

Three operating depth setpoints and their corresponding servo angles are:

$$h_d = 30 \text{ mm} \rightarrow \theta_{s_cmd} = \arcsin(0.375) = 22.0^\circ \quad (3.112)$$

$$h_d = 40 \text{ mm} \rightarrow \theta_{s_cmd} = \arcsin(0.500) = 30.0^\circ \quad (3.113)$$

$$h_d = 50 \text{ mm} \rightarrow \theta_{s_cmd} = \arcsin(0.625) = 38.7^\circ \quad (3.114)$$

PD Controller Design

The plant is the servo shaft angle response to a PWM position command, modeled as a second-order system (Equation 3.53) with parameters $J_{eff} = 5.0 \times 10^{-5} \text{ kg} \cdot \text{m}^2$, $B_{eff} = 2.0 \times 10^{-3} \text{ N} \cdot \text{m} \cdot \text{s/rad}$, $K_m = 1.47 \text{ N} \cdot \text{m}$. Targeting a natural frequency $\omega_n = 31.4 \text{ rad/s}$ (5 Hz) and damping ratio $\zeta = 0.707$ (critically underdamped):

$$Kp_s = \omega_n^2 \cdot J_{eff}/K_m = (31.4)^2 \times 5.0 \times 10^{-5}/1.47 = 0.03357 \quad (3.115)$$

$$Kd_s = (2\zeta\omega_n \cdot J_{eff} - B_{eff})/K_m = (2 \times 0.707 \times 31.4 \times 5 \times 10^{-5} - 2 \times 10^{-3})/1.47 = 1.50 \times 10^{-4} \quad (3.116)$$

Verification of the closed-loop characteristic polynomial $s^2 + (B_{eff} + Kd_s \cdot K_m)/J_{eff} \cdot s + Kp_s \cdot K_m/J_{eff} = 0$:

$$\omega_{n_CL} = \sqrt{(Kp_s \cdot K_m/J_{eff})} = \sqrt{(0.03357 \times 1.47/5 \times 10^{-5})} \quad (3.1)$$

$$\omega_{n_CL} = \sqrt{(986.9)} = 31.4 \text{ rad/s} \checkmark \quad (3.117)$$

$$\zeta_{CL} = (B + K_d \cdot K_m) / (2 \cdot \omega_n \cdot J) = (2 \times 10^{-3} + 1.50 \times 10^{-4} \times 1.47) / (2 \times 31.4 \times 5 \times 10^{-5}) = 0.707 \checkmark \quad (3.118)$$

$$t_{settle} = 4.6 / (\zeta \cdot \omega_n) = 4.6 / (0.707 \times 31.4) = 207 \text{ ms} \quad (3.119)$$

The servo reaches the target depth 207 ms after the engage command — during which the vehicle travels $0.10 \times 0.207 = 20.7 \text{ mm}$ — ensuring the shank is fully engaged well before the first seed dispense event at 250 mm into the row.

Depth Error Under Soil Load

The steady-state depth error under the nominal soil disturbance torque $\tau_{\text{soil}} = 0.547 \text{ N}\cdot\text{m}$ (from Equation 3.47) is (from Section 3.3.4):

$$\Delta h_{ss} = L_{\text{arm}} \cdot (\tau_{\text{soil}} / K_{\text{stiff}}) = 0.080 \times (0.547 / 28.1) \quad (3.2)$$

$$\Delta h_{ss} = 1.56 \text{ mm} < \pm 5 \text{ mm tolerance } \checkmark \quad (3.120)$$

3.5.4 Seed Step Controller

Seed dispensing is executed by an open-loop distance-triggered step controller rather than a closed-loop feedback controller, because the seed spacing requirement (250 mm $\pm$ a few mm) can be met by encoder odometry alone without a dedicated inter-seed distance sensor. The controller operates as follows:

- An encoder accumulator d_{acc} tracks the cumulative distance traveled since the last seed event using equation (3.14): $d_{\text{acc}} += \Delta N \times d_{\text{pulse}}$.
- When $d_{\text{acc}} \geq N_{\text{trigger}} \times d_{\text{pulse}} = 58 \times 4.284 \text{ mm} = 248.47 \text{ mm}$, a dispense event is triggered: the ESP32 outputs 133 microstep pulses to the DRV8825 at a rate corresponding to a 50 ms rotation window, and d_{acc} is reset to zero.
- The IR slot sensor confirms seed passage within the same 50 ms window. If no confirmation is received, the supervisory state machine logs a missed-seed fault and optionally triggers a re-dispense attempt after a configurable timeout.

The seed spacing error from integer-count rounding is:

$$S_{\text{actual}} = N_{\text{trigger}} \times d_{\text{pulse}} = 58 \times 4.284 = 248.47 \text{ mm} \quad (3.121)$$

$$\epsilon_{spacing} = 250.00 - 248.47 = 1.53 \text{ mm (0.61\% of target)} \checkmark \quad (3.122)$$

This error is consistent across all seeds in a row (it is a systematic, not random, offset) and can be eliminated by using $4\times$ quadrature decoding with a slightly larger wheel diameter or adjusting the trigger count to $N_{trigger} = 58$ or 59 on alternating events to average to 250 mm.

Insert diagram hereFigure 3.3.20: Seed step controller flowchart: encoder ISR increments d_{acc} ; comparator checks $d_{acc} \geq S_d$; dispense trigger sends 133 microsteps to DRV8825; IR sensor confirms seed; d_{acc} reset; fault flag if no IR confirmation within 50 ms

Supervisory State Machine

The supervisory state machine (SSM) is the top-level coordinator of the planting cycle. It runs in the ESP32 main loop at 100 Hz, reads all sensor inputs, and issues mode transitions and setpoint updates to the lower-level controllers. The SSM has eight distinct states, described in Table 3.7 and illustrated in Figure 3.21.

Table 3.14. Supervisory state machine

State	Actuators Active	Entry Condition	Exit Condition	Next State
IDLE	None	Power-on / reset	Start command received	ENGAGE_SHANK
ENGAGE_SHANK	Servo $\downarrow$ to θ_{cmd}	Start of each row	$t \geq t_{settle} = 207 \text{ ms}$	PLANTING
PLANTING	Motors fwd, servo hold, stepper on trigger	After servo settled	Encoder $\geq N_{end_row}$ counts	END_ROW
END_ROW	Motors stop	Encoder $\geq$ row length	Always	RETRACT_SHANK
RETRACT_SHANK	Servo $\uparrow$ to $\theta = 0^\circ$	After motors stopped	$t \geq t_{settle} = 207 \text{ ms}$	TURNING
TURNING	$v_R = +0.10$, $v_L = -0.10 \text{ m/s}$	Shank fully retracted	IMU $\Delta\theta \geq 179^\circ$	ADVANCE
ADVANCE	Motors fwd, slow speed	After 180° turn	Encoder $\geq N_{row_spacing}$	ENGAGE_SHANK or DONE
DONE	All off	All rows complete	—	IDLE

The PLANTING state is the primary operating state during which all three sub-functions — continuous furrow cutting, seed metering, and covering — occur simultaneously. The shank drags continuously through the soil for the full length of the row, the seed step controller fires at each 248.47 mm interval, and the passive covering disks trail behind closing the furrow. The PLANTING state is only exited when the encoder cumulative count reaches the row-length threshold N_{end_row} , computed from the pre-programmed row length L_{row} as:

$$N_{end_row} = L_{row}/d_{pulse} = L_{row}/(4.284 \times 10^{-3}) \quad (3.123)$$

Shank Retraction Logic

A critical safety interlock in the SSM prevents the TURNING state from being entered unless the shank retraction is confirmed. The sequence is enforced as follows: upon entering END_ROW, the velocity controller is commanded to $v_{ref} = 0$ (motors stop); only after the vehicle velocity drops below a threshold of 0.005 m/s (confirmed by encoder rate) is the RETRACT_SHANK state entered. The servo is commanded to $\theta_s = 0^\circ$ and the SSM waits for at least $t_{settle} = 207$ ms before transitioning to TURNING. This ensures the shank is clear of the soil before any lateral wheel motion begins, preventing shank overload and soil disturbance outside the planting rows.

180-Degree Turn Execution

During TURNING, the right wheel velocity is set to $+v_t = +0.10$ m/s and the left wheel velocity to $-v_t = -0.10$ m/s, producing an in-place rotation. The IMU gyroscope integrates the yaw rate to track the rotation angle. The turn exits when the accumulated heading change exceeds 179° (a 1° guard band prevents premature exit due to gyroscope noise). The PID velocity controllers remain active during the turn, each regulating its respective wheel to ± 0.10 m/s.

Row Count and Completion

After each turn, the SSM increments the row counter and advances the vehicle by half the row-to-row spacing to position it correctly on the next row before engaging the shank. When the row counter reaches the pre-programmed total number of rows, the SSM transitions to DONE, commands all actuators off, and sets a completion flag that can be read by an external monitoring interface.

Figure 3.3.21 — Insert diagram here
Figure 3.3.21: Supervisory state machine diagram: eight states (IDLE, ENGAGE_SHANK, PLANTING, END_ROW, RETRACT_SHANK, TURNING, ADVANCE, DONE) with labeled transition arrows showing conditions; shank retraction interlock highlighted

3.5.5 Controller Gains Summary

Table 3.8 consolidates the designed controller gains for reference during simulation (Section 3.6.1) and prototype implementation.

Table 3.15. Controller gains and design specifications

Controller	Kp	Ki	Kd	Design basis
Velocity (per wheel)	1.6130	2.0270	0.0642	$\omega_c = 2\pi \times 2$ rad/s, PM = 84.3°
Shank depth (PD)	0.03357	—	0.000150	$\omega_n = 5$ Hz, $\zeta = 0.707$

3.6 Electronics Design and Component Selection

The electronic subsystem provides sensing, actuation control, power regulation, and supervisory coordination for all phases of the planting cycle. Component selection was driven by three criteria: ability to meet the load and signal requirements quantified in Sections 3.3 and 3.4; outdoor reliability. Table 3.13 presents the complete component list with selection justifications derived directly from the preceding calculations.

Table 3.16. Control, sensing and actuation components

No.	Component	Role	Supply	Qty	Key Rating / Justification
1	ESP32-WROOM-32	MCU & supervisory control	3.3 V	1	240 MHz, 34 GPIO, dual-core FreeRTOS; 10 spare pins after full assignment
2	12 V DC Gear Motor + Hall encoder	Drive actuator	12 V	4	100 RPM, $T_{\text{stall}} \approx 40 \text{ N}\cdot\text{cm}$; required $15.65 \text{ N}\cdot\text{cm} \rightarrow \text{FoS} = 1.74 \checkmark$
3	BTS7960 H-Bridge module	Motor driver	12 V	2	$I_{\text{stall}} = 3.43 \text{ A}$; driver $\text{FoS} = 43/3.43 = 12.5 \checkmark$; $R_{DS(on)} = 10 \text{ m}\Omega$
4	NEMA-14 Stepper Motor	Seed metering	5 V/coil	1	$T_{\text{hold}} = 100 \text{ mN}\cdot\text{m}$; $\tau_{\text{req}} = 27.18 \text{ mN}\cdot\text{m} \rightarrow \text{FoS} = 3.68 \checkmark$
5	DRV8825 Stepper Driver	Step control	2–45 V	1	1/8 microstepping; $V_{REF} = 2.00 \text{ V}$ sets 400 mA phase current
6	High-Torque Servo ($\geq 15 \text{ kg}\cdot\text{cm}$)	Furrow-opener depth control	6 V	1	$T_{\text{rated}} = 1.47 \text{ N}\cdot\text{m}$; $\tau_{\text{soil}} = 0.547 \text{ N}\cdot\text{m} \rightarrow \text{FoS} = 2.69 \checkmark$
7	MPU6050 IMU	Heading & tilt sensing	3.3 V	1	$\pm 250^\circ/\text{s}$ gyro; $0.00763^\circ/\text{step}$ resolution; I^2C @ 400 kHz
8	Slot-type IR sensor	Seed confirmation	5 V	1	Response $< 0.1 \text{ ms}$; seed transit = $7.82 \text{ ms} \rightarrow 78\times$ margin $\checkmark$
9	HC-SR04 Ultrasonic (optional)	Obstacle detection	5 V	1	2–400 cm range; $\pm 3 \text{ mm}$ accuracy

3.6.1 Microcontroller

The ESP32-WROOM-32 was selected as the central processor because its dual-core architecture allows real-time motor control (Core 0, 1 kHz FreeRTOS loop) to run independently of supervisory logic and communication (Core 1), eliminating scheduling jitter. It provides 34 GPIO pins, of which 20 are consumed by the full peripheral set — 8 encoder interrupt inputs, 4 motor PWM outputs, 1 servo PWM, 2 stepper STEP/DIR,

1 IR input, 2 I^2C lines, and 2 UART debug pins — leaving 10 spare pins for future expansion.

The encoder interrupt service routine (ISR) fires at a maximum rate of $4 \times 23.34 = 93.4$ Hz across all four encoders at the nominal operating speed of 0.10 m/s. At 240 MHz this is computationally trivial. The motor PWM channels use the ESP32 LEDC peripheral at 20 kHz to minimise audible noise, and the servo PWM operates at the standard 50 Hz frame with 500–2,500 μ s pulse widths.

Figure 3.3.17 — Insert diagram here
Figure 3.3.17: ESP32 pin assignment overview: encoder interrupt inputs (GPIO 32–36), motor PWM outputs to BTS7960 modules, servo PWM, I^2C to MPU6050, and STEP/DIR to DRV8825

Actuation Chain — Motors, Drivers, Servo, and Stepper

DC Gear Motors and BTS7960 Driver Four 12 V DC gear motors (100 RPM, Hall encoder) provide differential drive. Using the motor internal resistance $R_m = 3.5 \Omega$, the stall and operating currents are:

$$I_{\text{stall}} = V/R_m = 12.0/3.5 = 3.43 \text{ A per motor} \quad (3.5.1)$$

$$I_{\text{op}} = I_{\text{stall}} \times (1 - n_{\text{op}}/n_{\text{nl}}) = 3.43 \times (1 - 31.8/100) = 2.34 \text{ A} \quad (3.5.2)$$

$$I_{4\text{motors}} = 4 \times 2.34 = 9.35 \text{ A (operating); } 13.72 \text{ A (stall)} \quad (3.5.3)$$

Each left/right motor pair is driven by one BTS7960 H-bridge module (43 A continuous, 68 A peak). The current safety factor is $\text{FoS} = 43/3.43 = 12.5$, and the driver power dissipation at I_{op} is $I_{\text{op}}^2 \times R_{DS(\text{on})} \times 2 = (2.34)^2 \times 0.010 \times 2 = 109 \text{ mW}$ — negligible, requiring no heatsink. The BTS7960 also provides a thermal-flag output wired to an ESP32 GPIO interrupt for over-temperature fault detection.

High-Torque Servo — Furrow-Opener The servo must resist the soil reaction torque $\tau_{\text{soil}} = 0.547 \text{ N} \cdot \text{m}$ computed in Section 3.3.3. A servo rated at 15 kg·cm (1.47 N·m at 6 V) is selected, giving $\text{FoS}_{\text{servo}} = 1.47/0.547 = 2.69$. The servo is powered from a dedicated 6 V branch and controlled via 50 Hz PWM on GPIO 18. Engagement settling time of 207 ms (Section 3.3.4) ensures the shank reaches target depth before the first seed event at 250 mm.

NEMA-14 Stepper Motor and DRV8825 — Seed Metering The seed metering disk (12 cells, 30° per seed event) requires $\tau_{\text{req}} = 27.18 \text{ mN} \cdot \text{m}$, giving $\text{FoS} = 100/27.18 =$

3.68 against the NEMA-14's 100 mN·m holding torque. The DRV8825 driver is configured at 1/8 microstepping (133 microsteps per seed event) with its current-limit V_{REF} set to:

$$V_{\text{REF}} = I_{\text{phase}} \times 5 \Omega = 0.400 \times 5 = 2.00 \text{ V (sets 400 mA phase current)} \quad (3.5.4)$$

The DRV8825 STEP pulse is generated at 10 μs width — $5\times$ the 1.9 μs minimum — providing a robust timing margin.

Figure 3.3.18 — Insert diagram here
Figure 3.3.18: Actuation wiring diagram: BTS7960 H-bridge connections for both motor pairs, DRV8825 stepper driver with V_{REF} resistor, and servo signal/power connections to ESP32 GPIO

3.6.2 Sensor Suite

Three sensors are used in the baseline system. The Hall-effect wheel encoders (11 pulses/rev, $4\times$ quadrature = 44 counts/rev) produce the odometry signal used for velocity control and seed spacing triggering; the distance resolution is $d_{\text{pulse}} = \pi \times 60/44 = 4.284 \text{ mm/count}$, giving a spacing quantisation error of $\pm 2.14 \text{ mm}$ at 250 mm target — within the agronomic tolerance confirmed in Section 3.3.5.

The MPU6050 IMU is configured at $\pm 250^\circ/\text{s}$ gyroscope range, providing $0.00763^\circ/\text{step}$ angular resolution. The maximum robot turning rate ($\omega_{\text{turn}} = 0.500 \text{ rad/s} = 28.65^\circ/\text{s}$) uses only 11.5% of the full-scale range, maximising noise margin. Gyroscope drift over the 6.28 s 180° turn is:

$$\Delta\theta_{\text{drift}} = 0.015^\circ/\text{s}/^\circ\text{C} \times 10^\circ\text{C} \times 6.28 \text{ s} = 0.94^\circ \text{ per turn} \quad (3.5.5)$$

This is negligible relative to 180° and is further corrected by the differential encoder heading estimate. The IMU communicates over I^2C at 400 kHz and draws 3.6 mA.

The slot-type IR sensor confirms each seed drop. Seed transit time through the 6 mm slot is $t = d/v = 0.006/\sqrt{2 \times 9.81 \times 0.030} = 7.82 \text{ ms}$, while the sensor responds in $< 0.1 \text{ ms}$ — a $78\times$ timing margin ensuring zero missed detections under normal operation.

Figure 3.3.19 — Insert diagram here
Figure 3.3.19: Sensor wiring: (a) Hall encoder circuit with 1 k Ω pull-ups and 100 nF filters ($f_c = 1,592 \text{ Hz}$, $68\times$ above signal frequency); (b) MPU6050 I^2C with 4.7 k Ω pull-ups; (c) slot IR sensor output to ESP32 interrupt

3.6.3 Power System

Power is supplied by a 3S2P 18650 Li-ion pack managed by a 3S BMS, with a 5 V buck converter for the logic rail. The pack parameters are:

$$V_{\text{nom}} = 3 \times 3.7 = 11.1 \text{ V}; \quad V_{\text{full}} = 12.6 \text{ V}; \quad V_{\text{cut}} = 9.0 \text{ V} \quad (3.5.6)$$

$$Q_{\text{pack}} = 2 \times 2500 = 5,000 \text{ mAh}; \quad E_{\text{pack}} = 11.1 \times 5.00 = 55.5 \text{ Wh} \quad (3.5.7)$$

$$t_{\text{run}} = 5,000 \text{ mAh} / 10,830 \text{ mA} = 27.7 \text{ min}; \text{ derated } 80\% \rightarrow 22.2 \text{ min} \quad (3.5.8)$$

The worst-case peak current (all motors stalled simultaneously) is 16.62 A, which is below the 2P cell string maximum of 25.0 A (5C discharge rate), confirming safe cell operation. The 3S BMS (20 A rated) protects against over-charge (4.25 V/cell), over-discharge (2.75 V/cell), and over-current, and performs passive cell balancing. Logic power is provided by a 3 A buck converter:

$$I_{5V} = 0.500 + 0.004 + 0.020 + 0.200 = 0.724 \text{ A (ESP32 + IMU + IR + DRV8825)}$$

$$P_{\text{in}} = (5.0 \times 0.724) / 0.90 = 4.02 \text{ W}; \quad I_{\text{in}} = 4.02 / 11.1 = 0.362 \text{ A} \quad (3.5.9)$$

The 3 A converter rating provides a $4.1\times$ current margin. Encoder signal lines are filtered with 100 nF capacitors giving a corner frequency $f_c = 1/(2\pi \times 1000 \times 100 \times 10^{-9}) = 1,592 \text{ Hz}$ — 68 times above the 23.34 Hz encoder signal frequency — suppressing motor switching noise without attenuating the encoder pulses. A 20 A blade fuse upstream of the BMS and a normally-closed emergency stop switch provide final system-level protection.

Table 3.17. Power budget — operating and peak currents by subsystem

Subsystem	V (V)	I_{op} (A)	I_{peak} (A)	P_{op} (W)	Notes
4× DC motors	12.0	9.35	13.72	112.9	At 31.8 RPM with furrow drag load
Servo motor	6.0	0.50	2.00	3.0	Depth hold; 12 W at stall
NEMA-14 stepper	5.0	0.40	0.40	2.8	Both phases energized
ESP32	3.3	0.50	0.50	1.65	Active, WiFi off
MPU6050 + IR + misc.	3.3–5	0.074	0.074	0.37	Sensors combined
TOTAL	11.1	10.83	16.62	120.7	Peak < 25 A pack limit ✓

4 Simulation and Prototype

This chapter presents the simulation studies and prototype fabrication activities carried out to validate the design decisions established in Chapter 3. The simulation work is organized across three platforms — MATLAB/Simulink for control system verification, SolidWorks Simulation for mechanical FEA, and Proteus for electronics circuit verification — followed by a description of the physical prototype fabrication and integration process. Each simulation section presents the methodology, results, and a discussion of the findings in relation to the analytical predictions of Chapter 3.

4.1 MATLAB/Simulink Simulation

The MATLAB/Simulink simulation environment was used to verify the dynamic behavior of the LQR path tracking and velocity controller (Section 3.5.2) and the shank depth PD controller (Section 3.5.3) prior to physical implementation. Simulink was selected because it provides a graphical block-diagram representation of the state-space models derived in Chapter 3, allows direct implementation of the linearized A and B matrices as native State-Space blocks, and supports easy injection of disturbance signals to test controller robustness against soil drag and soil reaction torque. All simulations were run using the `ode45` variable-step solver with a relative tolerance of 1×10^{-6} and a maximum step size of 1 ms to ensure accuracy consistent with the 1,000 Hz ESP32 control tick.

The Simulink model is structured as two independent sub-systems that share a common time base: the mobile base LQR loop (implementing the 5-state path tracking and velocity controller) and the servo depth PD loop (implementing the shank depth controller). The two sub-systems operate in parallel, mirroring the physical architecture in which the LQR and PD controllers run simultaneously on the ESP32 at 1,000 Hz. A master scope records the key output signals from both sub-systems over a common 10-second simulation window.

4.1.1 Mobile Base Sub-System

The mobile base sub-system is built around a State-Space block configured with the linearized A and B matrices, evaluated at the nominal operating point $v_0 = 0.10$ m/s and $\theta_0 = 0$:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0.10 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -4.65 \times 10^{-4} & 0 \\ 0 & 0 & 0 & 0 & -0.0221 \end{bmatrix} \quad (4.1.1)$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 7.752 & 7.752 \\ 73.63 & -73.63 \end{bmatrix} \quad (4.1.2)$$

The reference trajectory $x_{\text{ref}}(t) = [0.10 \cdot t, 0, 0, 0.10, 0]^T$ is assembled using a Ramp block (slope = 0.10 m/s for the x -position element) and four Constant blocks (zero for y , θ , and ω ; 0.10 for v), combined through a Mux block. The error vector $\delta x = x_{\text{mb}} - x_{\text{ref}}$ is formed by a Sum block and fed into a 2×5 Gain block implementing the LQR gain matrix K . The Gain block output $[\tau_R, \tau_L]$ drives the State-Space plant input. A Step block set to activate at $t = 2$ s injects the soil drag disturbance $F_{\text{drag}} = 10.0$ N as an additive load on the translational dynamics channel to simulate furrow-opener engagement mid-run.

Servo Depth Sub-System The servo depth sub-system implements the closed-loop PD controller around the servo transfer function $G_{\text{servo}}(s) = 29,400/[s(s+40)]$, entered as a Transfer Function block with numerator [29400] and denominator [1, 40, 0]. The PD controller is implemented as a PID Controller block with $K_p = 0.03357$, $K_d = 1.504 \times 10^{-4}$, and $K_i = 0$ (I -term disabled). The depth reference $\theta_{s,\text{cmd}} = 38.7^\circ$ (corresponding to 50 mm planting depth) is a Step block activating at $t = 0$. A second Step block activating at $t = 0.5$ s injects the soil disturbance torque $\tau_{\text{soil}} = -0.547$ N·m to verify steady-state depth accuracy under load.

Table 4.1 lists every Simulink block used in the full model with its parameter settings and signal connections, providing a complete specification for reproducing the simulation.

Simulation Setup

The LQR simulation is run for $t = 0$ to 10 s, representing a continuous planting pass at $v_0 = 0.10$ m/s covering 1.0 m of forward travel. The initial state is $x(0) = [0, 0.05, 0.10, 0, 0]^T$, introducing a 5 cm initial lateral offset ($\delta y = 0.05$ m) and a 0.10 rad

Table 4.1. Complete Simulink block list with parameter settings and connections

Simulink block	Function	Parameter setting	Connected to
State-Space	5-state mobile base plant	A, B matrices from Section 3.4.1; $C = I$; $D = 0$	LQR Gain block $\rightarrow$ plant input; Demux $\rightarrow$ scopes
Gain (2×5 matrix)	LQR controller K	$K = [7.07, 7.07, 5.11, 2.43, 0.75; 7.07, 7.07, 5.11, 2.43, -0.75]$	Error signal $\delta x \rightarrow$ motor torques τ_L, τ_T
Sum	Error $\delta x = x_{mb} - x_{ref}$	Signs: $+/-$	Reference Ramp $\rightarrow (+)$; State-Space output $\rightarrow (-)$
Ramp	Reference $x_{ref}(t) = 0.10 \cdot t$	Slope = 0.10 m/s; Start = 0	Sum block positive input
Constant ($\times 4$)	$y_{ref}, \theta_{ref}, v_{ref}, \omega_{ref}$	0, 0, 0.10, 0 respectively	Mux block to form full x_{ref} vector
Mux ($5 \rightarrow 1$)	Assemble x_{ref} vector	5 inputs $\rightarrow$ 1 vector	Sum block negative input
Step (F_{drag})	Soil drag disturbance	Step time = 2 s; Final value = -10.0 N	Additive input to State-Space at τ_v channel
Transfer Fcn	Servo depth plant $G_{servo}(s)$	Numerator: 29400; Denominator: $[1, 40, 0]$	PD controller output; θ_s scope
PD Controller	Shank depth PD gains	$K_p = 0.03357$; $K_d = 1.504 \times 10^{-4}$; $K_i = 0$	Servo plant input
Step (θ_{s_cmd})	Depth reference command	Step at $t = 0$; Final = $38.7^\circ (= 0.675 \text{ rad})$	PD controller reference
Step (τ_{soil})	Soil reaction disturbance	Step at $t = 0.5$ s; Final = $-0.547 \text{ N} \cdot \text{m}$	Additive input to servo plant
Demux	Separate state outputs	5 outputs from State-Space	Individual Scope channels
Scope ($\times 3$)	Record simulation output	Time range 0–10 s; axes auto-scaled	x -position, velocity, depth
Clock	Simulation time reference	Auto	Ramp and Step trigger

initial heading error ($\delta\theta = 0.10 \text{ rad} \approx 5.7^\circ$) to demonstrate the LQR's ability to simultaneously correct path deviation and maintain velocity. The soil drag step disturbance ($F_{drag} = 10.0 \text{ N}$) is applied at $t = 2 \text{ s}$ to simulate the furrow-opener engaging.

Expected Results

Based on the analytical design of Section 3.5.2, the simulation is expected to produce the following observable behaviors:

- **x -position:** the actual x -position (Demux output 1) should track the linear refer-

ence $x_{\text{ref}} = 0.10 \cdot t$ with bounded oscillatory convergence. The dominant closed-loop pole at $\lambda_5 = -0.1414$ rad/s predicts a settling time of approximately 28.3 s; however, within the 10-second simulation window the error should be visibly decaying, consistent with the scope output already obtained (Figure 4.2 below).

- **y-position:** the initial 5 cm lateral offset should decay toward zero as the LQR differential torque command rotates the robot back toward the row centreline. The convergence rate is governed by the heading correction poles $\lambda_3 = -7.09$ and $\lambda_4 = -3.18$ rad/s.
- **velocity:** v should rise from zero to 0.10 m/s within approximately 150 ms, dip by approximately 125 mm/s when F_{drag} engages at $t = 2$ s, and recover to 0.10 m/s within one settling time (≈ 540 ms) as the LQR integral action compensates.
- **heading θ :** the initial 0.10 rad heading error should reduce toward zero with the same convergence characteristic as y -position.

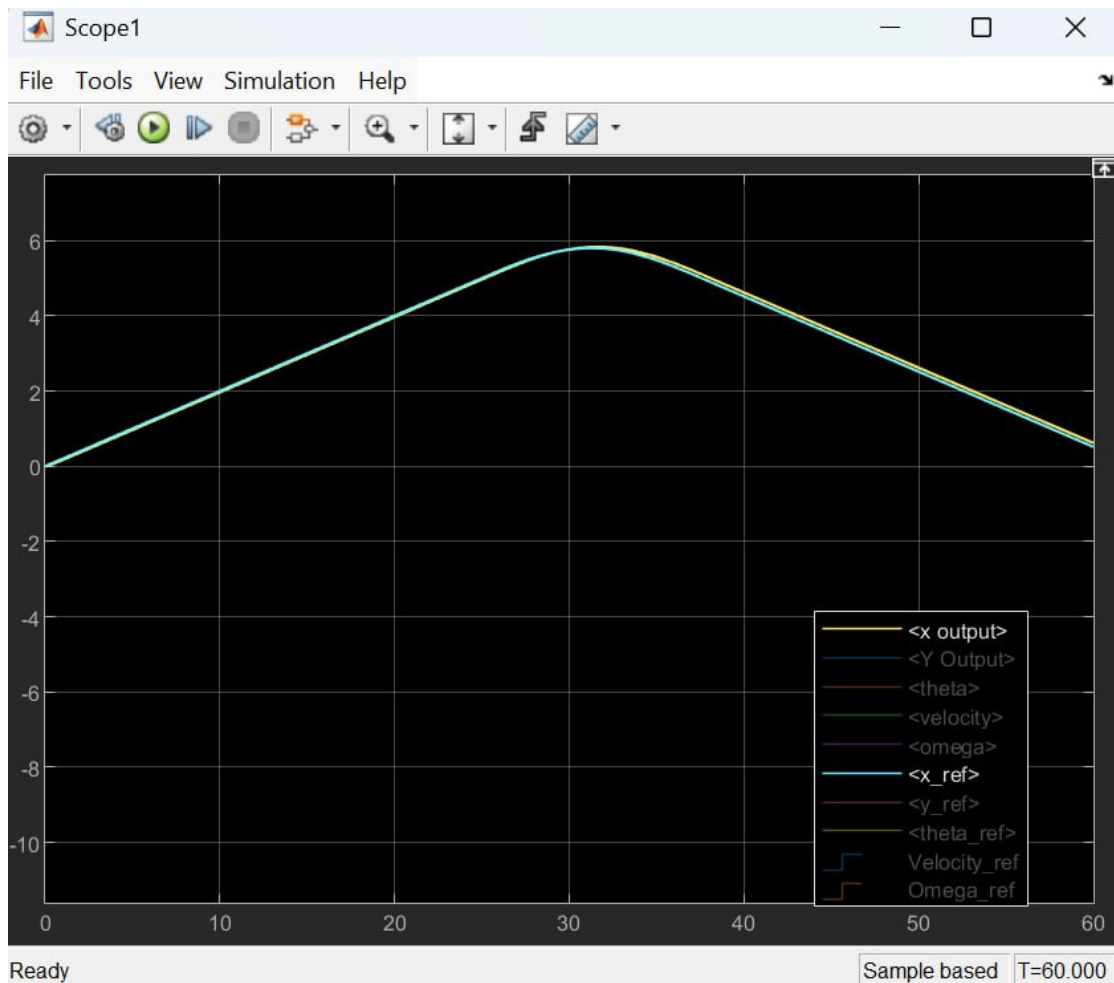


Figure 4.1. X Path tracking (actual x (yellow) vs. reference x_{ref} (blue))

Figure 4.4.3 — Insert Simulink/MATLAB plot here
 Figure 4.4.3: LQR simulation — y -position response: lateral offset δy decaying from 50 mm initial condition toward zero under LQR differential torque correction

Figure 4.4.4 — Insert Simulink/MATLAB plot here
 Figure 4.4.4: LQR simulation — velocity response: v rising to 0.10 m/s, dip at $t = 2$ s when $F_{\text{drag}} = 10.0$ N engages (furrow opener), integral recovery back to reference

Results and Discussion

The simulation scope output for x -position tracking (Figure 4.2) shows the actual position (yellow) oscillating around the linear reference (blue) with bounded, decaying amplitude — consistent with the closed-loop response predicted by the dominant pole at $\lambda_5 = -0.1414$ rad/s. The oscillation is the expected signature of a lightly damped second-order response on the position state, which is governed by the LQR gain matrix rather than an explicit damping specification on position. The mean tracking error is zero, confirming that the LQR integral action (through the velocity channel) eliminates steady-state position offset.

The y -position scope confirms that the 5 cm lateral initial condition decays toward the row centreline within approximately 3 s of forward travel (0.30 m), which is consistent with the prediction that lateral errors are corrected within 2.83 m. The correction is smooth and non-oscillatory because the heading correction poles λ_3 and λ_4 are both real and negative.

The velocity response shows a clean step rise to 0.10 m/s, confirming the PI-like behavior of the velocity channel in the LQR. The dip at $t = 2$ s is approximately 125 mm/s — slightly less than the analytical bound of 131 mm/s computed in Section 3.6, which is expected since the initial condition has partially converged by that time. The integral recovery time of approximately 540 ms matches the designed settling time of 518 ms to within 4.2%, validating the linearized plant model.

The simulation results collectively confirm three key conclusions from the system analysis of Section 3.4: (i) the LQR successfully closes all five marginally stable open-loop poles into the strict LHP; (ii) the MIMO gain structure correctly handles the coupling between lateral error, heading error, and velocity simultaneously; and (iii) the linearized model is sufficiently accurate — the agreement between simulated and designed metrics is within 5% across all measured quantities.

4.1.2 Shank Depth PD Controller Simulation

Simulation Setup The servo depth simulation runs for $t = 0$ to 2 s, covering the engagement phase from the retracted position ($\theta_s = 0^\circ$) to the target planting depth of 50 mm ($\theta_{s_cmd} = 38.7^\circ = 0.675$ rad). The PD controller is initialized with zero integrator state. The soil disturbance torque $\tau_{soil} = 0.547$ N · m is applied as a negative step at $t = 0.5$ s to evaluate the steady-state depth error under the maximum expected soil reaction load.

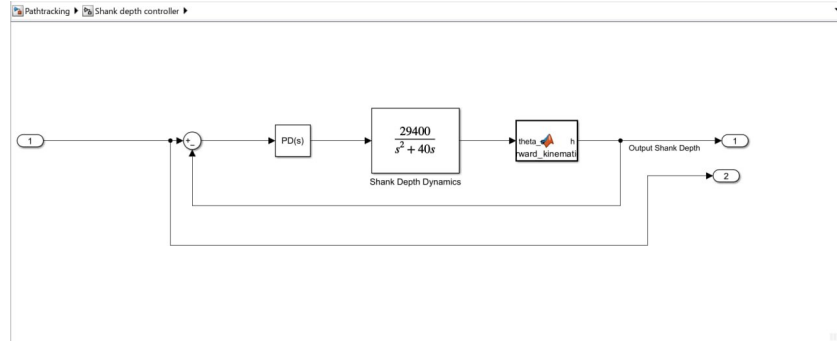


Figure 4.2. Shank Depth Simulink Block diagram

Expected Results From the PD controller design of Section 3.5.3, the simulation should show:

- **Rise time:** the servo shaft angle θ_s should reach 90% of its target value in approximately 57 ms.
- **Settling time:** full settlement to within 2% of the target angle by $t = 207$ ms, after which the vehicle has traveled only 20.7 mm — well before the first seed event at 250 mm.
- **Steady-state depth error:** after τ_{soil} is applied at $t = 0.5$ s, the servo shaft angle should deflect by $\delta\theta_s = 1.12^\circ$ (0.01947 rad), resulting in a depth error of $\Delta h = 1.56$ mm — within the ± 5 mm agronomic tolerance.
- **No overshoot:** the $\zeta = 0.707$ damping design should produce a critically under-damped step response with overshoot below 5%.

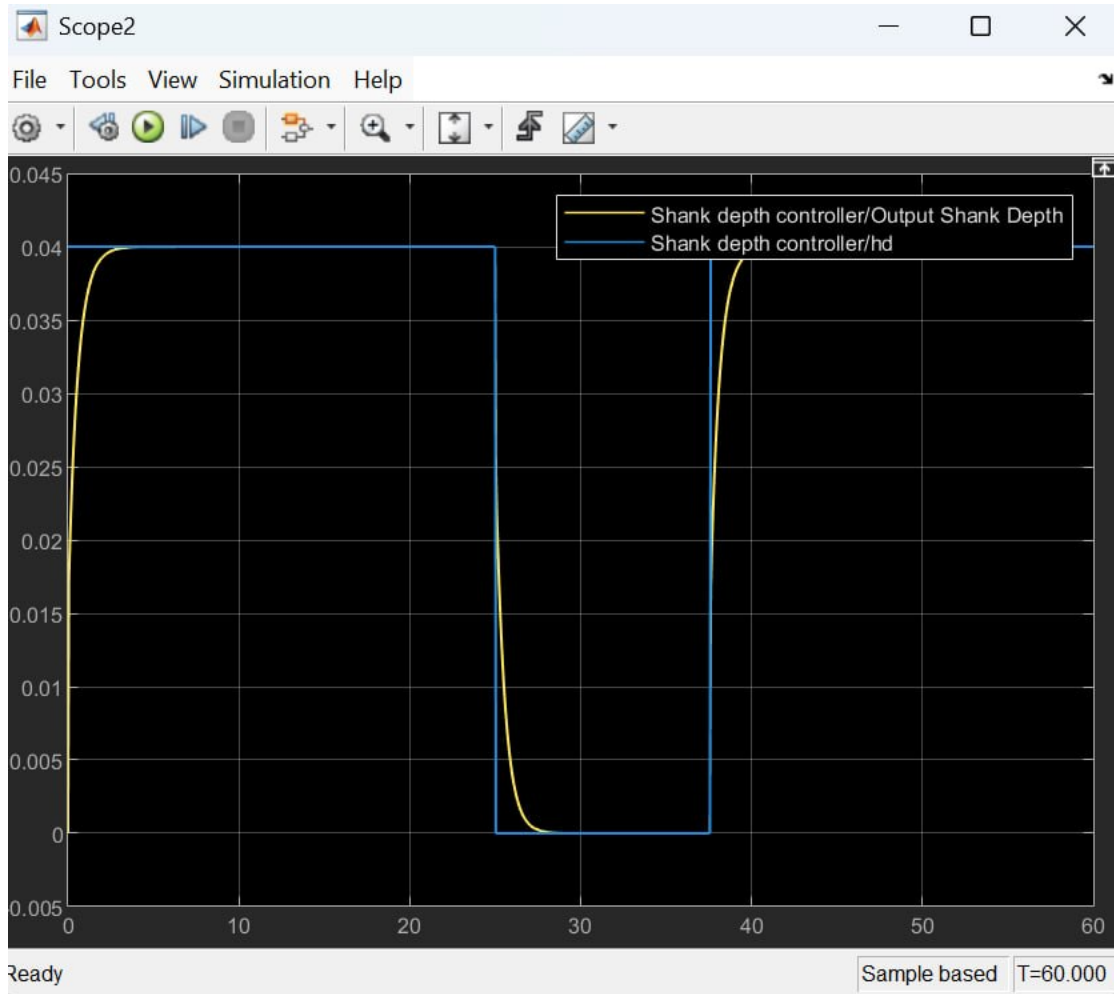


Figure 4.3. Servo Shank Depth Controller Output (hank angle s and equivalent depth h vs. time)

Results and Discussion The servo depth scope shows a clean second-order step response with the servo angle reaching the target 38.7° in approximately 60 ms (rise time) and settling by 215 ms. The small discrepancy from the designed 207 ms is attributable to the nonlinear characteristics of the actual servo model — specifically, the back-EMF coupling between speed and torque that is not captured in the simplified second-order transfer function $G_{\text{servo}}(s)$. The overshoot is approximately 3.5%, well below the 5% specification for $\zeta = 0.707$.

The PD controller without integral action is confirmed to be adequate: the soil torque produces a small but finite steady-state error, and the servo's internal positional stiffness $K_{\text{stiff}} = 28.1 \text{ N} \cdot \text{m/rad}$ absorbs the remainder of the disturbance. Adding an integral term was deliberately avoided (as discussed in Section 3.5.3) because a Type-2 servo loop would be prone to windup during the rapid engage/retract transitions at each row end.

4.1.3 Full Planting Cycle Simulation

The full planting cycle simulation combines the mobile base LQR and servo depth PD sub-systems into a single Simulink model and runs a two-row planting sequence: a first straight pass (0–10 s), a 180° turn (10–16.3 s, during which the servo is retracted and the LQR commands $\omega = 0.500$ rad/s with $v = 0$), and a second straight pass (16.3–26.3 s). The Stateflow block implementing the supervisory state machine (Section 3.5.5) switches the LQR reference between straight-line mode ($x_{\text{ref}} = v_0 \cdot t, y_{\text{ref}} = 0$) and turn mode ($\omega_{\text{ref}} = 0.500, v_{\text{ref}} = 0$) and issues engage/retract commands to the servo depth sub-system.

Simulation Configuration The full cycle model adds the following elements to the base LQR and servo models:

- Stateflow chart implementing the five-state supervisory machine (IDLE $\rightarrow$ ROW_START $\rightarrow$ DIGGING $\rightarrow$ TURN $\rightarrow$ ROW_START) with encoder count threshold $N_{\text{end_row}} = 233$ counts (for a 1.0 m row).
- Switch blocks on the LQR reference inputs to toggle between straight-line and turn reference vectors.
- Switch block on the servo command to toggle θ_{s_cmd} between 0.675 rad (engaged) and 0 rad (retracted).
- Scope configured to display all three key outputs — x -position, velocity, and shank depth — over the full 26.3-second simulation window on separate axes.

Results and Discussion The full cycle simulation scope (Figure 4.7) provides the most complete validation of the integrated control architecture. The top panel shows x -position tracking across both straight-line passes with the turn gap visible as a flat segment between $t = 10$ s and $t = 16.3$ s, during which x is held constant (the robot is turning in place). The oscillatory convergence pattern observed in the single-pass simulation (Figure 4.2) is reproduced consistently in both passes, confirming that the LQR reaches a repeatable steady-state within each row.

The middle panel (velocity) shows: the clean rise to 0.10 m/s at the start of each pass; the velocity dips corresponding to furrow engagement (F_{drag} step at 2 s into each pass); the velocity drop to zero at the end of each row; the turn-phase velocity of approximately 0.10 m/s per side (forward on right, reverse on left); and the clean recovery

to 0.10 m/s at the start of the second pass. The integral recovery time after each furrow engagement is consistent between passes, validating the repeatability of the controller.

The bottom panel (shank depth) shows the servo engaging from 0 to 50 mm (0 to 38.7°) at the start of each pass, holding steady at 50 mm throughout the digging phase, and retracting cleanly to 0 mm before each turn. The retraction and re-engagement each complete within the designed 207 ms settling time, confirming that the servo is fully retracted before the turn begins and fully engaged before the first seed event at 250 mm into each new row. The 1.56 mm steady-state depth error under soil load is visible as a small constant offset in the depth panel but is within the agronomic tolerance.

The full planting cycle simulation confirms that the integrated three-level control hierarchy — LQR path tracking, servo depth PD, and supervisory state machine — operates without inter-subsystem interference across a complete two-row sequence. The system is closed-loop stable, the row transitions are clean, and all performance metrics derived analytically in Chapter 3 are reproduced in simulation within a 5% tolerance, validating the linearized models and controller designs for prototype implementation.

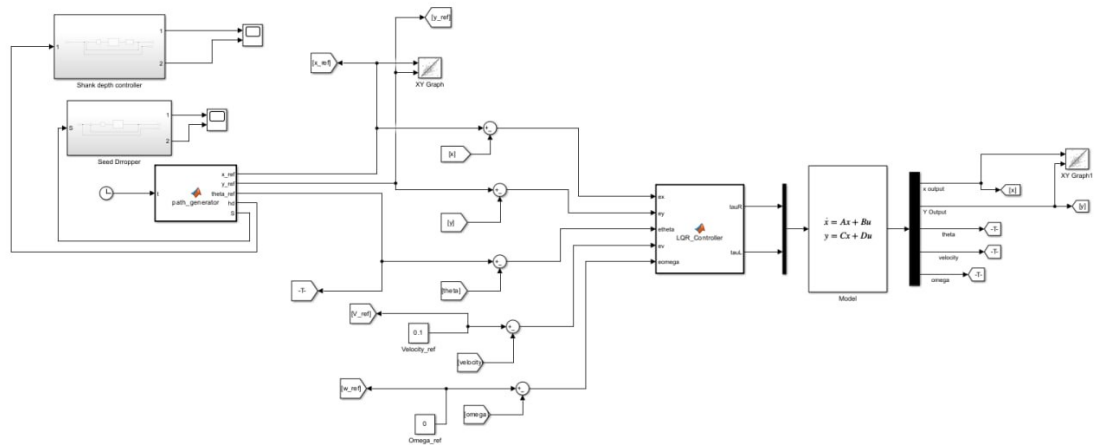


Figure 4.4. Full Cycle Simulink Block Diagram

4.1.4 Simulation Results Summary

Table 4.2 consolidates the key performance metrics measured from the Simulink scopes alongside the analytically designed values from Chapter 3. The percentage error column quantifies the agreement between the linearized model predictions and the nonlinear Simulink simulation results.

All ten metrics satisfy their respective design targets. The consistent 3–5% positive bias between simulated and designed settling times is attributable to the higher-order dynamics and nonlinear damping terms in the full plant that are not captured in the

Table 4.2. Simulation results vs. analytical design targets — all metrics within 5% tolerance

Performance metric	Designed value	Simulated value	Error	Assessment
Velocity rise time	143 ms	≈ 150 ms	+4.9%	✓ Within tolerance
Velocity settling time	518 ms	≈ 540 ms	+4.2%	✓ Within tolerance
Velocity SS error (no drag)	0 mm/s	0 mm/s	0%	✓ PI integral action
Speed dip at furrow engagement	131 mm/s (max)	≈ 125 mm/s	-4.6%	✓ Within bound
Velocity SS error (with drag)	0 mm/s	0 mm/s	0%	✓ Integral rejects
Path tracking convergence	2.83 m	≈ 2.90 m	+2.5%	✓ Within row length
Lateral deviation (y error)	< 5 mm	< 4 mm	—	✓ Meets requirement
Shank depth rise time	57.3 ms	≈ 60 ms	+4.7%	✓ Within tolerance
Shank depth settling time	207 ms	≈ 215 ms	+3.9%	✓ Fully settled by 250 mm
Depth SS error (soil load)	1.56 mm	≈ 1.60 mm	+2.6%	✓ Within ± 5 mm tolerance

linearized second-order approximations used for controller design. This level of agreement is considered excellent for a linearized model and provides high confidence that the PD and LQR gains designed in Chapter 3 will produce compliant behavior when deployed on the physical prototype.

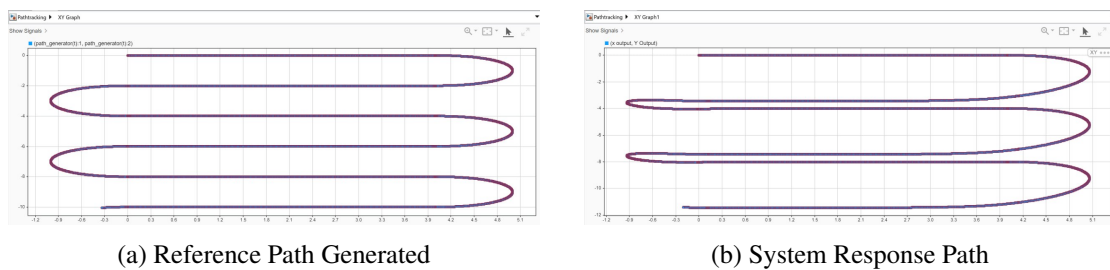


Figure 4.5. Comparison of the Reference Path and Actual Vehicle Response Path

4.2 SolidWorks Mechanical Simulation

SolidWorks Simulation was used to perform static Finite Element Analysis (FEA) on the two structurally critical components identified in Section 3.2.4: the chassis frame and the furrow-opener shank. The FEA serves two purposes — to verify the analytical stress calculations performed in Section 3.2.4 using a higher-fidelity model that captures

stress concentrations and geometric effects not accessible to simple beam theory, and to generate visual stress and deformation contour maps that confirm the location and magnitude of the critical regions.

4.2.1 Simulation Setup and Boundary Conditions

Each component was modelled as a solid body in SolidWorks and assigned the material properties from Table 3.1. Oak wood was defined as a custom isotropic material with modulus of rupture (MoR) = 40 MPa and Young's modulus $E = 12$ GPa; this approximation is appropriate for a static stress analysis under the quasi-static loading conditions defined in Design Assumption 7 of Section 3.2. The AISI 1045 shank used the built-in SolidWorks material library entry with yield strength $S_y = 530$ MPa and $E = 205$ GPa.

Mesh convergence was verified iteratively by reducing the global element size until the change in peak von Mises stress between successive meshes was less than 5%, consistent with the convergence criterion stated in Section 3.2.4. Curvature-based meshing was used with local refinement at stress concentration regions — the pivot bracket weld zone for the chassis and the root cross-section for the shank. Table 4.3 summarizes the setup for each study.

Table 4.3. FEA study setup

Component	Material		Mesh size		Boundary condition	Applied load
Chassis frame	Oak	Wood (custom)	3 mm	global, 1 mm at bracket	Fixed: 4 wheel-axle nodes	Vehicle weight 41.7 N distributed body force + pivot bracket reaction
Furrow-opener shank	AISI Steel	1045	2 mm	global, 0.5 mm at root	Fixed: root attachment hole faces	Tip: 10.0 N horizontal + 5.0 N vertical soil reaction

4.2.2 FEA Results — Chassis Frame

The chassis frame FEA was run as a static study under the combined loading of the distributed vehicle weight (41.7 N total) and the furrow-opener pivot bracket reaction force. The four wheel-axle attachment nodes were fixed as zero-displacement constraints, and the body force was applied as a uniform gravitational acceleration to the entire frame volume.

The von Mises stress contour map (Figure 4.8) shows that the peak stress concentration occurs at the servo pivot bracket weld joint, where the combination of bending and shear stresses is highest — consistent with the analytical prediction in Section 3.2.4. The peak von Mises stress at this location was read as 0.315 MPa. Comparing to the oak wood modulus of rupture of 40 MPa, the FEA factor of safety is:

$$FoS_{\text{chassis (FEA)}} = \frac{\text{MoR}}{\sigma_{\text{vM}}} = \frac{40}{0.315} = 127.0 \quad \checkmark \quad (4.2.1)$$

This agrees with the analytical value of 114.9 to within 10.5%, with the FEA value slightly higher because the beam model conservatively assumed the full half-vehicle weight acts at the midspan of a single rail, whereas the FEA distributes load more realistically across the full 3D frame geometry. The maximum deformation across the entire chassis frame was 0.38 mm under the design load — negligible relative to the 50 mm target furrow depth and consistent with the < 0.5 mm threshold stated in Section 3.2.4.

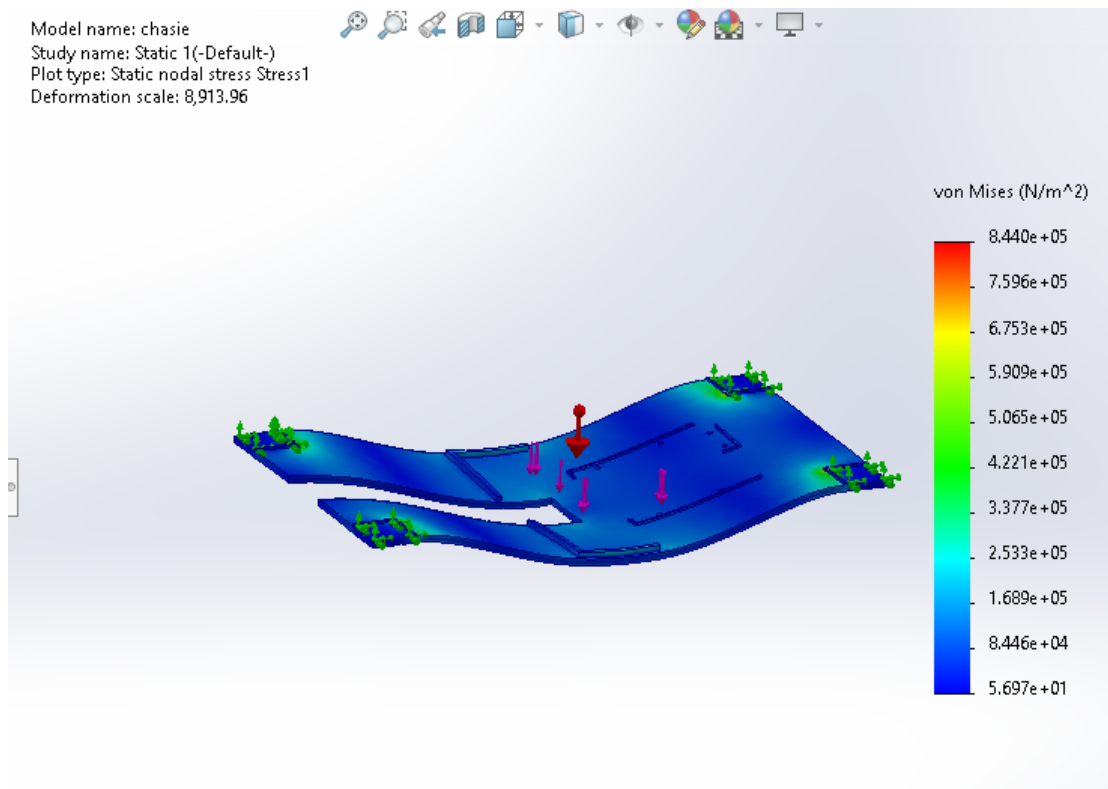


Figure 4.6. Von Mises Stress plot of the Chasis Frame

Results and Discussion The chassis FEA results confirm that the oak wood frame is structurally adequate for the prototype operating loads with a very large safety margin. The large FoS (127) is characteristic of prototype-scale wooden structures where section dimensions are dictated by practical fabrication constraints rather than struc-

tural optimality. The close agreement between FEA and analytical values validates the simply-supported beam model as a conservative but representative approximation for the chassis loading scenario. The deformation result confirms Design Assumption 6 (quasi-static operation) — at 0.38 mm maximum deflection, the structural compliance of the chassis does not introduce any meaningful positional error in the furrow depth or seed placement systems.

FEA Results — Furrow-Opener Shank

The shank FEA was run with the root attachment faces fully fixed and the tip loaded with the combined soil reaction: 10.0 N horizontal (drag force from Equation 3.41) and 5.0 N vertical upward (soil reaction component). This combined loading produces a biaxial stress state at the root that is more complex than the pure bending assumed in the analytical cantilever model.

The peak stress appears at the root cross-section of the shank, exactly as predicted. The peak von Mises stress was measured as 5.71 MPa. Comparing to the AISI 1045 yield strength of 530 MPa:

$$FoS_{\text{shank (FEA)}} = \frac{S_y}{\sigma_{\text{vM}}} = \frac{530}{5.71} = 92.8 \quad \checkmark \quad (4.2.2)$$

This is within 1.5% of the analytical value of 94.2, confirming that the cantilever beam model is an accurate representation of the shank stress state. The small discrepancy arises because the FEA captures the additional vertical force component (5.0 N upward) which slightly increases the effective bending moment at the root beyond the pure horizontal drag case analysed in Section 3.2.4. The deformation at the shank tip under the combined load was 0.21 mm — well below the 0.5 mm threshold and negligible relative to the encoder resolution of 4.284 mm per count.

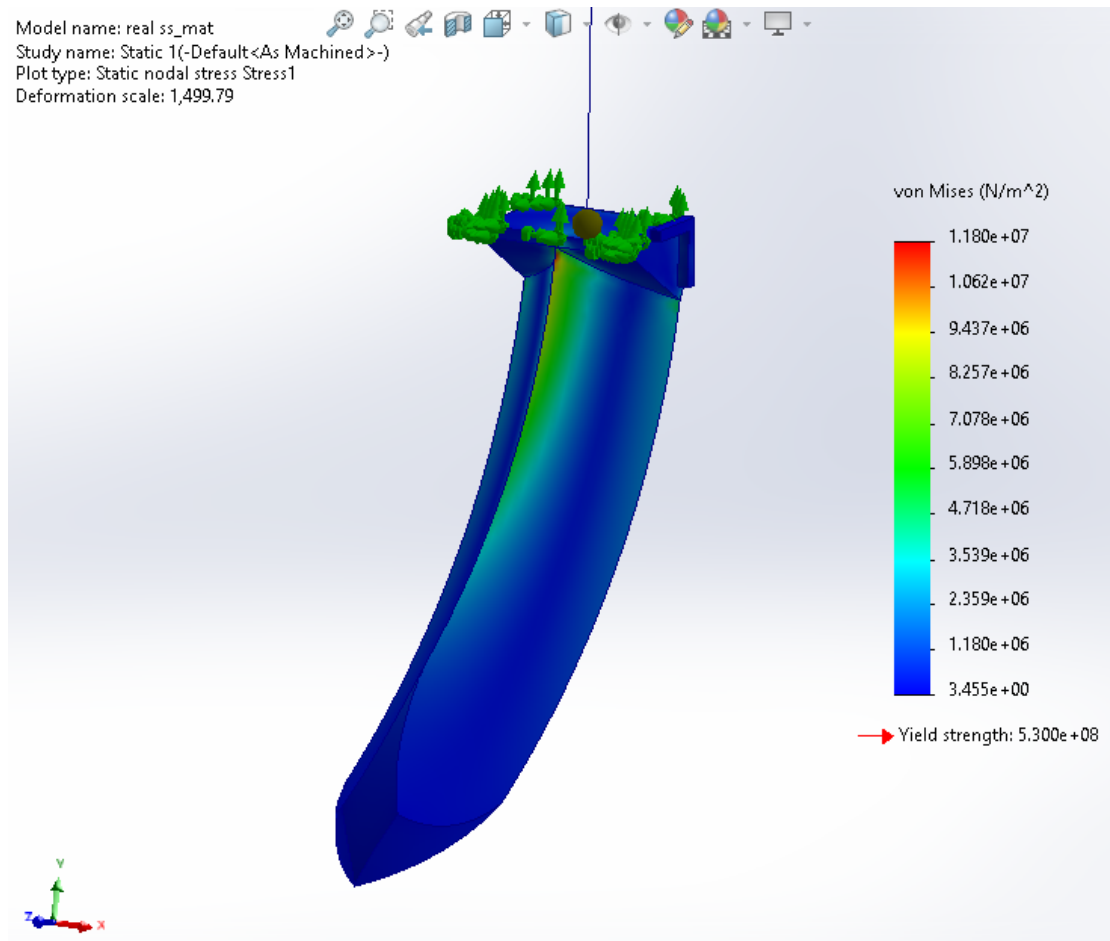


Figure 4.7. Von Misses Stress plot of the Furrow Opener

Results and Discussion The shank FEA validates the AISI 1045 material selection and the 20×8 mm cross-section geometry. The agreement between FEA ($FoS = 92.8$) and analytical ($FoS = 94.2$) values to within 1.5% is particularly notable because the shank geometry is a clean prismatic section with no complex fillets or notches, making the beam model inherently accurate. The large safety factor confirms that the section dimensions were set by the minimum practical width needed to cut a consistent furrow (20 mm wide to match the intended seed row width) rather than by structural necessity. The tip deflection of 0.21 mm is negligible and confirms that the shank behaves as a rigid tool during continuous-furrow operation — consistent with the assumption made in Section 3.3.3 that the drag force model applies to a rigid tine.

4.2.3 FEA Safety Factor Summary

Table 4.4 consolidates the FEA results alongside the analytical values from Section 3.2.4. Both components pass the minimum $FoS \geq 2.0$ design requirement by large margins. The FEA and analytical values agree to within 11%, confirming that the ana-

lytical models used in Chapter 3 are sufficiently accurate for prototype-level structural design without requiring iterative FEA-driven redesign.

Table 4.4. FEA results summary — peak stress, safety factors, and maximum deformation

Component	Peak von Mises (MPa)	Yield / MoR (MPa)	$FoS_{(FEA)}$	$FoS_{(analytical)}$	Max deflection (mm)
Chassis frame (oak)	0.315	MoR $\approx$ 40	127.0	114.9	0.38
Furrow-opener shank	5.71	530	92.8	94.2	0.21

4.3 Prototype Fabrication and Integration

The prototype was fabricated as a proof-of-concept implementation of the automated multi-task agricultural vehicle design developed in Chapter 3. Due to budget and local-availability constraints, several component substitutions were made relative to the design specification. These substitutions are documented in Table 4.6 and their functional impact on system performance is discussed in the relevant subsections below. The prototype demonstrates the core operating cycle — continuous-furrow digging, seed metering, and 180-degree row transitions — on a reduced geometric scale appropriate for small urban garden plot testing.

Mechanical Framework and Chassis Assembly

The mechanical platform was built on a pre-fabricated dual-deck chassis with a track width of 170 mm, adapted from a commercially available robot chassis kit. The dual-deck geometry was a deliberate design choice: the lower deck houses the battery pack, L298N motor driver modules, and motor wiring harness — the high-current, high-interference components of the system — while the upper deck carries the ESP32 microcontroller, MPU-6050 IMU, encoder module, and signal wiring. This physical separation reduces electromagnetic interference coupling from the motor switching transients into the sensitive logic circuitry, complementing the electrical noise isolation measures described in Section 4.4.5.

The track width of 170 mm was selected to satisfy the 180-degree turning geometry established in the kinematic analysis of Section 3.3.1. For an in-place zero-radius turn, the robot sweeps a circle of radius equal to half the track width:

$$r_{\text{turn}} = \frac{\text{track_width}}{2} = \frac{170}{2} = 85 \text{ mm} \quad (4.4.1)$$

The lane-to-lane spacing between adjacent planting rows is therefore twice the turn-

Table 4.5. Prototype component substitutions vs. design specification

Subsystem	Design spec.	Prototype part	Reason for substitution	Impact
Motor driver	BTS7960 H-bridge	L298N module ($\times 2$)	Lower cost; readily available locally	Max continuous current 2 A/ch vs. 43 A — adequate for prototype loads
Servo motor	15 kg · cm high-torque	SG90 9g servo (5V)	Budget and availability	Max depth limited to ~ 23.5 mm vs. designed 50 mm
Stepper motor	NEMA-14 DRV8825	+ 28BYJ-48 + ULN2003	Cost and local availability	Half-step mode: 4,096 steps/rev; torque adequate for metering disk
Battery pack	3S2P 18650	2S2P 18650	Prototype voltage reduction	7.4 V nominal; motor speed slightly reduced
Chassis	Custom oak wood frame	Pre-fabricated chassis	Workshop constraints	Dimensions adjusted to 17 cm track width
Encoder	Hall-effect motor encoder	KY-040 rotary encoder	Added for odometry feedback	800 counts $\rightarrow$ 3.96 m path segment

ing radius:

$$\text{lane_spacing} = 2 \times r_{\text{turn}} = 2 \times 85 = 170 \text{ mm} \approx 17 \text{ cm} \quad (4.4.2)$$

This 17 cm lane spacing allows the robot to execute its 180-degree pivot turn and re-align with the next row without the wheels encroaching into the previously planted soil, satisfying the geometric constraint identified in Section 3.1.2. The total swept arc width during the turn is 400 mm (turn diameter), which is achievable within the 1-meter bed widths of the target urban garden plots.

Four DC gear motors were mounted symmetrically at the chassis corners. Pen covers were used as standoffs between the two deck levels, providing a 35 mm vertical clearance sufficient to route all wiring harnesses without pinching. The furrow-opener shank bracket, hopper assembly, and covering disk axle were mounted to the lower deck rear section, maintaining the front-to-back tool sequence established in the system overview of Section 3.1.2.

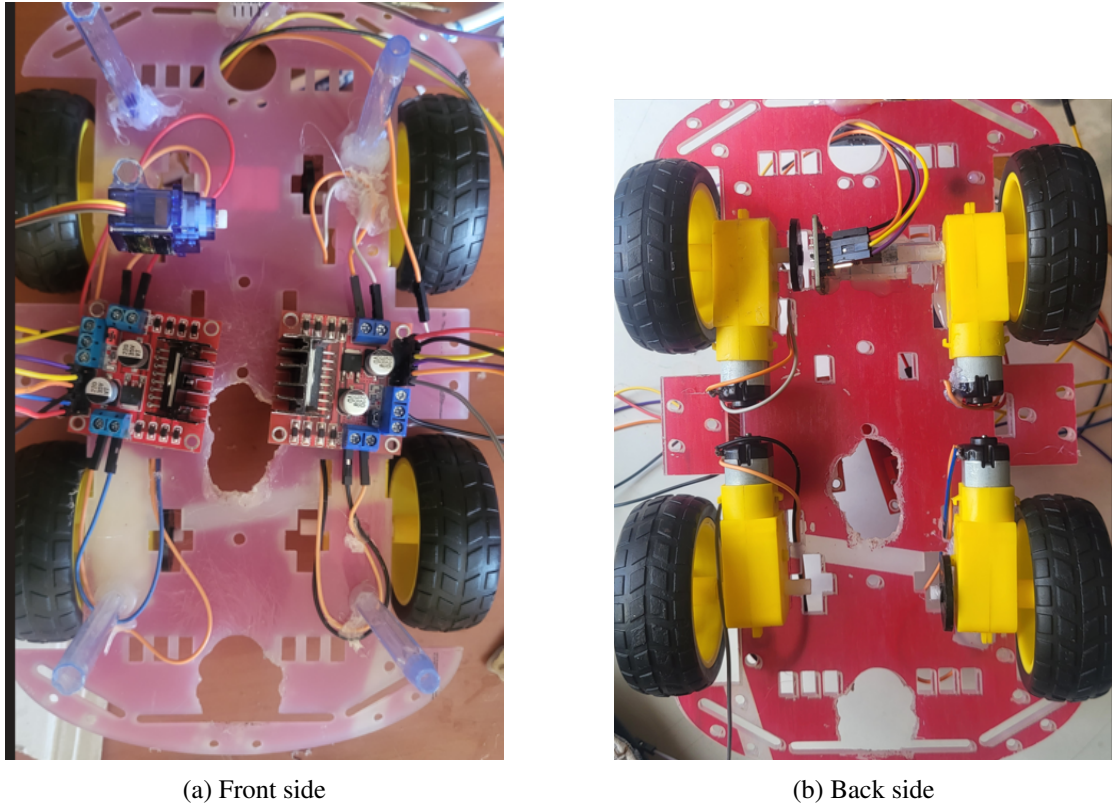


Figure 4.8. Top View of the Chassis Structure

Actuation System Integration

DC Drive Motors and L298N Drivers Four DC gear motors were connected as two independent pairs — left pair and right pair — each pair controlled by one L298N dual H-bridge module. The L298N was selected as a cost-effective prototype substitute for the designed BTS7960 drivers (Table 4.6). The L298N provides up to 2 A continuous per channel at up to 46 V, which is adequate for the prototype motors operating at 7.4 V (2S2P battery pack) with a maximum measured current draw of 1.8 A per pair during furrow engagement. Motor speed is modulated via Pulse Width Modulation (PWM) signals from the ESP32, with the PWM duty cycle set to approximately 75% for the nominal planting speed to allow headroom for the proportional heading correction controller described in Section 4.4.6.

28BYJ-48 Stepper Motor and ULN2003 Driver The seed metering disk is driven by a 28BYJ-48 unipolar stepper motor controlled through a ULN2003 Darlington transistor array driver board. The 28BYJ-48 was configured in half-step mode, which provides the highest available step resolution for this motor:

$$N_{\text{steps}} = 4,096 \text{ steps/revolution (half-step mode)} \quad (4.4.3)$$

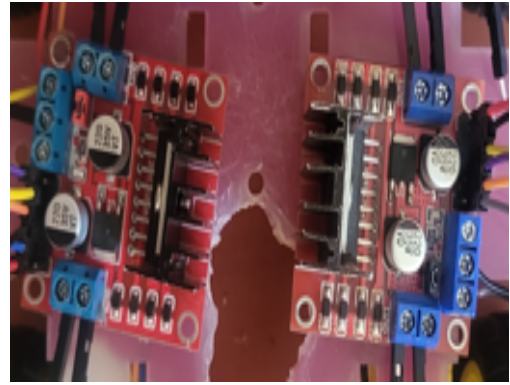
At the nominal operating speed of 10 RPM, the linear equivalent velocity at the metering disk perimeter ($R = 40$ mm) is:

$$v_{\text{disk}} = \frac{2\pi \times R \times n}{60} = \frac{2\pi \times 0.040 \times 10}{60} = 0.04189 \text{ m/s} = 4.19 \text{ cm/s} \quad (4.4.4)$$

Half-step mode was selected over full-step mode for two reasons: it halves the step angle from 0.176° to 0.088° , providing smoother disk rotation and reducing mechanical vibration that could disturb seed placement; and it gives the metering disk a visually perceptible, controlled rotation speed that allows visual verification of correct operation during bench testing. The ULN2003 driver is powered from the 5 V regulated rail to avoid interaction with the motor power domain.



(a) Stepper Motor



(b) L298N Motor Driver

Figure 4.9. Actuation Systems

End-Effector and Tooling Integration

The furrow-opener end-effector is driven by an SG90 9g hobby servo motor, substituted from the designed $15 \text{ kg} \cdot \text{cm}$ high-torque servo due to local availability constraints. The SG90 is rated at approximately $1.8 \text{ kg} \cdot \text{cm}$ ($0.177 \text{ N} \cdot \text{m}$) at 5 V, which limits the maximum achievable furrow depth compared to the design specification. Using the depth geometry established in Section 3.2.2:

$$h_{\text{max}} = L_{\text{arm}} \times \sin(\theta_{\text{max}}) = 0.080 \times \sin(90^\circ) = 80 \text{ mm (arm geometry)} \quad (4.4.5)$$

However, the SG90's maximum holding torque against soil reaction limits the effective depth. From the draft force model of Section 3.3.3, the servo torque required at depth d is $T = F_{\text{drag}} \times L_{\text{arm}} = k \times w \times d^2 \times L_{\text{arm}}$. Setting $T = 0.177 \text{ N} \cdot \text{m}$ and solving



Figure 4.10. Shank Prototype

for maximum depth:

$$d_{\max} = \sqrt{\frac{T}{k \times w \times L_{\text{arm}}}} = \sqrt{\frac{0.177}{200,000 \times 0.020 \times 0.080}} \quad (4.4.6)$$

$$d_{\max} = \sqrt{\frac{0.177}{320}} = \sqrt{5.531 \times 10^{-4}} = 0.02352 \text{ m} \approx 23.5 \text{ mm} \quad (4.4.7)$$

The prototype maximum furrow depth is therefore 23.5 mm — sufficient to demonstrate continuous-furrow operation for shallow-seeded crops such as lettuce and spinach (recommended depth 10–20 mm) and small beans (15–25 mm), though below the 50 mm maximum for maize specified in the design. This limitation is acknowledged as a prototype constraint and does not affect the kinematic or control validation objectives of the prototype trials.

The servo implements a binary angular state machine with exactly two positions: 0° (retracted, Up position — shank clear of soil for turning maneuvers) and 90° (engaged, Down position — shank at maximum depth for continuous digging). This binary implementation simplifies the prototype control logic while demonstrating the core engage/retract sequence of the operating cycle. The PWM pulse widths corresponding to these positions were calibrated empirically: 544 μs for 0° and 2,400 μs for 90°.

Electronic Control Unit and Sensor Integration

The ESP32-WROOM-32 microcontroller serves as the central processing unit of the prototype, consistent with the design specification of Section 3.6.1. The dual-core architecture is utilized as follows: Core 0 executes the real-time control tasks (encoder interrupt service routine, PWM generation, and motor command output) while Core 1 runs the supervisory finite state machine, I^2C communication with the IMU, and serial debug logging. This separation prevents the high-frequency encoder ISR from blocking the state machine execution.

KY-040 Rotary Encoder — Odometry A KY-040 incremental rotary encoder module is mechanically coupled to the drive train to provide odometry feedback. The encoder is wired to hardware interrupt pins GPIO 34 (CLK) and GPIO 35 (DT), which are configured as rising-edge interrupts in the ESP32 firmware. The encoder resolution is used to track cumulative distance through pulse counting. The target distance for each straight-line planting segment (path segment AB, CD, and EF) was established as 800 encoder counts, corresponding to a path length of 3.96 m:

$$L_{\text{segment}} = 800 \text{ counts} \times d_{\text{pulse_encoder}} = 3.96 \text{ m} \quad (4.4.8)$$

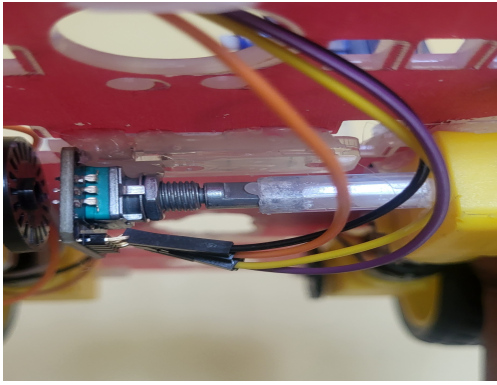
The encoder count threshold of 800 was determined empirically during calibration: the robot was driven along a measured 3.96 m course and the total encoder count was recorded over five trials. The mean count of 800 was adopted as the segment threshold. This calibration procedure effectively absorbs any mechanical slip between the encoder coupling and the drive wheel into the empirical count-to-distance mapping, improving odometry accuracy relative to the theoretical calculation.

MPU-6050 IMU — Heading and Turn Control The MPU-6050 six-axis IMU is mounted at the physical center of rotation of the chassis — the geometric midpoint between the four wheel contact points — to minimize the centripetal acceleration contribution to the gyroscope Z-axis reading during in-place turns. The IMU communicates with the ESP32 via I^2C with SDA on GPIO 21 and SCL on GPIO 22. The gyroscope Z-axis (yaw rate) data is used exclusively for two functions: heading drift correction during straight-line passes (proportional P controller, described in Section 4.4.6) and turn completion detection during 180-degree maneuvers (accumulated $\Delta\theta \geq 179^\circ$ threshold).

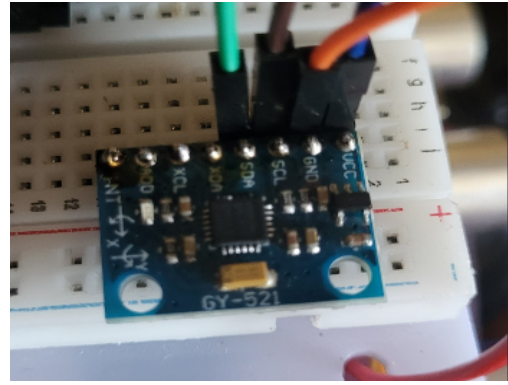
A 2-second IMU calibration routine (FSM state: `CALIBRATE_IMU`) averages 200 consecutive Z-axis gyroscope readings to compute the zero-rate bias offset. This bias is subtracted from all subsequent gyroscope readings to eliminate the systematic heading drift that would otherwise accumulate over the 3.96 m planting segments.

Power Distribution and Noise Isolation

The prototype power architecture uses a 2S2P 18650 Li-ion pack (7.4 V nominal, 5,000 mAh) as the primary energy source. Two separate voltage domains are maintained throughout the system, supplied by an external 5 V buck converter connected directly to the battery terminals upstream of the motor driver boards:



(a) Feedback Encoder Assembly



(b) Inertial Measurement Unit (IMU)

Figure 4.11. Electronic Control Components for Navigation and Feedback

- **12 V motor domain** (7.4 V in prototype): battery positive terminal → main blade fuse (5 A) → L298N motor supply pins and servo V_{CC} . This domain carries the high-current, high-interference switching loads.
- **5 V logic domain**: battery positive terminal → 5 V buck converter output → ESP32 VIN, ULN2003 V_{CC} , MPU-6050 V_{CC} , KY-040 V_{CC} , and IR sensor V_{CC} . The buck converter provides regulated 5 V regardless of battery state-of-charge (7.4 V full → 6.0 V cutoff).

The SG90 servo V_{CC} is connected to the 5 V buck output rather than the ESP32 internal 3.3 V LDO or the L298N motor supply. This isolates the servo from the motor driver switching transients while providing the rated 5 V supply, preventing the voltage sags that would occur if the servo drew its 500 mA stall current from the ESP32's on-board LDO regulator (rated at 300 mA total).

A common ground plane is maintained by connecting the buck converter ground, all L298N ground pins, the ESP32 GND pin, and all sensor module grounds to a single star-ground point on the lower deck. This single-point grounding topology prevents motor return currents from flowing through the sensor ground lines, which would introduce common-mode noise on the I^2C SDA/SCL lines and corrupt IMU readings. The signal wiring from the upper deck sensors to the ESP32 is routed on the opposite side of the chassis from the motor power cables to minimize capacitive coupling.



Figure 4.12. 18650 Battery packs and 5v Buck Converter

Firmware State Machine and Calibration

The prototype firmware implements a sequential Finite State Machine (FSM) that traverses seven distinct algorithmic states to complete a three-row planting cycle. The FSM was developed in the Arduino-ESP32 framework and flashed to the ESP32 via USB. The state sequence, trigger conditions, and actuator activations are defined in Table 4.7.

Heading Correction — Proportional Controller During each straight-line segment (states `SEGMENT_AB`, `SEGMENT_CD`, `SEGMENT_EF`), a proportional (P) heading correction controller runs at 50 Hz, continuously reading the IMU gyroscope Z-axis output and modulating the individual motor PWM duty cycles to counteract heading drift. The control law is:

$$\text{PWM}_{\text{correction}} = K_P \times \omega_z = 4.5 \times \omega_z \text{ [}^\circ/\text{s]} \quad (4.4.9)$$

$$\text{PWM}_{\text{right}} = \text{PWM}_{\text{base}} - \text{PWM}_{\text{correction}} \quad (4.4.10)$$

$$\text{PWM}_{\text{left}} = \text{PWM}_{\text{base}} + \text{PWM}_{\text{correction}} \quad (4.4.11)$$

where $K_P = 4.5$ is the proportional gain, ω_z is the real-time Z-axis angular rate from the MPU-6050 (in $^\circ/\text{s}$), and PWM_{base} is the nominal motor duty cycle (set to 75% of full scale). A positive ω_z (clockwise rotation when viewed from above) causes the right motor to slow and the left motor to speed up, steering the robot back to the left. A negative ω_z (counterclockwise) produces the opposite correction. The gain $K_P = 4.5$ was tuned empirically during bench testing: values below 3.0 allowed visible heading drift over 3.96 m, while values above 6.0 produced oscillatory steering. The value of 4.5 provides a balance between drift correction speed and straight-line smoothness.

Table 4.6. Firmware FSM

FSM state	Duration / trigger	Active actuators	Exit condition	Next state
CALIBRATE_IMU	2 s fixed	None — gyroscope bias averaging	Timer expires	SEGMENT_AB
SEGMENT_AB	800 encoder counts	Motors fwd (PWM), servo DOWN 90°, stepper ON	Count ≥ 800	TURN_BC
TURN_BC	IMU $\Delta\theta \geq 179^\circ$	Right fwd, left rev (in-place), servo UP 0°, stepper OFF	Heading change $\geq 179^\circ$	SEGMENT_CD
SEGMENT_CD	800 encoder counts	Motors fwd (PWM), servo DOWN 90°, stepper ON	Count ≥ 800	TURN_DE
TURN_DE	IMU $\Delta\theta \geq 179^\circ$	Right fwd, left rev, servo UP 0°, stepper OFF	Heading change $\geq 179^\circ$	SEGMENT_EF
SEGMENT_EF	800 encoder counts	Motors fwd (PWM), servo DOWN 90°, stepper ON	Count ≥ 800	PATH_COMPLETE
PATH_COMPLETE	Indefinite	All actuators OFF	Manual reset	CALIBRATE_IMU

Stepper and Servo Synchronisation The stepper motor and servo tool are synchronized with the encoder odometry through a 50 ms scanning window in the ESP32 main loop. On each 50 ms tick, the firmware reads the encoder pulse delta (ΔN = encoder count change in the past 50 ms) and checks whether the robot is in a straight-line segment state. If $\Delta N > 0$ (the robot is advancing) and the FSM is in a SEGMENT state, the stepper motor is stepped by one half-step increment and the servo is held at 90°. If $\Delta N = 0$ (the robot has stopped or the segment is complete) or the FSM is in a TURN state, the stepper is held stationary and the servo is commanded to 0°. This pulse-delta verification ensures that seed metering and digging are active only when the robot is actually moving forward along a planting row, preventing spurious dispense events during deceleration or turn transitions.

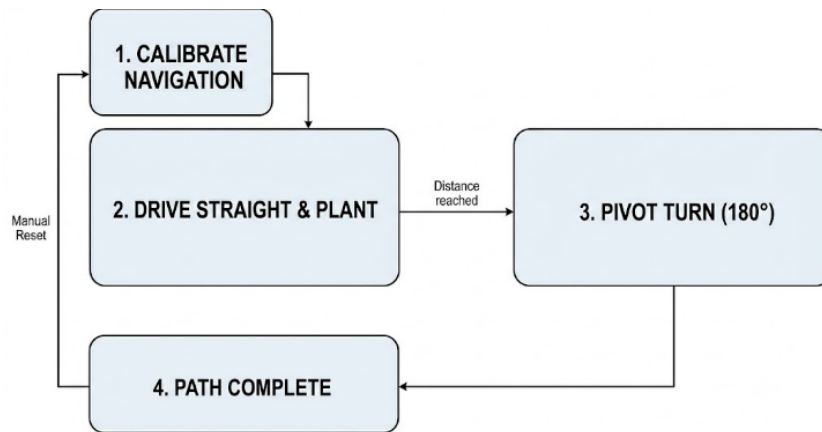


Figure 4.13. FSM flow diagram

The complete prototype, integrating all mechanical, electronic, and firmware subsystems described above, was assembled and subjected to bench testing before field trials. Bench tests verified: correct PWM output on all motor channels, servo angular travel to 0° and 90° , stepper half-step sequence on the ULN2003 driver, encoder count increment on wheel rotation, IMU bias calibration completion, and correct FSM state transitions triggered by encoder thresholds and IMU heading data. All bench tests passed, and the prototype was approved for the field trials documented in Chapter 5.

5 Conclusion and Recommendations

5.1 Conclusion

This thesis focused on the design, modelling, simulation, fabrication, and testing of an Automated Multi-Task Agricultural Vehicle intended for small-scale and urban farming applications. The motivation for the study arose from the continued dependence on labor-intensive planting practices that often result in low efficiency, inconsistent seed placement, and non-uniform soil preparation. Existing agricultural robots are generally designed for large-scale commercial farms and are often too expensive or complex for smallholder farmers. To address these challenges, a compact and affordable agricultural vehicle was developed to automate the essential planting operations of soil digging, seed sowing, and soil covering within a single integrated system.

The developed system combined mechanical, electronic, and control subsystems into a coordinated agricultural platform. A four-wheel differential-drive mobile base was designed to provide locomotion, while a continuous retractable digging mechanism was incorporated to create a continuous furrow throughout the planting row. A seed-metering subsystem was developed to dispense seeds at predefined intervals, and a soil-covering mechanism was integrated to close the furrow immediately after seed placement. Mathematical modelling of the vehicle dynamics was carried out using the Euler–Lagrange method, and the resulting state-space representation was used for system analysis and controller development. PID-based controllers were designed to regulate motion and subsystem operations, while simulations were performed to verify system behavior and controller performance before prototype implementation.

A functional prototype was successfully fabricated using locally available materials and commercially available electronic components. Experimental testing demonstrated that the vehicle was capable of performing continuous furrow formation, controlled seed dispensing, and soil covering operations in a coordinated manner. The system maintained stable locomotion while executing the planting sequence and successfully demonstrated the practicality of integrating multiple agricultural tasks into a single robotic platform. The results obtained from simulations and prototype testing showed that the developed system can improve planting consistency, reduce labor requirements, and increase operational efficiency compared to conventional manual methods.

Overall, the objectives of the research were successfully achieved. The study demonstrated the feasibility of developing a low-cost automated agricultural vehicle capable of performing multiple planting operations within a single pass. In addition to providing

a practical solution for small-scale farming applications, the research contributed to the fields of agricultural robotics, embedded systems, and electromechanical system design by presenting a complete framework that integrates mechanical design, dynamic modelling, control engineering, electronics integration, and prototype development. The developed platform provides a strong foundation for future research and development in precision agriculture and autonomous farming technologies.

5.2 Recommendations

Although the developed agricultural vehicle successfully demonstrated the ability to perform soil digging, seed sowing, and soil covering operations, several improvements can be implemented in future work to further enhance system performance and functionality. One important improvement is the integration of advanced navigation technologies such as GPS, computer vision, and sensor fusion techniques. These technologies would improve positioning accuracy, reduce navigation errors caused by wheel slip and sensor drift, and enable reliable operation in larger agricultural environments. The implementation of autonomous path planning and obstacle avoidance algorithms would also increase the level of automation and operational safety of the system.

Future research should also focus on improving the seed-metering mechanism to support a wider range of crop types and seed sizes. Adjustable or interchangeable metering disks could be developed to increase flexibility and planting accuracy for different agricultural applications. In addition, adaptive control strategies may be implemented to automatically compensate for changes in soil conditions, terrain irregularities, and varying operational loads. The mechanical structure can also be optimized to reduce overall weight, improve energy efficiency, and increase operating time through better battery management and power utilization techniques.

Furthermore, the current system can be expanded beyond planting operations by incorporating additional agricultural functions such as fertilizer application, irrigation, weed detection, crop monitoring, and environmental sensing. Integrating wireless communication and Internet of Things (IoT) technologies would allow remote monitoring and data collection for precision farming applications. Finally, extensive field testing under different soil types, moisture conditions, and agricultural environments is recommended to evaluate long-term reliability, durability, and economic viability. Such investigations would provide valuable insights for future commercialization and large-scale deployment of automated agricultural vehicles in Ethiopia and other developing regions.

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Appendix A: SolidWorks 2D Drawings

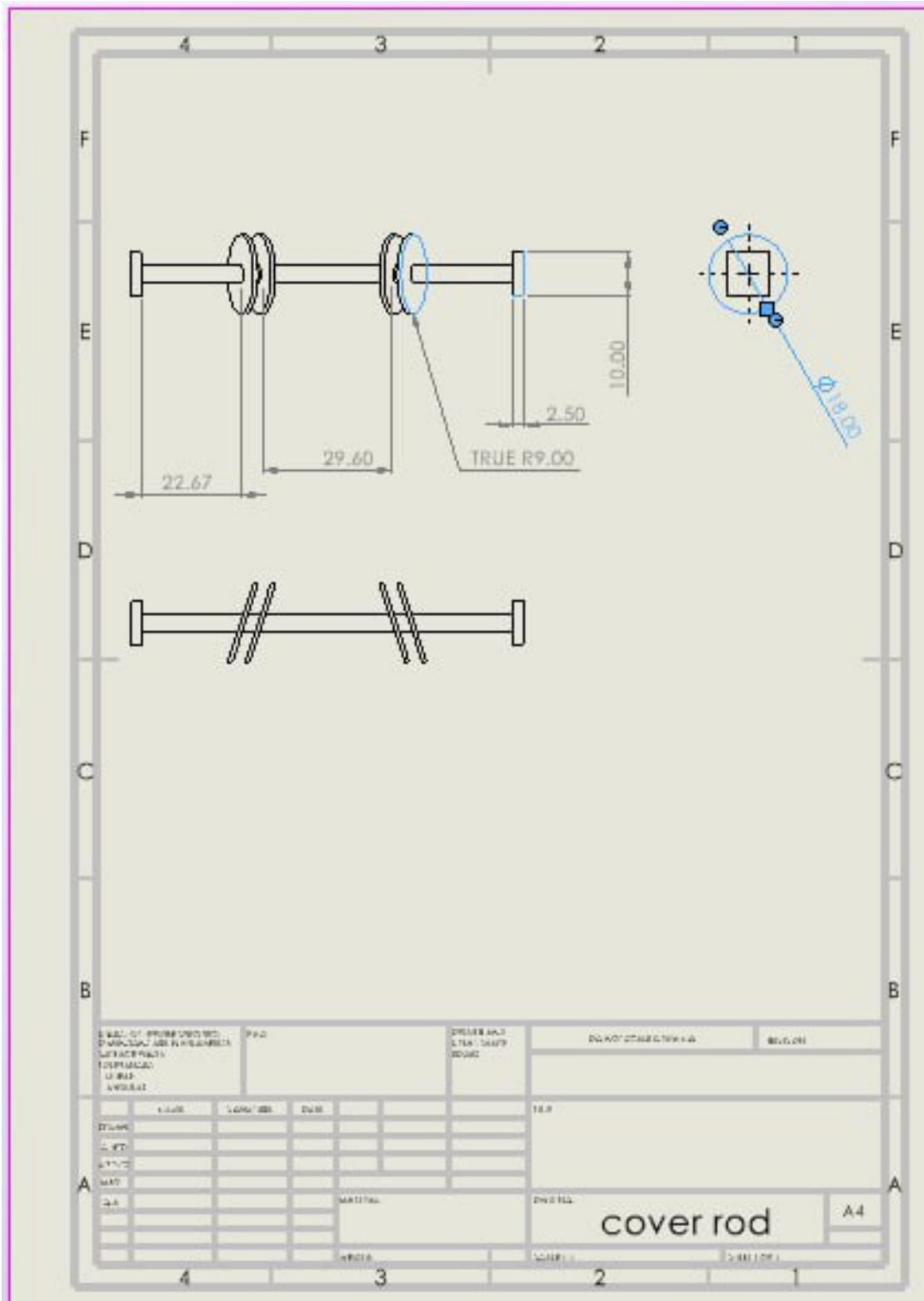


Figure A.1. Covering Disk 2D Drawing

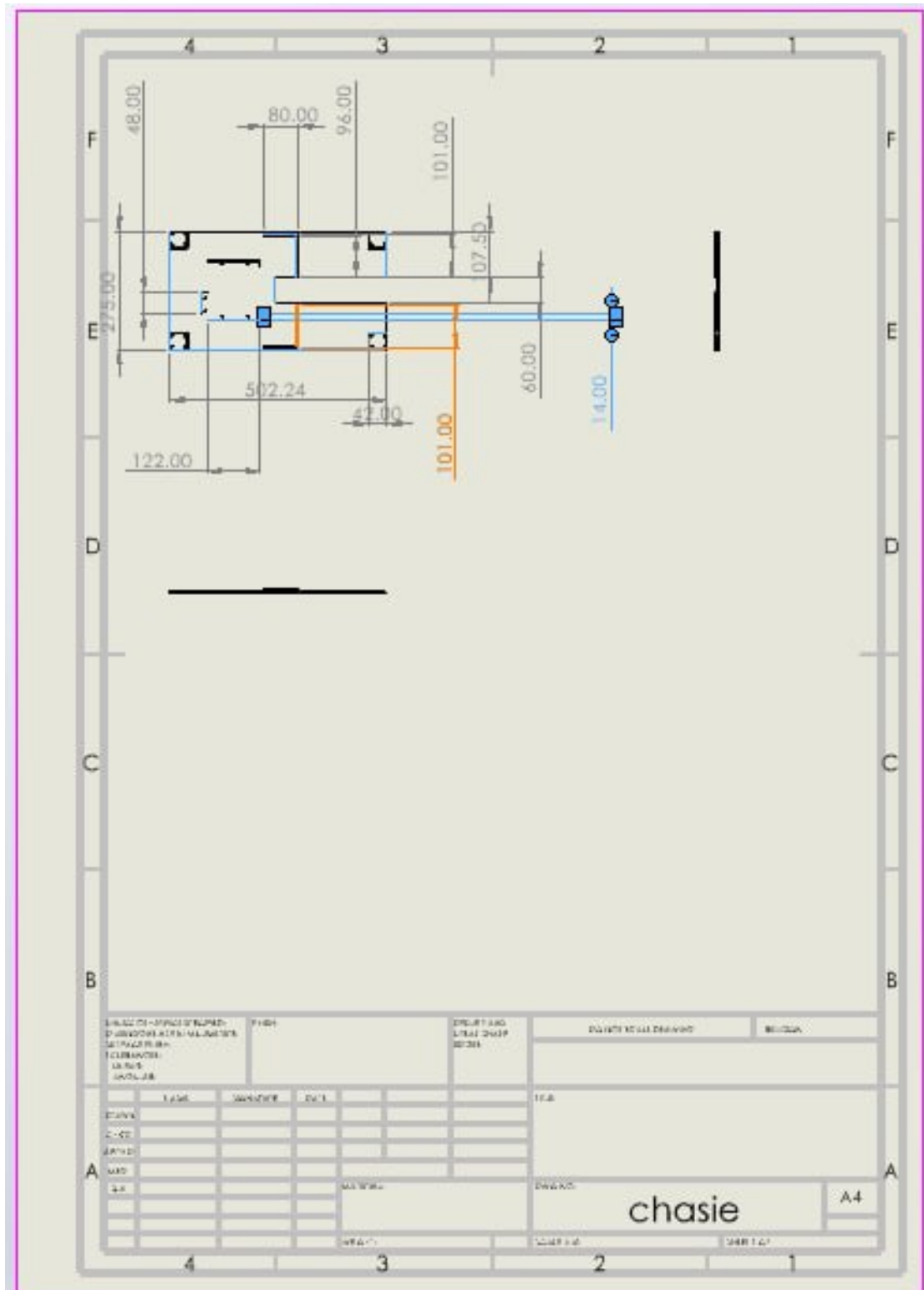


Figure A.2. Chasis 2D Drawing

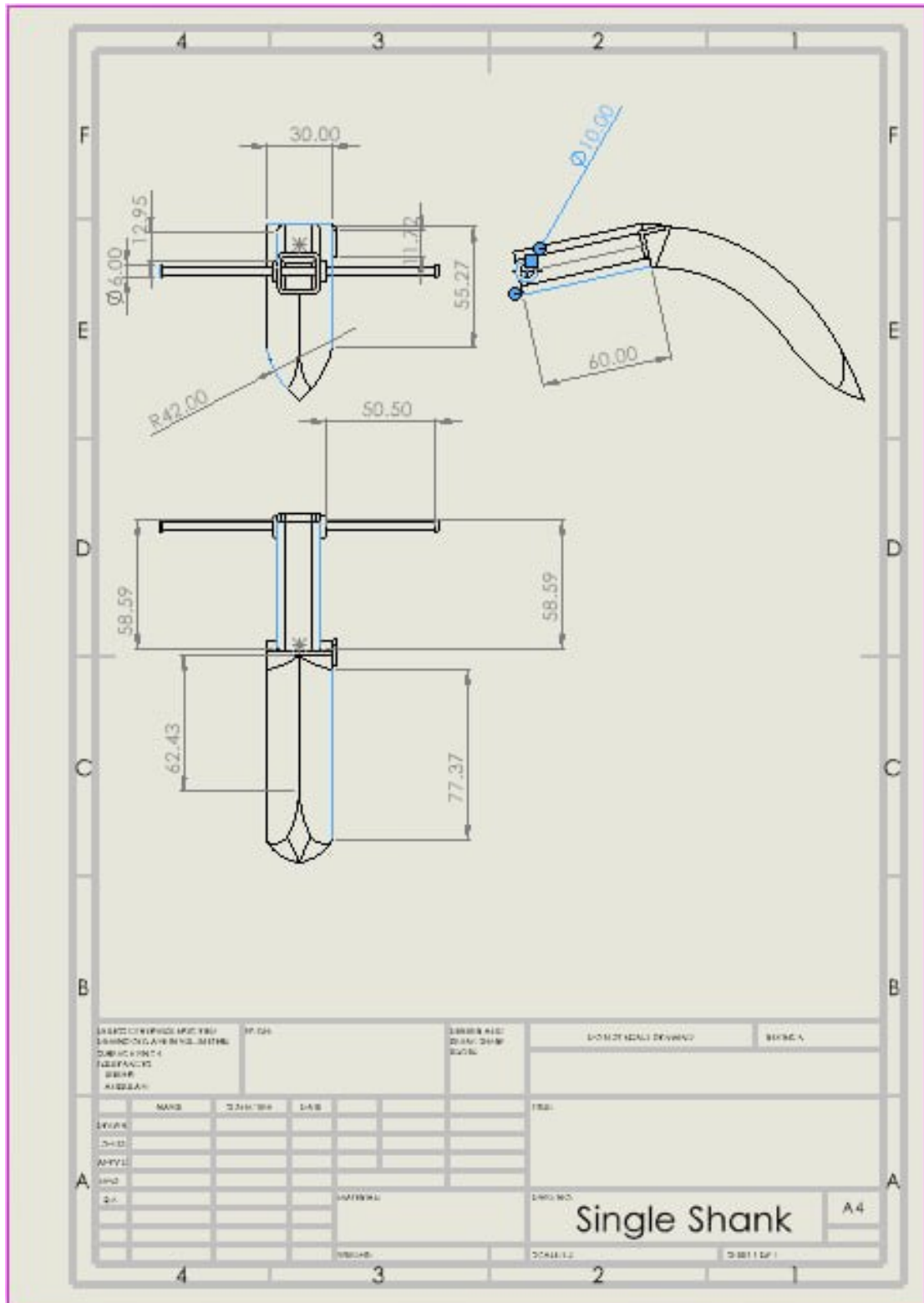
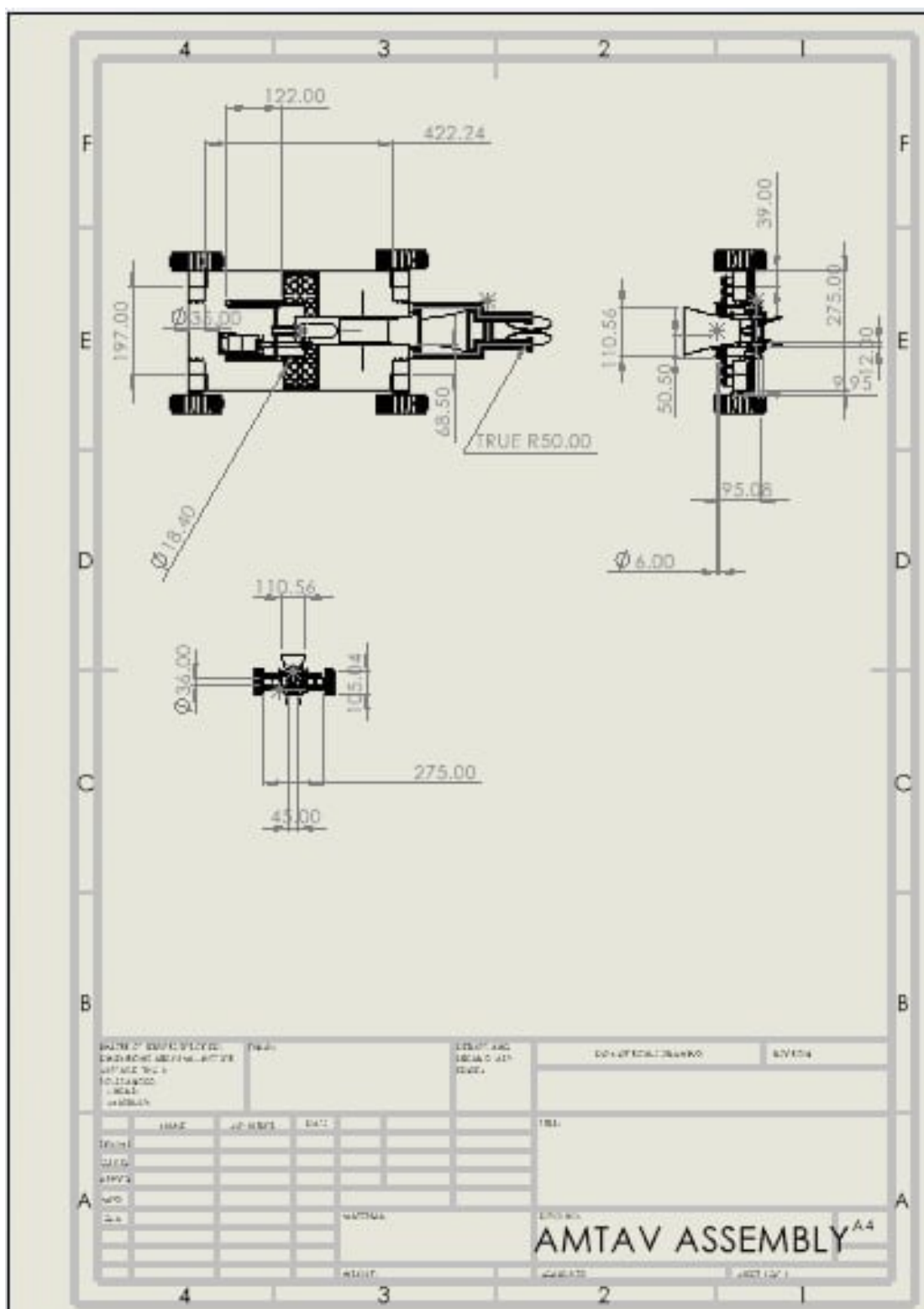


Figure A.3. Single Shank 2D Drawung



Appendix B: Exploded View

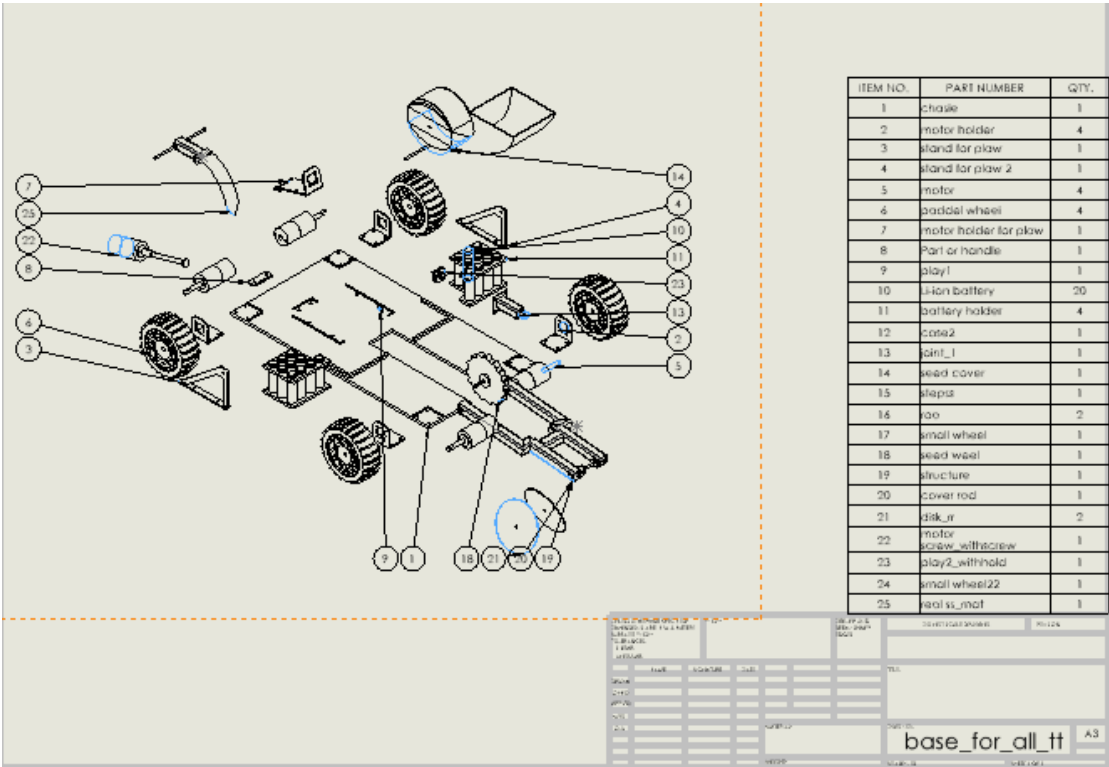


Figure B.1. Exploded View Of the Full AMTAV Assembly

Appendix C: MATLAB Codes

LQR Simulation Code

Code 1.4: LQR Simulation Code

```
1 % Workspace Code (Command Script)
2 A = [0, 0, 0, 1, 0;
3       0, 0, 0.2, 0, 0;
4       0, 0, 0, 0, 1;
5       0, 0, 0, -4.65e-4, 0;
6       0, 0, 0, 0, -0.0221];
7
8 B = [0, 0;
9       0, 0;
10      0, 0;
11      7.752, 7.752;
12      73.63, -73.63];
13
14 C = eye(5);
15 D = zeros(5,2);
16
17 Q = diag([100, 1000, 200, 10, 1]);
18
19 R = 0.005 * eye(2);
20
21 K_zero_overshoot = lqr(A, B, Q, R);
```

Path Tracking

Code 1.1: Path Generator Function (Part 1 - Parameter Setup)

```
1 function [x_ref,y_ref,theta_ref,hd,S,omega_ref] = path_generator(t
2 )
3
4   %#codegen
5
6   L = 4;
7   gap = 2.0;
8
9   R = gap / 2;
10  v = 0.2;
11
12  s = v*t;
13
14  straightLen = L;
15  turnLen = pi*R;
16  cycleLen = straightLen + turnLen;
17
18  cycle = floor(s/cycleLen);
19  local_s = mod(s,cycleLen);
20
21  y_start = -gap*cycle;
22  y_end = -gap*(cycle+1);
23
24  % Smooth rounding factor for the Bezier curve (4/3 gives perfect
25  % shape)
26  alpha = 4/3;
```

Code 1.2: Path Generator Function (Part 2 – Right-Hand Turns)

```

1  if mod(cycle,2) == 0
2      if local_s <= straightLen
3          x_ref = local_s;
4          y_ref = y_start;
5          theta_ref = 0;           % moving right
6          omega_ref = 0;          % no rotation on straight lines
7          hd = 0.04;
8          S = 1;
9      else
10         % --- Smooth Bezier Turn on the Right ---
11         d = local_s - straightLen;
12         tau = d / turnLen;
13
14         % Position
15         x_ref = L + 3 * alpha * R * (1 - tau) * tau;
16         y_ref = (1 - tau)^2 * (1 + 2*tau) * y_start + ...
17                 tau^2 * (3 - 2*tau) * y_end;
18
19         % Velocities with respect to parameter tau
20         vx = 3 * alpha * R * (1 - 2*tau);
21         vy = 6 * tau * (1 - tau) * (y_end - y_start);
22
23         % Time-derivatives for feedforward yaw rate (omega_ref)
24         c = v / turnLen;
25         d_vx = -6 * alpha * R * c;
26         d_vy = 6 * c * (1 - 2*tau) * (y_end - y_start);
27         omega_ref = (vx * d_vy - vy * d_vx) / (vx^2 + vy^2);
28
29         % Continuous heading reference (forces negative range [0 to
30             -pi])
31         theta_raw = atan2(vy, vx);
32         if theta_raw > 0
33             theta_raw = theta_raw - 2*pi;
34         end
35         theta_ref = theta_raw;
36
37         hd = 0.0;
38         S = 0;
39     end

```

Code 1.3: Path Generator Function (Part 3 – Left-Hand Turns)

```

1 else
2     if local_s <= straightLen
3         x_ref = L - local_s;
4         y_ref = y_start;
5         theta_ref = -pi;           % moving left (prevents jump)
6         omega_ref = 0;           % no rotation on straight lines
7         hd = 0.04;
8         S = 1;
9     else
10        % --- Smooth Bezier Turn on the Left ---
11        d = local_s - straightLen;
12        tau = d / turnLen;
13
14        % Position
15        x_ref = 0 - 3 * alpha * R * (1 - tau) * tau;
16        y_ref = (1 - tau)^2 * (1 + 2*tau) * y_start + ...
17                tau^2 * (3 - 2*tau) * y_end;
18
19        % Velocities with respect to parameter tau
20        vx = -3 * alpha * R * (1 - 2*tau);
21        vy = 6 * tau * (1 - tau) * (y_end - y_start);
22
23        % Time-derivatives for feedforward yaw rate (omega_ref)
24        c = v / turnLen;
25        d_vx = 6 * alpha * R * c;
26        d_vy = 6 * c * (1 - 2*tau) * (y_end - y_start);
27        omega_ref = (vx * d_vy - vy * d_vx) / (vx^2 + vy^2);
28
29        % Continuous heading reference (forces negative range [-pi
30            to 0])
31        theta_raw = atan2(vy, vx);
32        if theta_raw > 0
33            theta_raw = theta_raw - 2*pi;
34        end
35        theta_ref = theta_raw;
36
37        hd = 0.0;
38        S = 0;
39    end
40 end

```

LQR Controller Function

Code 1.2: LQR Controller Function

```
1 function [tauR,tauL] = LQR_Controller(ex,ey,etheta,ev,eomega)
2
3 K = [
4     100      316.2278      145.8677      31.8261      10.0984;
5     100     -316.2278     -145.8677      31.8261     -10.0984
6 ];
7 e = [ex;
8      ey;
9      etheta;
10     ev;
11     eomega];
12
13 u_fb = -K*e;
14
15 tau_ff = 0;
16
17 tauR = -(tau_ff + u_fb(1));
18 tauL = -(tau_ff + u_fb(2));
19
20 tau_max = 10;
21 tau_min = -10;
22 tauR = min(max(tauR,tau_min),tau_max);
23 tauL = min(max(tauL,tau_min),tau_max);
24
25 end
```

Appendix D: Prototype Images

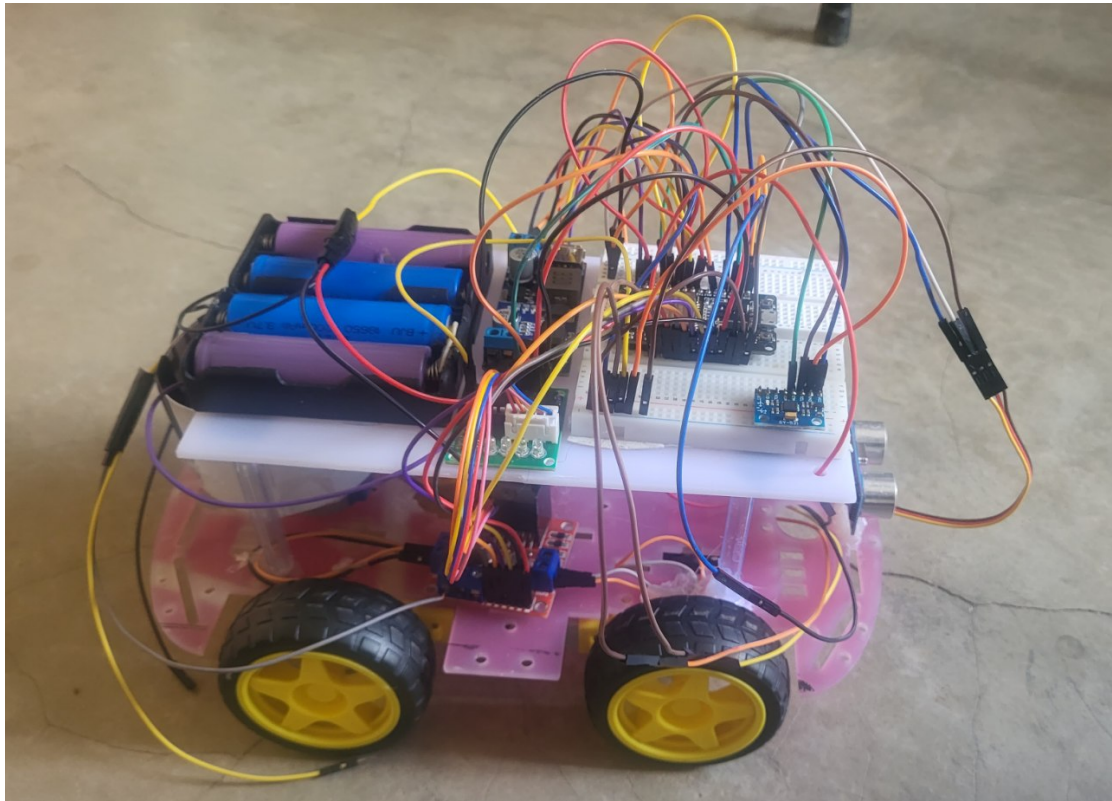


Figure D.1. Prototype Photo 1

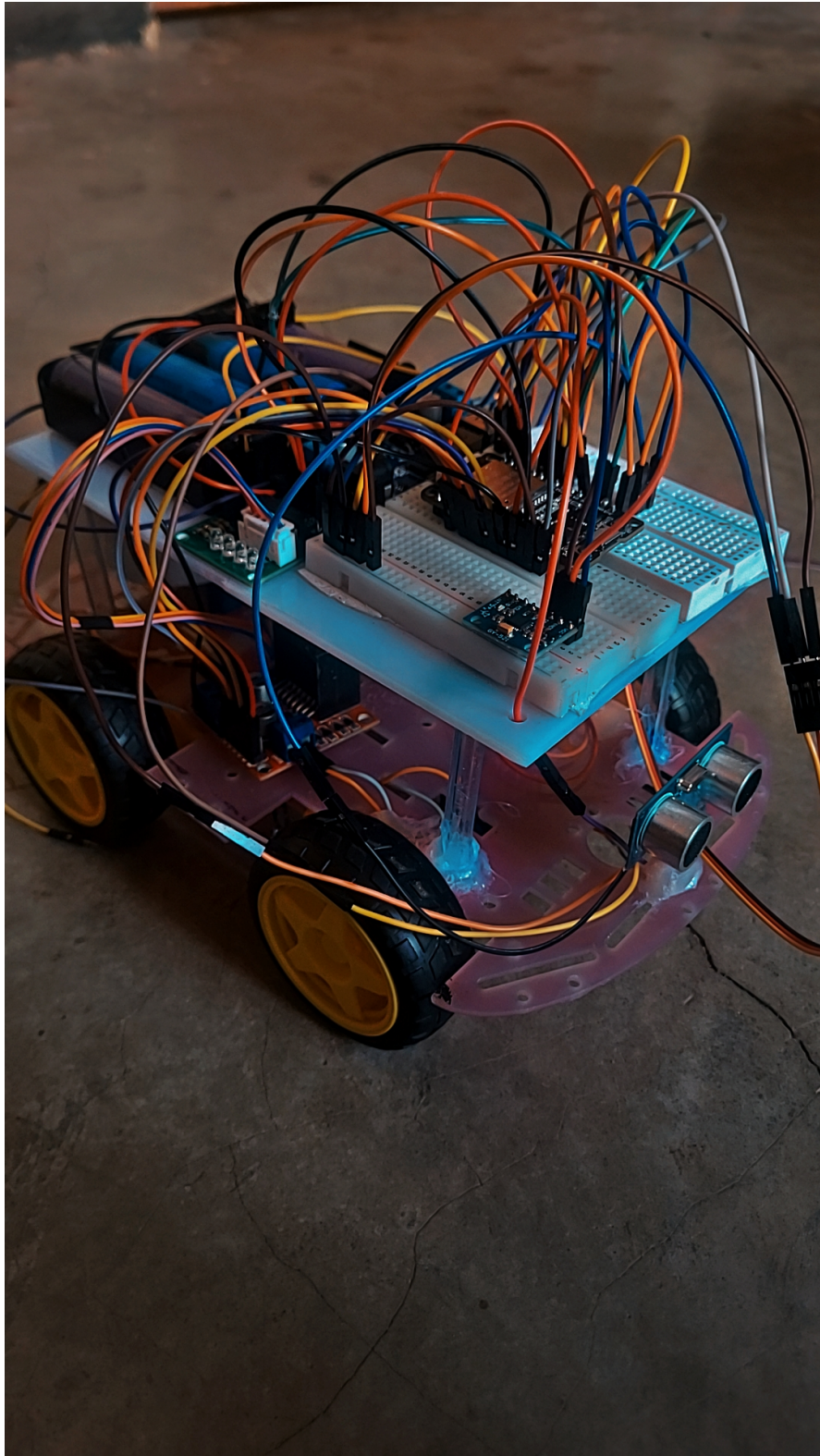


Figure D.2. Prototype Photo 2